Tradeoffs Between Rate and Conditional Content with Encoder-Observed Context

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Abstract

An encoder observes a jointly Gaussian pair (Y, V) and describes Y at rate R for a decoder that sees the description alone; the correlated variable V is the context. A third party retains a noisy copy $S = V + U$ of the context and never decodes. We characterize the conditional content $L = I(X; \hat{X} \mid S)$ of the stored description relative to the retained context: Steinberg's common-reconstruction functional applied to the augmented source $X = (Y, V)$ with distortion on Y alone. Under an ideal conditional reset, L equals the asymptotic erasure work per symbol in units of $k_B T \ln 2$. Three results are primary. First, a closed form: $L(D) = \frac{1}{2} \log_2 g^\star$, with $g^\star$ the larger root of an explicit quadratic whose coefficients are set by (D, ρ^2, τ^2), when Y and V are scalar linear functionals of an arbitrary-dimensional jointly Gaussian source. Second, the exact rate–content region: its Pareto frontier is a coupled two-water-level system, nondegenerate exactly when $0 < \rho^2 < 1$ within the strictly noisy model $\tau^2 > 0$, with the strict tradeoff persisting at the clean-context boundary. Third, non-determination: two instances whose (Y, S) laws are equivalent up to bijective scaling have different L, so replacing the encoder's observation of (Y, V) by its reduced relation to S changes the operational problem, and the augmentation is load-bearing. The closed form recovers the classical rate–distortion function at $\rho = 0$ and Gray's conditional function as $\tau^2 \to 0$, and meets Steinberg's scalar Gaussian formula as $\rho^2 \to 1$. We further settle exactly when the natural determinant lower bound is attained, and give the joint rate–content frontier of the doubly symmetric binary case through a tilt equation anchored at Gray's function and the marginal rate–distortion function.

Index Terms

Rate–distortion theory, side information, common reconstruction, conditional mutual information, Landauer principle, Gaussian sources.

I. Introduction

A stored description of data has two prices. The first is its rate, the space the description occupies, and rate–distortion theory prices it against the fidelity the decoder requires. The second is paid at the end of the description's life: erasing a stored description irreversibly costs work, and when the mechanism performing the reset retains correlated information, the work is governed not by the description's entropy but by its conditional entropy given what the mechanism holds. The two prices are set by different parties. The decoder never sees the reset mechanism's information, and the reset mechanism never serves the decoder. This paper solves the source coding problem that this asymmetry creates, for Gaussian and binary sources, in closed form.

The setting is as follows. An encoder observes a jointly Gaussian pair (Y, V): Y is the variable the decoder must reproduce, and V is a correlated variable, which we call the context ($\rho = \mathbb{E}[YV]$ after normalization). The encoder emits a description under a distortion constraint $\mathbb{E}(Y - \hat{Y})^2 \le D$ that binds the unconditional error; the decoder reconstructs from the description alone. A third party holds $S = V + U$, a noisy observation of the context with noise variance τ^2. The quantity of interest is $L(D) = \min I(X; \hat{X} \mid S)$ over test channels satisfying the distortion constraint and the Markov chain $\hat{X} - X - S$. We call $L(D)$ the minimal conditional content of an admissible description; by the operational theorems of Section II and, for the Gaussian source, Appendix B, it is the least asymptotically achievable value of the normalized conditional entropy $H(M \mid S^n)/n$ of the stored index M among codes meeting the distortion constraint. The thermodynamic reading of this quantity, as minimal ideal erasure work per symbol, is stated once, as Remark 6.

The single-letter functional itself is not new. With the augmented source $X = (Y, V)$ and a distortion that ignores the V coordinate, $\min I(X; \hat{X} \mid S)$ is exactly Steinberg's common-reconstruction functional [12], and Remark 8

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of Lapidoth, Malär, and Wigger [13] notes precisely this augmentation device: an encoder observing an additional sequence correlated with the side information is handled by enlarging the source alphabet and letting the distortion ignore the extra coordinate. What this paper adds is the solution of the augmented problem, and three contributions are primary. First, the closed form: we evaluate the augmented Gaussian functional exactly, $L(D) = \frac{1}{2}\log_2 g^\star$ with $g^\star$ the larger root of an explicit quadratic together with the achieving channel (Theorem 13), after Theorem 9 reduces the problem, for Y and V scalar linear functionals of an arbitrary-dimensional jointly Gaussian source, to the pair (ρ, τ^2); the known Gaussian solutions [12], [13] are scalar and unaugmented. Second, the exact rate–content region with its strict misalignment dichotomy: the Pareto frontier is traced by a coupled two-water-level stationarity system, and within the strictly noisy model $\tau^2 > 0$ it is nondegenerate, so that rate and content genuinely trade off, exactly when $0 < \rho^2 < 1$, with the tradeoff persisting at the clean-context boundary (Theorem 16 and Corollary 19). Third, non-determination by the reduced marginal (Corollary 21): replacing the encoder's observation of the pair (Y, V) by its reduced relation to S changes the operational problem even when the reduced (Y, S) laws are equivalent up to invertible scaling. That is the one-line answer to why the function is not Steinberg's scalar formula: the encoder's access to the clean context V, which neither the decoder nor the third party shares, enters the function itself. The operational layer supports these results rather than leading them: coding theorems characterize $L(D)$ as the least asymptotically achievable normalized conditional entropy $H(M \mid S^n)/n$ of the stored index (Section II, and Appendix B for the Gaussian source), and the thermodynamic reading of $L(D)$ (Remark 6) is one consequence of that characterization.

The remaining results present as structural consequences, extensions, and consistency anchors, and the bracketed words in the theorem titles carry the roles. Section VI settles when the natural determinant lower bound is attained, correcting a plausible conjecture: the attainment condition involves the encoder-accessible conditional covariance, not the third party's. Three extensions follow: the binary case closes completely, for a doubly symmetric binary source with a binary symmetric context channel, through an elementary tilt equation anchored at Gray's binary conditional function and the marginal rate–distortion function (Section VIII); the frontier extends to vector context (Section VII); and the operational region theorem is proved directly for the Gaussian source (Appendix B). The consistency anchors are the three classical corners of the closed form, Shannon's rate–distortion function at $\rho = 0$ [1], Gray's conditional function as $\tau^2 \to 0$ [4], and Steinberg's scalar Gaussian formula at the induced correlation as $\rho^2 \to 1$, together with the strict suboptimality of marginalization and an exact information-bottleneck identity for the weighted objective.

Relation to prior formulations

Table I places each main result against its nearest published neighbor. The single-letter functional is inherited: it is Steinberg's common-reconstruction functional [12], applied unchanged to the augmented source. The augmentation device is also inherited, as Remark 8 of [13]; what is contributed is the evaluation of the reduced problem, and the closed form, the two coupled water levels, the misalignment dichotomy, and the non-determination phenomenon of Corollary 21 are properties of the solution rather than of the reduction. The physical reading leaves the mathematics unchanged: the coordinate $H(M \mid S^n)/n$ is characterized by a proved coding theorem (Section II and Appendix B), and the thermodynamic reading is confined to Remark 6, one consequence of that theorem. The proof machinery is classical, but the solved augmented optimization and its joint region are, to our knowledge, new: we are unaware of a prior derivation of this particular coupled frontier, in which two levels are tied by a common distortion constraint, and the quadratic of Section IV does not appear in the literature we surveyed.

Related work

Conditional rate–distortion theory is due to Gray [3], [4]. In that problem both terminals observe the conditioning variable, and $R_{Y|S}(D)$ prices description under shared context. Here no terminal observes S; Gray's function appears only as the clean-context limit $\tau^2 \to 0$ of our function (Section IV). Wyner and Ziv [5] place the side information at the decoder, where it participates in reconstruction. Our decoder holds nothing but the description, and Wyner–Ziv binning enters the operational theorem of Section II only as a proof device, through a decoder that no party in the model operates.

Steinberg [12] introduced the common-reconstruction constraint: the decoder holds side information, but the encoder must be able to reproduce the decoder's output exactly. The resulting functional is $\min I(X; \hat{X} \mid S)$ under

TABLE I
THEOREM-LEVEL COMPARISON WITH THE NEAREST PRIOR RESULTS.

Result here	Nearest prior	What the prior establishes	What is new here
Closed form (Theorem 13)	Steinberg [12]; Lapidoth–Malär–Wigger [13]	Scalar Gaussian CR formula; scalar formula with general jointly Gaussian SI. Both scalar, encoder observes the source alone	The augmented source (Y, V), with Y and V scalar linear functionals of an arbitrary-dimensional source; the quadratic $g^\star$; the achieving channel
Rate–content region (Theorem 16)	[12], [13]; Gaussian IB [30]	Rate under a CR constraint; unconstrained IB Lagrangian with its eigenmode structure	The joint (R, L) region; a two-water-level frontier under a hard distortion constraint
Misalignment dichotomy (Corollary 19, Remark 20)	none known to us		Strict tradeoff iff $0 < \rho^2 < 1$ under $\tau^2 > 0$; persistence at $\tau^2 = 0$
Non-determination by the reduced marginal (Corollary 21)	Remark 8 of [13]	The augmentation reduction exists	The value genuinely depends on the augmentation: same (Y, S) law up to bijective scaling, different L
Determinant-bound attainment (Theorem 23)	Xiao–Luo [23]; Chen et al. [24]	Rate-only exactness of the determinant bound, without a conditioner	A conditioner present; attainment governed by $\Sigma_{Y\|V}$, not $\Sigma_{Y\|S}$
Binary function (Theorem 27)	Lu *et al.* [19]; Steinberg [12]; Ahmadi et al. [14]	Binary Heegard–Berger with encoder side information and common reconstruction [19]; decoder-SI CR values [12], [14]	Third-party conditioner; the joint (R, L) frontier and its tilt parametrization interpolating Gray at $q = 0$ to the marginal RDF at $q = \frac{1}{2}$; the conditional endpoint is compared with [19] in Appendix D

the Markov chain $\hat{X} - X - S$, the single-letter object of this paper, and Steinberg solved its scalar Gaussian case. Lapidoth, Malär, and Wigger [13] relaxed exact to approximate reproduction and solved the Gaussian problem with general jointly Gaussian side information; their Remark 8 notes that an encoder observing an additional correlated sequence is incorporated by augmenting the source and letting the distortion ignore the extra coordinate. Our problem is exactly that augmented problem, with $X = (Y, V)$ and distortion on Y alone; Table I delineates what is new relative to this line.

Side information held by the encoder alone is classically inert: when it does not enter the distortion measure, it leaves the rate–distortion function unchanged [2]. Kaspi [10] analyzed the configurations in which the encoder is informed of the side information and the decoder may or may not hold it. In our problem the encoder-held context V matters despite the classical fact, because the priced coordinate $I(X; \hat{X} \mid S)$ is conditional on S: the encoder never sees S, but V is correlated with it, and access to V strictly lowers the price (Section IV). Heegard and Berger [11] serve decoders with and without side information from one description; Ahmadi, Tandon, Simeone, and Poor [14] imposed the common-reconstruction constraint on the Heegard–Berger and cascade problems and computed Gaussian and binary erased-side-information cases. Section VIII delineates our binary solution from their calculations and from Steinberg's binary example.

The Gaussian multi-terminal literature under individual distortion constraints supplies the exactness anchor of Section VI: Xiao and Luo [23] derived the joint rate–distortion function of a bivariate Gaussian under individual mean-square constraints, Lapidoth and Tinguely [25] re-derived it, and Nayak, Tuncel, Gündüz, and Erkip [26] and Stylianou, Charalambous, and Charalambous [27] extended the line to successive refinement and to tuples of vector sources. These results enter this paper only at one corner of the attainment condition in Section VI. The closest common-reconstruction counterpart is Lu and Xu's vector Gaussian Heegard–Berger problem with common reconstructions [20], posed under positive diagonal error-covariance constraints; its single-decoder, unpenalized-coordinate limit is the natural point of contact with the conditioner-governed attainment of Section VI, a correspondence we note but do not resolve here.

The priced coordinate also has a security reading. Because L is the normalized conditional entropy $H(M \mid S^n)/n$ of the stored description given the third party's observation, it is the equivocation of that description at a party holding the noisy context, and rate–distortion problems under a secrecy constraint form a long line. Yamamoto priced distortion against the equivocation of the source in the Shannon cipher system [15], Villard and Piantanida characterized secure source coding with side information at the eavesdropper [16], and Schieler and Cuff developed a rate–distortion theory for secrecy systems [17]. Two structural differences separate the present problem from that line. Those results price the equivocation of the *source* and, under a secrecy requirement, seek to make it large, whereas L is the equivocation of the *description* and is made small, as the cost of a conditional reset rather than a secrecy guarantee. And the encoder here observes the clean context V, an access the eavesdropper-side-information models do not grant. The exact Gaussian rate–content region (Theorem 16), the strict-misalignment dichotomy (Corollary 19), and the non-determination by the reduced marginal (Corollary 21) are, to our knowledge, new relative to this line.

The thermodynamic reading of L rests on Landauer's principle [32] and its conditional sharpenings [34], [35]; Remark 6 states the reading and its scope, the paper's only physical claim. Task-based quantization [28] and semantic and goal-oriented communication [29] also price descriptions by a downstream objective rather than by reconstruction alone; the structural difference here is the party that holds the conditioning variable and the party that pays for it. A non-common-reconstruction baseline is remote source coding with two-sided information [21], in which the reconstruction may depend on decoder side information; that freedom is exactly what the common-reconstruction constraint removes, and it is what makes S a pure conditioner here rather than a reconstruction resource. The relation to the information bottleneck [31] is exact and worth stating; Meidlinger, Winkelbauer, and Matz [22] connected the Gaussian information bottleneck to mean-square rate–distortion quantization, a connection on the rate coordinate alone that does not carry the shared hard-distortion constraint or the conditional second coordinate priced here. For any channel with $\widehat{Y} - (Y,V) - S$, writing $T = (Y,V)$, the weighted objective traced in Section V collapses by the Markov chain:

$$J_\alpha := \alpha I(T;\widehat{Y}) + (1-\alpha) I(T;\widehat{Y} \mid S) = I(T;\widehat{Y}) - (1-\alpha) I(S;\widehat{Y}), \tag{1}$$

an information-bottleneck Lagrangian with tradeoff parameter $1-\alpha$, augmented by the hard distortion constraint $\mathbb{E}(Y-\widehat{Y})^2 \le D$. Gaussian IB [30] establishes the optimality of Gaussian linear-noise representations and the eigenmode structure of the unconstrained Lagrangian. The hard Y-distortion constraint changes the program: it binds a reconstruction error, not a mutual information, and the two-water-level frontier with its distortion price is what the constraint adds. Our Gaussian exhaustion lemma (Section IV) plays the role here that the Gaussian-optimality theorem plays there.

Finally, [36] is a preliminary related manuscript by the author, archived at the repository record of the title footnote. The operational region theorem for discrete sources (Section II) and its single-letterization appeared there, as did the scalar Gaussian corner used in Section IV. The material from Section III onward is developed here: pair sufficiency, the closed form, the region and its frontier, the determinant-bound exactness, vector context, the binary joint frontier, and the Gaussian operational theorem of Appendix B. Two of these have close predecessors that we credit in place: the pair-sufficiency reduction of Section III is a distortion-preserving specialization of Armstrong's source-side sufficiency for the information bottleneck [18], and the conditional endpoint of the binary case has a near neighbor in Lu *et al.* [19], compared in Appendix D. The present paper supersedes the source-coding content of that record, which remains the archived evidence artifact.

II. Problem Formulation and the Operational Theorem

All logarithms are base two. Random variables have finite alphabets unless a Gaussian specialization is stated. The model has one source and three parties, and Fig. 1 shows it. Let $(X,S) \sim p_{XS}$, and let (X^n, S^n) be i.i.d. copies. In the Gaussian sections the source letter is the pair $X = (Y,V)$, with Y the described variable and V the context, and the side information is $S = V + U$ with U independent of X; we call S the *conditioner* when its role in the conditional content is in view (the terms side information and conditioner are synonyms for the variable, as third party and reset mechanism are for the party holding it). The operational theorem of this section needs no such structure. The encoder observes X^n alone and emits the description M. The decoder observes M alone and outputs the reproduction; the general discrete theorem below reproduces the full letter, $\widehat{X}^n$, while in the Gaussian sections the reproduction is a scalar estimate $\widehat{Y}$ of Y alone. The third party of Section I, called the *reset mechanism* when its erasure role is in view (the two terms are synonyms throughout), retains S^n, which remains unchanged by the reset, and accesses the stored index M in order to erase it; it never decodes.

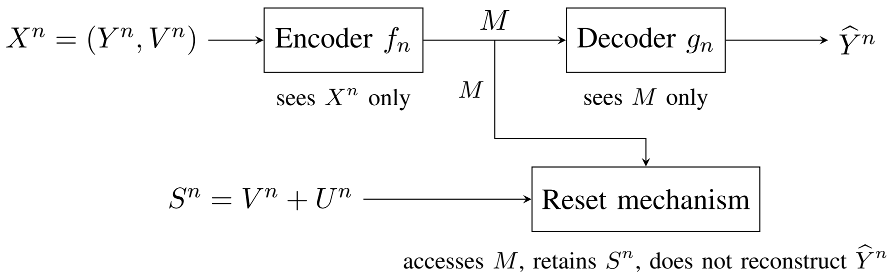

Fig. 1. The system. The encoder observes the source X^n alone and stores the description M at rate R_n; the decoder reconstructs from M alone; the third party (the reset mechanism) retains the side information S^n and accesses the stored index M in order to erase it; it never reconstructs. The figure shows the Gaussian instantiation, where $X = (Y, V)$, $S = V + U$ with U independent of X, and the reproduction is $\widehat{Y}^n$, an estimate of Y^n alone; the general discrete theorem reproduces the full letter.

A distortion measure is a bounded function

$$d : \mathcal{X} \times \widehat{\mathcal{X}} \longrightarrow [0, d_{\max}]. \tag{2}$$

An $(n, \mathcal{M}_n)$ observation code consists of an encoder and decoder

$$f_n : \mathcal{X}^n \to \mathcal{M}_n, \qquad g_n : \mathcal{M}_n \to \widehat{\mathcal{X}}^n. \tag{3}$$

We write $M = f_n(X^n)$ and $\widehat{X}^n = g_n(M)$. Randomized encoders do not improve the region below, but allowing them in the converse is harmless.

The three operational quantities are

$$R_n = \frac{1}{n} \log |\mathcal{M}_n|, \tag{4}$$

$$D_n = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\, d(X_i, \widehat{X}_i), \tag{5}$$

$$L_n = \frac{1}{n} H(M \mid S^n). \tag{6}$$

The dimensionless quantity L_n is the residual description content: the uncertainty about the stored index that remains given the retained side information. We name it the *Landauer content* of the memory relative to the retained side information; the term is ours, chosen for the thermodynamic reading consolidated in Remark 6 after the coding theorem. Every statement and proof of the paper treats L_n as the coding quantity (6), in bits per source symbol.

Definition 1 (Achievability)**.** *A triple (R, L, D) is achievable if there is a sequence of observation codes such that*

$$\limsup_{n\to\infty} R_n \le R, \qquad \limsup_{n\to\infty} L_n \le L, \qquad \limsup_{n\to\infty} D_n \le D. \tag{7}$$

The closure of the achievable set at distortion D is denoted $\mathcal{RW}(D)$.

For a test channel $p(\hat{x} \mid x)$, the Markov chain $\widehat{X} - X - S$ holds because side information is unavailable when the observation is formed. Define

$$\mathcal{T}(D) = \left\{ p(\hat{x} \mid x) : \mathbb{E} d(X, \widehat{X}) \le D \right\}. \tag{8}$$

The two single-letter functions of the paper are the classical rate–distortion function and the content endpoint,

$$R_{\min}(D) = \min_{p(\hat{x}|x) \in \mathcal{T}(D)} I(X; \widehat{X}), \qquad L(D) = \min_{p(\hat{x}|x) \in \mathcal{T}(D)} I(X; \widehat{X} \mid S). \tag{9}$$

Both minima exist by compactness of $\mathcal{T}(D)$ and continuity of the two coordinates. The coding theorem below gives both functions their operational meaning. The following box fixes the two notational scopes of the paper.

Definition 2 (Problem statement: discrete and Gaussian scopes)**.** General letters*, used for the discrete theorem of this section: source $(X, S) \sim p_{XS}$, i.i.d. copies (X^n, S^n), test channel $p(\hat{x} \mid x)$ with the Markov chain $\widehat{X} - X - S$,* *constraint set* $\mathcal{T}(D)$*, and the two functions of* (9). Gaussian letters*, used from Section III on: the source letter is the jointly Gaussian pair* $T = (Y, V)$ *with* $\operatorname{Var} Y = \operatorname{Var} V = 1$ *and* $\mathbb{E}[YV] = \rho$*; the context copy is generated as* $S = V + U$ *with* $U \sim \mathcal{N}(0, \tau^2)$ *independent of everything else; the test channel is* $P_{\widehat{Y}|T}$*, the reproduction* $\widehat{Y}$ *does not see* S*, and the Markov chain is* $\widehat{Y} - T - S$*; the constraint is* $\mathbb{E}(Y - \widehat{Y})^2 \le D$*; and*

$$R_{\min}(D) = \min I(T; \widehat{Y}), \qquad L(D) = \min I(T; \widehat{Y} \mid S),$$

both minima over test channels meeting the constraint.

Lemma 3 (Convexity in the test channel)**.** *For fixed* p_{XS}*, both coordinates* $I(X; \widehat{X})$ *and* $I(X; \widehat{X} \mid S)$ *are convex functions of the test channel* $q(\hat{x} \mid x)$*.*

Proof. Convexity of $I(X; \widehat{X})$ in the channel at fixed input is classical. For the conditional coordinate, the Markov chain $\widehat{X} - X - S$ gives $p(\hat{x} \mid x, s) = q(\hat{x} \mid x)$, so

$$I(X; \widehat{X} \mid S) = \sum_s p(s)\, I_{p(x|s)}(q), \tag{10}$$

a nonnegative combination of mutual informations, each with fixed input distribution $p(x \mid s)$ and the common channel q; each term is convex in q, hence so is the sum. We note that the decomposition $I(X; \widehat{X} \mid S) = I(X; \widehat{X}) - I(S; \widehat{X})$ does *not* yield this conclusion: $I(S; \widehat{X})$ is itself convex in q (the induced channel $p(\hat{x} \mid s) = \sum_x p(x \mid s) q(\hat{x} \mid x)$ is linear in q), so the difference is a priori indefinite. □

Lemma 4 (Closure and right-continuity of the single-letter region)**.** *For a finite-alphabet source* (X, S) *and bounded distortion* d*, define*

$$\mathcal{R}(D) := \bigcup_{q \in \mathcal{T}(D)} \big\{(R, L) : R \ge I_q(X; \widehat{X}),\ L \ge I_q(X; \widehat{X} \mid S)\big\}.$$

Then $\mathcal{R}(D)$ *is closed and nondecreasing in* D*, and it is right-continuous:*

$$\mathcal{R}(D) = \bigcap_{D' > D} \mathcal{R}(D').$$

Equivalently, if $(R_j, L_j) \in \mathcal{R}(D_j)$ *with* $(R_j, L_j) \to (R, L)$ *and* $\limsup_j D_j \le D$*, then* $(R, L) \in \mathcal{R}(D)$*.*

Proof. Monotonicity is the nesting of the constraint sets $\mathcal{T}(D)$. For the sequential form, pick $q_j \in \mathcal{T}(D_j)$ with $R_j \ge I_{q_j}(X; \widehat{X})$ and $L_j \ge I_{q_j}(X; \widehat{X} \mid S)$. The channels live in the compact simplex of conditional distributions, so a subsequence converges to some q; the distortion $\mathbb{E}_q\, d(X, \widehat{X})$ is continuous in the channel, so $\mathbb{E}_q\, d = \lim_j \mathbb{E}_{q_j} d \le \limsup_j D_j \le D$ and $q \in \mathcal{T}(D)$. Both mutual-information coordinates are continuous in the channel on finite alphabets, so $I_q(X; \widehat{X}) = \lim_j I_{q_j}(X; \widehat{X}) \le R$ and likewise $I_q(X; \widehat{X} \mid S) \le L$: the limit (R, L) lies in q's quadrant, hence in $\mathcal{R}(D)$. Closedness is the sequential form with $D_j \equiv D$. For right-continuity, the inclusion $\mathcal{R}(D) \subseteq \bigcap_{D' > D} \mathcal{R}(D')$ is monotonicity, and conversely a point of the intersection is a constant sequence with $(R_j, L_j) \equiv (R, L) \in \mathcal{R}(D'_j)$ along any $D'_j \downarrow D$, so it lies in $\mathcal{R}(D)$ by the sequential form. □

Architecture of the coding theorem: The proof of Theorem 5 below has six steps, and one possible confusion is worth preempting at the outset: the reconstruction decoder never sees S^n; the binning of Wyner–Ziv type appears only inside an accounting argument. (i) The description rate comes from ordinary covering: a rate–distortion codebook whose index M the decoder alone uses. (ii) Reconstruction is from the full index M. (iii) A *virtual* bin index $B = b(M)$ is introduced for conditional-entropy accounting only; no party transmits or stores B. (iv) A Fano bound controls the residual uncertainty about M at a *hypothetical* decoder holding (B, S^n); this hypothetical decoder is a proof device, not a change of the model. (v) An expurgation step selects one deterministic code satisfying the covering and the accounting requirements simultaneously. (vi) The Gaussian source is treated separately, in Appendix B; no Gaussian result of Sections III–VIII relies on the finite-alphabet theorem below.

Theorem 5 (Rate–content–distortion region)**.** *For a discrete memoryless source* (X, S) *and bounded distortion* d*,*

$$\boxed{\mathcal{RW}(D) = \bigcup_{p(\hat{x}|x) \in \mathcal{T}(D)} \Big\{(R, L) : R \ge I(X; \widehat{X}),\ L \ge I(X; \widehat{X} \mid S)\Big\}.} \tag{11}$$

By Lemma 3 both coordinates are convex in the common test channel, so the union is already convex, and it is closed and right-continuous in the distortion level (Lemma 4); consequently no time sharing is required, and every boundary point is attained by a single test channel.

Proof. We first prove the converse. Let a code produce M and $\widehat{X}^n$. Since $\widehat{X}^n$ is a function of M,

$$\begin{aligned}\log|\mathcal{M}_n| &\geq H(M) \geq I(X^n;M) \geq I(X^n;\widehat{X}^n)\\ &\geq \sum_{i=1}^{n} I(X_i;\widehat{X}_i),\end{aligned}\tag{12}$$

where the last step follows from memorylessness and conditioning reducing entropy (Appendix A). Likewise,

$$\begin{aligned}H(M \mid S^n) &\geq I(X^n;M \mid S^n) \geq I(X^n;\widehat{X}^n \mid S^n)\\ &\geq \sum_{i=1}^{n} I(X_i;\widehat{X}_i \mid S_i).\end{aligned}\tag{13}$$

Each induced per-letter channel $p(\hat{x}_i \mid x_i)$ satisfies the Markov constraint $\widehat{X}_i - X_i - S_i$: $\widehat{X}_i$ is a function of $(X_i, X_{\neq i})$ and S_i is independent of $X_{\neq i}$ given X_i for i.i.d. pairs. Let $\bar{q}(\hat{x} \mid x) = \frac{1}{n}\sum_{i=1}^{n} p(\hat{x}_i = \hat{x} \mid X_i = x)$ be the mixture channel; the constraint survives the mixture because $p(s \mid x)$ is common across i. Its distortion is D_n, so $\bar{q} \in \mathcal{T}(D_n)$, and by Lemma 3,

$$R_n \;\geq\; \frac{1}{n}\sum_i I(X_i;\widehat{X}_i) \;\geq\; I_{\bar{q}}(X;\widehat{X}), \qquad L_n \;\geq\; \frac{1}{n}\sum_i I(X_i;\widehat{X}_i \mid S_i) \;\geq\; I_{\bar{q}}(X;\widehat{X} \mid S),$$

which places (R_n, L_n) in the (already convex and closed) union of (11) at distortion D_n. The right side of (11) is the set $\mathcal{R}(D_n)$ of Lemma 4, so, for an achievable triple with $\limsup D_n \leq D$, the sequential form of that lemma places every limit point of (R_n, L_n), and with it (R, L), in the right side of (11) at D.

For achievability, fix $p(\hat{x} \mid x) \in \mathcal{T}(D)$ and $\epsilon > 0$. Generate a random codebook $\mathcal{C}$ of $2^{n(I(X;\widehat{X})+2\epsilon)}$ independent $\hat{x}^n$ codewords drawn from the $\widehat{X}$-marginal, and, independently, a random binning b assigning each codeword index uniformly to one of $2^{n(I(X;\widehat{X}|S)+3\epsilon)}$ bins. The covering encoder selects a codeword jointly (strongly) typical with x^n, declaring failure otherwise, and stores its index M; the decoder reconstructs the codeword from M alone. Two failure events. First, E_{cov}: no jointly typical codeword exists; $\Pr(E_{\mathrm{cov}}) \to 0$, averaged over $(X^n, \mathcal{C})$, by the covering lemma, and on $E_{\mathrm{cov}}^{\mathrm{c}}$ the distortion is at most $D+\epsilon$ by typicality. Second, E_{bin}: the side-information proof device fails. Because $\widehat{X} - X - S$,

$$I(X;\widehat{X} \mid S) = I(X;\widehat{X}) - I(S;\widehat{X}),\tag{14}$$

so each bin holds $2^{n(I(S;\widehat{X})-\epsilon)}$ codewords on average, a per-bin codeword rate ϵ below the packing threshold $I(S;\widehat{X})$, and the Wyner–Ziv packing argument [5] (the Markov lemma for the typicality of the selected codeword with S^n, and the packing bound for wrong codewords in the bin) gives a decoder that recovers M from (B, S^n), $B = b(M)$, with $\Pr(E_{\mathrm{bin}}) \to 0$ averaged over $(X^n, S^n, \mathcal{C}, b)$. This decoder is a proof device; the actual decoder of the code still receives no side information.

Selection of one deterministic code. Let $\epsilon_n := \mathbb{E}_{\mathcal{C},b}\big[\Pr(E_{\mathrm{cov}} \mid \mathcal{C}) + \Pr(E_{\mathrm{bin}} \mid \mathcal{C}, b)\big] \to 0$. By Markov's inequality there is a deterministic pair $(\mathcal{C}, b)$ with $\Pr(E_{\mathrm{cov}}) + \Pr(E_{\mathrm{bin}}) \leq 2\epsilon_n$. Fix it. Its distortion satisfies $D_n \leq D + \epsilon + 2\epsilon_n d_{\max}$, and by Fano's inequality applied to the proof device on this fixed code, $H(M \mid B, S^n) \leq 1 + 2\epsilon_n \log|\mathcal{M}_n| = 1 + 2\epsilon_n\, n\big(I(X;\widehat{X}) + 2\epsilon\big)$, so

$$\begin{aligned}H(M \mid S^n) &\leq H(B) + H(M \mid B, S^n)\\ &\leq n\big(I(X;\widehat{X} \mid S) + 3\epsilon\big) + 1 + 2\epsilon_n\, n\big(I(X;\widehat{X}) + 2\epsilon\big) \;=\; n\big(I(X;\widehat{X} \mid S) + 3\epsilon + o(1)\big).\end{aligned}\tag{15}$$

The selected deterministic code thus satisfies the distortion and the conditional-entropy requirements simultaneously: the stored index has ordinary description rate $I(X;\widehat{X})+2\epsilon$, distortion at most $D+\epsilon+o(1)$, and conditional Landauer content at most $I(X;\widehat{X} \mid S) + 3\epsilon + o(1)$. Letting $\epsilon \downarrow 0$ along a diagonal sequence of blocklengths exhibits the corner $\big(I(X;\widehat{X}), I(X;\widehat{X} \mid S)\big)$ at distortion D; since the union in (11) is already convex and closed (Lemma 3), this exhibits the entire region. $\square$

The binning step in the proof has a simple interpretation. The full codeword index is required by the decoder. Given the side information, the codeword is identified up to a bin of asymptotic size $2^{nI(X;\widehat{X}|S)}$; only that residual uncertainty remains in the stored index. The step has also been verified numerically at finite blocklength (Appendix C).

Remark 6 (Thermodynamic reading)**.** *This remark states the paper's only physical claim; everywhere else, L_n and $L(D)$ are the coding quantities* (6) *and* (9)*. The reset model is the ideal conditional-erasure protocol of [34], [35], specialized to classical side information: the third party may apply ideal joint operations to the record and a work reservoir, conditioned on the classical side information S^n, which is read without disturbance and remains unchanged; only the record M is required to return to a standard state. Under these protocols, the minimal average work to reset M while the third party retains S^n is*

$$W_n^{\min} \;=\; k_{\mathrm{B}}T\ln 2\, H(M \mid S^n) + o(n) \;=\; n\, k_{\mathrm{B}}T\ln 2\,\big(L_n + o(1)\big), \tag{16}$$

Landauer's bound [32] conditioned on the retained side information, tight up to sublinear terms in the classical asymptotic regime [34], [35]. Hence, by Theorem 5, and by Theorem 30 for the Gaussian source, $L(D)$ is the least asymptotic erasure work per symbol, in units of $k_{\mathrm{B}}T\ln 2$, over codes meeting the distortion constraint. The scope of the claim: (16) *prices only the logically irreversible reset of the volatile index M; it does not count the work of acquiring X^n, driving a channel, correcting errors, or finishing in finite time [32], [33], and the codebook, wiring, and any permanent lookup tables are part of the apparatus and are not erased in each cycle. Physical units are restored by multiplying any content value in bits per symbol by $k_{\mathrm{B}}T\ln 2$.*

The content-only endpoint of the region is the minimum Landauer content, the function $L(D)$ defined in (9). As an information quantity, $L(D)$ is Steinberg's common-reconstruction rate–distortion function for side information S [12]: the reproduction must be determined without access to S. The operational roles differ. In the common-reconstruction problem, S is decoder side information that must not corrupt reproducibility; here S is held by a reset mechanism that never serves the decoder. The remainder of the paper characterizes $L(D)$, and the region around it, for the source of Section III.

Remark 7 (The Gaussian pair specialization)**.** *From Section III on, the source letter is the jointly Gaussian pair $X = (Y, V)$ introduced there, and the distortion is $d(x, \hat{x}) = (y - \hat{y})^2$ on the Y coordinate: the decoder is judged on Y alone, while the encoder observes the pair.*

Remark 8 (Scope of the coding theorem)**.** *Theorem 5 is stated and proved for finite alphabets and bounded per-letter distortion. For the jointly Gaussian source of Section III with quadratic distortion on the described variable, the corresponding operational statement is Theorem 30 in Appendix B: the converse is proved there in full, and the achievability by reduction to quantized side information, with two classical probabilistic ingredients imported under precise citations. The Gaussian closed forms of Sections III–VI are evaluations of the single-letter coordinates in* (11)*, and their operational meaning is inherited through Theorem 30.*

Table II collects the notation used from here on.

III. Pair Sufficiency

Let $X \sim \mathcal{N}(0, \Sigma_x)$ on $\mathbb{R}^d$, i.i.d. across source symbols. The described variable is the linear functional $Y = w^{\mathsf{T}}X$; the third party's copy is a second linear functional observed in noise, $S = \gamma^{\mathsf{T}}X + U$. Normalize $\mathrm{Var}(w^{\mathsf{T}}X) = 1$ (rescaling D), write $V = \gamma^{\mathsf{T}}X/\sigma_\gamma$, the normalized context, with $\rho := \mathbb{E}[YV]$, $|\rho| < 1$, and rescale S (invariantly for all mutual informations) to $S = V + U$, $U \sim \mathcal{N}(0, \tau^2)$, $\tau^2 > 0$. Write $s = 1 + \tau^2$, the context-noise parameter (a scalar, not to be confused with the random variable S). Throughout, X, U, and all channel randomness (including any resampling randomness below) are *mutually* independent. This joint form of the independence assumption is load-bearing: it is exactly what the S-kernel argument in the next proof uses.

From here on the Gaussian scope of Definition 2 is in force: the ambient source is X, the pair is $T = (Y, V)$, the reproduction is $\widehat{Y}$, and once Theorem 9 below is proved, all information quantities are written on $(T, \widehat{Y}, S)$. Reproductions may be taken scalar without loss of optimality: replacing $\widehat{X}$ by $\widehat{Y} := w^{\mathsf{T}}\widehat{X}$ leaves the distortion unchanged and can only decrease $I(X; \cdot \mid S)$ (conditional data processing), and $\widehat{Y} - X - S$ persists.

TABLE II
NOTATION.

Symbol	Meaning
$X = (Y, V)$	source letter: described variable Y, context V (normalized in Section III)
$S = V + U$	the third party's copy; $U \sim \mathcal{N}(0, \tau^2)$ independent of X
$s = 1 + \tau^2$	the context-noise parameter, the variance of S after normalization (scalar conditioner; the vector-context section uses Σ_U, with conditioner covariance $\Sigma_S = \Sigma_V + \Sigma_U$)
$\rho = \mathbb{E}[YV]$	correlation of the described variable and the context
D	distortion level, $\mathbb{E}(Y - \widehat{Y})^2 \le D$
M, R_n, L_n	stored index; rate $\frac{1}{n} \log \lvert\mathcal{M}_n\rvert$; Landauer content $\frac{1}{n} H(M \mid S^n)$
$R_{\min}(D)$, $L(D)$	the two single-letter functions of (9)
$\mathsf{B}_{\det}$, Δ, β	determinant lower bound, distortion matrix $\operatorname{diag}(D_A, D_B)$, and $\operatorname{Cov}(Y, V)$ of Section VI
Derived quantities	
(a, b), n	regression coordinates of the test channel $\widehat{Y} = aY + bV + N$, and the noise variance $n = \operatorname{Var} N$ (Section IV)
Q_0, Q_1	unconditional and S-conditional signal parts of the reproduction, $\operatorname{Var}(aY + bV)$ and $\operatorname{Var}(aY + bV \mid S)$ (Section IV)
$h(a, b)$	noise-free distortion contribution $\operatorname{Var}\big((1-a)Y - bV\big)$, so that $\mathbb{E}(Y - \widehat{Y})^2 = h + n$ (23)
$\mu_c = (a\rho + b)/s$	regression coefficient of the reproduction on S (proof of Theorem 13)
$P(g)$, $g^\star$	the quadratic (20) and its larger root; $L(D) = \frac{1}{2} \log_2 g^\star$
α; γ_0, γ_1	frontier weight (rate versus content); the two water levels $n/(Q_0 + n)$ and $n/(Q_1 + n)$ (Theorem 16)
$L^\star_R(D)$, $R^\star_L(D)$	conditional content of the rate-optimal channel; rate of the content-optimal channel (Corollary 19)
Λ, c, y_0	whitened conditional covariance, regression vector, and the unit vector with $Y = y_0^\mathsf{T} Z$ (37) (Sections V, VII)

Standing assumptions and degeneracies: The Gaussian sections run under $0 < D < 1$, $|\rho| < 1$, and $\tau^2 > 0$, and every boundary is handled where it arises rather than excluded silently. For $D \ge 1$ the zero reproduction gives $R = L = 0$, and $D = 0$ is the extended-real limit; both are recorded with the Gaussian operational theorem (Appendix B). The boundaries $\rho^2 \in \{0, 1\}$ are the anchor corners, the classical function at $\rho = 0$ and Steinberg's scalar formula as $\rho^2 \to 1$ (Corollary 14), and the frontier collapse at both is computed in Corollary 19. The clean-context boundary $\tau^2 = 0$ has its own branch wherever it matters: Proposition 10 and the $\tau^2 \to 0$ anchor for the function, Remark 20 for the frontier, and Lemma 22 for the determinant bound; positivity $\tau^2 > 0$ is exactly what makes $\operatorname{Cov}((T, S)) \succ 0$, the hypothesis of the exhaustion lemma (Lemma 11). Degenerate covariances reduce: $\Sigma_T \succ 0$ is assumed, with linearly dependent components dropped where noted, and singular channel residuals Σ_N are dispatched by Remark 12. The determinant lower bound of Section VI can be negative at large distortions and is quoted clipped, $[\mathsf{B}_{\det}]^+$. The closed form's root satisfies $g^\star > 1$ throughout $0 < D < 1$, $\tau^2 > 0$, by the bracketing $P(1) = (D - 1)\tau^2 < 0$ (Theorem 13).

Theorem 9 (Pair sufficiency)**.** *In the problem $L(D) = \min\{I(X; \widehat{Y} \mid S) : \widehat{Y} - X - S,\ \mathbb{E}(Y - \widehat{Y})^2 \le D\}$, it is without loss of optimality to let the channel depend on X only through $T = (Y, V)$, and then $I(X; \widehat{Y} \mid S) = I(T; \widehat{Y} \mid S)$. Consequently $L(D)$ depends on $(\Sigma_x, w, \gamma, \tau^2)$, for the scalar linear functionals $Y = w^\mathsf{T} X$ and $V = \gamma^\mathsf{T} X / \sigma_\gamma$ of a source of arbitrary dimension d, only through (ρ, τ^2).*

Proof. Given an admissible $p(\hat{y} \mid x)$, draw $\widehat{Y}'$ from the conditional law $p(\hat{y} \mid T = t)$ using fresh randomness. (a) $(T, \widehat{Y}') \overset{d}{=} (T, \widehat{Y})$, so the distortion (a functional of $(Y, \widehat{Y})$) is unchanged. (b) $S = V + U$ with U independent of $(X, \widehat{Y})$ and of $(X, \widehat{Y}')$, so in both joints $p(s \mid t, \hat{y}) = p_U(s - v)$: hence $(T, \widehat{Y}', S) \overset{d}{=} (T, \widehat{Y}, S)$. (c) Given T, $\widehat{Y}'$ is independent of (X, S), so $I(X; \widehat{Y}' \mid S) = I(T; \widehat{Y}' \mid S) + I(X; \widehat{Y}' \mid T, S) = I(T; \widehat{Y} \mid S) \le I(X; \widehat{Y} \mid S)$, the last step because T is a function of X. (d) $\widehat{Y}' - X - S$ holds since $\widehat{Y}'$ is a function of $(X, \text{fresh noise})$ with the noise independent of (X, U). The same computation applied to a T-channel directly gives $I(X; \widehat{Y} \mid S) = I(T; \widehat{Y} \mid S)$. $\square$

This reduction is a distortion-preserving specialization of source-side sufficiency for the information bottleneck. Armstrong [18] establishes that a deterministic sufficient statistic $Z = \phi(T)$ with $C - Z - T$ leaves the bottleneck tradeoff unchanged at every blocklength; Theorem 9 is the case $Z = T = (Y, V)$, with the addition that the T-measurable mean-square distortion constraint is preserved, so the reduction holds for the constrained functional and not only the unconstrained bottleneck.

IV. The Function: Scalar Context

One further reduction identifies the class the closed form must beat. *(Marginalized class = scalar corner.)* If the channel depends on X only through Y (that is, $\widehat{Y} - Y - (X,S)$), then given (Y,S), $\widehat{Y}$ is independent of X, so $I(X;\widehat{Y} \mid S) = I(Y;\widehat{Y} \mid S)$, and the class optimum is exactly the scalar-corner value $\frac{1}{2}\log_2\big[(\sigma^2(1-\rho_{YS}^2) + \rho_{YS}^2 D)/D\big]$ of the pair (Y,S), where $\sigma^2 = \operatorname{Var} Y$ and ρ_{YS} is the correlation of that pair, $\rho_{YS}^2 = \rho^2/s$ after the normalization of Section III [12]; the corner's converse applies to every $p(\hat{y} \mid y)$, discrete or not.

Proposition 10 (Anchor: marginalization is strictly suboptimal)**.** *Let $d = 2$, $X \sim \mathcal{N}(0, I_2)$, $Y = X_1$, and $S = X_1 + X_2$ (so $\rho_{YS}^2 = \frac{1}{2}$ and S is X-measurable). For $0 < D < 1$:*

(i) *the marginalized-class optimum is the scalar corner $L_{\mathrm{marg}}(D) = \frac{1}{2}\log_2 \frac{1+D}{2D}$;*

(ii) *the vector problem has the exact value*

$$L_{\mathrm{vec}}(D) \;=\; \tfrac{1}{2}\log_2^+ \frac{1}{2D} \;=\; R_{Y|S}(D),$$

Gray's conditional rate–distortion function of (Y,S) [4] ($\operatorname{Var}(Y \mid S) = \frac{1}{2}$), attained by the linear channel $\widehat{Y} = (1-D)X_1 + DX_2 + N$, $N \sim \mathcal{N}(0, D(1-2D))$, for $D \le \frac{1}{2}$, and by $\widehat{Y} = S/2$ (zero conditional content) for $D \ge \frac{1}{2}$;

(iii) *hence the strict gap $L_{\mathrm{marg}}(D) - L_{\mathrm{vec}}(D) = \frac{1}{2}\log_2(1+D) > 0$ on $(0, \frac{1}{2}]$, and equal to the full marginalized value on $[\frac{1}{2}, 1)$ (where $L_{\mathrm{vec}} = 0$; the absolute gap is decreasing there, but it is all of L_{marg}).*

Proof. (i) is the reduction above, evaluated at $\sigma^2 = 1$, $\rho_{YS}^2 = \frac{1}{2}$.

(ii) *Converse.* For any admissible $\widehat{Y}$ (Markov $\widehat{Y} - X - S$), $I(X;\widehat{Y} \mid S) \ge I(Y;\widehat{Y} \mid S)$, and the induced conditional channel $p(\hat{y} \mid y, s)$ is feasible for Gray's conditional problem at the same distortion, so $I(Y;\widehat{Y} \mid S) \ge R_{Y|S}(D) = \frac{1}{2}\log_2^+\big(\operatorname{Var}(Y \mid S)/D\big) = \frac{1}{2}\log_2^+(1/(2D))$. *Achievability.* For $D \le \frac{1}{2}$ take $\widehat{Y} = (1-D)X_1 + DX_2 + N$: distortion $D^2 + D^2 + D(1-2D) = D$; $\operatorname{Var}(\widehat{Y}) = 1 - D$, $\operatorname{Cov}(\widehat{Y}, S) = 1$, $\operatorname{Var}(\widehat{Y} \mid S) = 1 - D - \frac{1}{2} = \frac{1}{2} - D$, so $I(X;\widehat{Y} \mid S) = \frac{1}{2}\log_2 \frac{(1-2D)/2}{D(1-2D)} = \frac{1}{2}\log_2 \frac{1}{2D}$. For $D \ge \frac{1}{2}$, $\widehat{Y} = S/2$ is a deterministic function of X, has distortion $\mathbb{E}\big((X_1 - X_2)/2\big)^2 = \frac{1}{2} \le D$, and is constant given S, so $I(X;\widehat{Y} \mid S) = 0$.

(iii) Subtract. The closed forms on both sides have been verified numerically. □

The closed form occupies the rest of this section. By Theorem 9 we restrict to T-channels, $T = (Y,V)$, with scalar reproduction $\widehat{Y}$. The working coordinates deserve a plain statement before the machinery. Every jointly Gaussian test channel takes the form $\widehat{Y} = aY + bV + N$ with N independent noise of variance n, and every admissible channel, Gaussian or not, is measured through the same three moments of its regression on T: the unconditional signal part $Q_0 = \operatorname{Var}(aY + bV)$, the S-conditional signal part $Q_1 = \operatorname{Var}(aY + bV \mid S)$, and the noise-free distortion contribution $h = \operatorname{Var}\big((1-a)Y - bV\big)$, so that $\mathbb{E}(Y - \widehat{Y})^2 = h + n$. Everything from here through Section VII is expressed in (a,b,n) and these moments; Table II collects them. The converse rests on one estimation-theoretic lemma. We state it for vector reproductions and vector conditioners, in the generality that Sections V, VI, and VII reuse.

Lemma 11 (Gaussian exhaustion and the determinant identity)**.** *Let (T,S) be jointly Gaussian and centered, $T \in \mathbb{R}^p$, $S \in \mathbb{R}^r$, with $\operatorname{Cov}\big((T,S)\big) \succ 0$. Let $\hat{\mathbf{Y}} \in \mathbb{R}^m$ be any centered random vector with finite second moments, and define*

$$A := \operatorname{Cov}(\hat{\mathbf{Y}}, T)\,\Sigma_T^{-1} \in \mathbb{R}^{m\times p}, \qquad N' := \hat{\mathbf{Y}} - AT, \qquad \Sigma_N := \operatorname{Cov}(N') = \operatorname{Cov}(\hat{\mathbf{Y}}) - A\Sigma_T A^{\mathsf{T}}.$$

Assume $\Sigma_N \succ 0$ and $\operatorname{Cov}(N', S) = 0$. Then

$$I(T;\hat{\mathbf{Y}} \mid S) \;\ge\; \tfrac{1}{2}\log_2 \frac{\det \Sigma_{T|S}}{\det \Sigma_e} \;=\; \tfrac{1}{2}\log_2 \frac{\det\big(A\Sigma_{T|S}A^{\mathsf{T}} + \Sigma_N\big)}{\det \Sigma_N}, \tag{17}$$

where $\Sigma_{T|S} = \Sigma_T - \Sigma_{TS}\Sigma_S^{-1}\Sigma_{ST}$ and Σ_e is the error covariance of the best linear estimator of T from $(\hat{\mathbf{Y}}, S)$. With S deleted throughout, the hypothesis $\operatorname{Cov}(N', S) = 0$ becomes vacuous and the same argument gives

$$I(T;\hat{\mathbf{Y}}) \;\ge\; \tfrac{1}{2}\log_2 \frac{\det\big(A\Sigma_T A^{\mathsf{T}} + \Sigma_N\big)}{\det \Sigma_N}. \tag{18}$$

The jointly Gaussian channel $\hat{\mathbf{Y}} = AT + N$, $N \sim \mathcal{N}(0, \Sigma_N)$ independent of (T, S), realizes the same second moments and attains both bounds with equality simultaneously.

Proof. Step 1: the max-entropy bound. Write $M := \operatorname{Cov}\big((\hat{\mathbf{Y}}, S)\big)$ and $J := \operatorname{Cov}\big(T, (\hat{\mathbf{Y}}, S)\big) \in \mathbb{R}^{p\times(m+r)}$. First, $M \succ 0$: by the Schur-complement criterion on

$$M = \begin{bmatrix} \operatorname{Cov}(\hat{\mathbf{Y}}) & \operatorname{Cov}(\hat{\mathbf{Y}}, S) \\ \operatorname{Cov}(S, \hat{\mathbf{Y}}) & \Sigma_S \end{bmatrix},$$

positivity is equivalent to $\Sigma_S \succ 0$ together with $\operatorname{Cov}(\hat{\mathbf{Y}}) - \operatorname{Cov}(\hat{\mathbf{Y}}, S)\Sigma_S^{-1}\operatorname{Cov}(S, \hat{\mathbf{Y}}) \succ 0$; the first holds because $\operatorname{Cov}((T, S)) \succ 0$, and the second is evaluated in Step 2 to equal $A\Sigma_{T|S}A^\mathsf{T} + \Sigma_N \succeq \Sigma_N \succ 0$. The best linear estimator of T from $(\hat{\mathbf{Y}}, S)$ is $\widehat{T} = JM^{-1}(\hat{\mathbf{Y}}; S)$, with error covariance $\Sigma_e = \Sigma_T - JM^{-1}J^\mathsf{T}$. Since (T, S) is jointly Gaussian, $h(T \mid S) = \frac{1}{2}\log_2\big((2\pi e)^p \det \Sigma_{T|S}\big)$ exactly. For the other term, translation invariance of differential entropy, conditioning reduces entropy, and the Gaussian maximum-entropy bound at fixed covariance give

$$h(T \mid \hat{\mathbf{Y}}, S) = h\big(T - \widehat{T} \bigm| \hat{\mathbf{Y}}, S\big) \;\le\; h\big(T - \widehat{T}\big) \;\le\; \tfrac{1}{2}\log_2\big((2\pi e)^p \det \Sigma_e\big)$$

(with the usual convention that $h = -\infty$ for laws without a density, in which case the bound is trivial). Subtracting the two displays inside $I(T; \hat{\mathbf{Y}} \mid S) = h(T \mid S) - h(T \mid \hat{\mathbf{Y}}, S)$ yields the inequality in (17).

Step 2: the determinant identity. Let $K := \operatorname{Cov}\big((T, S, \hat{\mathbf{Y}})\big)$. The residual N' is uncorrelated with T by construction of the regression and with S by hypothesis, and the change of variables $(T, S, \hat{\mathbf{Y}}) \mapsto (T, S, N')$ is block lower triangular with unit diagonal, hence determinant preserving:

$$\det K = \det \operatorname{Cov}\big((T, S, N')\big) = \det \begin{bmatrix} \Sigma_T & \Sigma_{TS} & 0 \\ \Sigma_{ST} & \Sigma_S & 0 \\ 0 & 0 & \Sigma_N \end{bmatrix} = \det \Sigma_S \, \det \Sigma_{T|S} \, \det \Sigma_N,$$

using in the last step the Schur factorization $\det \operatorname{Cov}((T, S)) = \det \Sigma_S \, \det \Sigma_{T|S}$. The same Schur factorization applied to K in the other block order gives $\det K = \det M \, \det \Sigma_e$, and applied to M itself gives

$$\det M = \det \Sigma_S \cdot \det\Big(\operatorname{Cov}(\hat{\mathbf{Y}}) - \operatorname{Cov}(\hat{\mathbf{Y}}, S)\Sigma_S^{-1}\operatorname{Cov}(S, \hat{\mathbf{Y}})\Big).$$

It remains to evaluate the conditional block. From $\hat{\mathbf{Y}} = AT + N'$ with $\operatorname{Cov}(N', T) = 0$ and $\operatorname{Cov}(N', S) = 0$, $\operatorname{Cov}(\hat{\mathbf{Y}}) = A\Sigma_T A^\mathsf{T} + \Sigma_N$ and $\operatorname{Cov}(\hat{\mathbf{Y}}, S) = A\Sigma_{TS}$, so

$$\operatorname{Cov}(\hat{\mathbf{Y}}) - \operatorname{Cov}(\hat{\mathbf{Y}}, S)\Sigma_S^{-1}\operatorname{Cov}(S, \hat{\mathbf{Y}}) = A\big(\Sigma_T - \Sigma_{TS}\Sigma_S^{-1}\Sigma_{ST}\big)A^\mathsf{T} + \Sigma_N = A\Sigma_{T|S}A^\mathsf{T} + \Sigma_N.$$

Combining the three displays,

$$\det \Sigma_e = \frac{\det K}{\det M} = \frac{\det \Sigma_{T|S} \, \det \Sigma_N}{\det\big(A\Sigma_{T|S}A^\mathsf{T} + \Sigma_N\big)},$$

which is the identity in (17); in particular $\Sigma_e \succ 0$, so Step 1's bound is finite. Deleting S (formally $r = 0$: $K = \operatorname{Cov}((T, \hat{\mathbf{Y}}))$, $M = \operatorname{Cov}(\hat{\mathbf{Y}})$) gives $\det \Sigma_{e0} = \det \Sigma_T \det \Sigma_N / \det(A\Sigma_T A^\mathsf{T} + \Sigma_N)$ and hence (18).

Step 3: equality. For the stated Gaussian channel the conditional law of T given $(\hat{\mathbf{Y}}, S)$ is Gaussian with covariance Σ_e for every realization, so $h(T \mid \hat{\mathbf{Y}}, S) = \frac{1}{2}\log_2\big((2\pi e)^p \det \Sigma_e\big)$ and (17) holds with equality; likewise for (18). The channel's moments match those of the given channel: $\operatorname{Cov}(AT + N, T) = A\Sigma_T$ and $\operatorname{Cov}(AT + N) = A\Sigma_T A^\mathsf{T} + \Sigma_N$. □

Remark 12 (Degenerate residuals)**.** *If Σ_N is singular, some combination $\eta^\mathsf{T}\hat{\mathbf{Y}}$ equals $\eta^\mathsf{T}AT$ almost surely. If $\eta^\mathsf{T}A \neq 0$ the reproduction reveals a noiseless linear functional of the continuous T, so $I(T; \hat{\mathbf{Y}} \mid S) = \infty$ and every bound below holds trivially; if $\eta^\mathsf{T}A = 0$ that combination of the reproduction is zero and may be dropped, reducing m. All uses of Lemma 11 below may therefore assume $\Sigma_N \succ 0$.*

Theorem 13 (Core: the closed form)**.** *With $s = 1 + \tau^2$, the context-noise parameter of Section III, for $0 < D < 1$,*

$$L(D) = \tfrac{1}{2}\log_2 g^\star, \qquad g^\star = \frac{(D + s - \rho^2) + \sqrt{(D + s - \rho^2)^2 - 4Ds(1 - \rho^2)}}{2Ds}, \tag{19}$$

the larger root of the quadratic

$$P(g) = Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2). \tag{20}$$

Since $P(1) = (D-1)\tau^2 < 0$, exactly one root exceeds 1, so $g^\star$ is well defined. The optimum is attained by the jointly Gaussian channel

$$\widehat{Y} = aY + bV + N, \qquad a = \frac{g^\star - 1}{g^\star}, \quad b = \frac{(g^\star - 1)\rho}{g^\star k}, \quad k = g^\star s - 1, \quad N \sim \mathcal{N}\Big(0,\ \frac{Q_1}{g^\star - 1}\Big), \tag{21}$$

where $Q_1 = a^2 + b^2 + 2ab\rho - (a\rho + b)^2/s$ is evaluated at these (a, b).

Proof. Moments. By Theorem 9 and the scalar-reproduction reduction of Section III, restrict to channels $p(\hat{y} \mid t)$. Fix an admissible channel with $\mathbb{E}(Y - \widehat{Y})^2 \le D$. Centering $\widehat{Y}$ leaves $I(T; \widehat{Y} \mid S)$ unchanged and can only decrease the distortion, so take $\mathbb{E}\widehat{Y} = 0$. Define the regression coordinates

$$\begin{pmatrix} a \\ b \end{pmatrix} := \Sigma_T^{-1} \begin{pmatrix} \mathbb{E}[Y\widehat{Y}] \\ \mathbb{E}[V\widehat{Y}] \end{pmatrix}, \qquad Q_0 := a^2 + b^2 + 2ab\rho = \operatorname{Var}(aY + bV), \qquad n := \operatorname{Var}\widehat{Y} - Q_0,$$

so that $\mathbb{E}[Y\widehat{Y}] = a + b\rho$ and $\mathbb{E}[V\widehat{Y}] = a\rho + b$; $n \ge 0$ is the residual variance of $\widehat{Y}$ after linear regression on T. If $n = 0$, Remark 12 applies: either $(a, b) \ne (0, 0)$ and $I(T; \widehat{Y} \mid S) = \infty$, or $(a, b) = (0, 0)$ and $\widehat{Y} = 0$ with distortion $1 > D$. So take $n > 0$.

The Markov moment identity. The residual $N' = \widehat{Y} - aY - bV$ is uncorrelated with T by construction. It is also uncorrelated with $S = V + U$: it is uncorrelated with V, a coordinate of T, and $\mathbb{E}[UN'] = 0$ because U is independent of $(X, \widehat{Y})$ by the mutual independence of X, U, and the channel randomness. Hence $\operatorname{Cov}(N', S) = 0$, which is the hypothesis of Lemma 11, and which restates the moment identity $\mathbb{E}[S\widehat{Y}] = \mathbb{E}[V\widehat{Y}]$.

The converse bound, written out. Here $p = 2$, $m = 1$, $r = 1$, $A = (a, b)$, $\Sigma_N = n$, and

$$\Sigma_T = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}, \qquad \Sigma_{TS} = \begin{bmatrix} \rho \\ 1 \end{bmatrix}, \qquad \Sigma_S = s,$$

$$\Sigma_{T|S} = \Sigma_T - \frac{1}{s}\,\Sigma_{TS}\Sigma_{TS}^{\mathsf{T}} = \begin{bmatrix} 1 - \rho^2/s & \rho\tau^2/s \\ \rho\tau^2/s & \tau^2/s \end{bmatrix}, \qquad \det \Sigma_{T|S} = \frac{\tau^2(1 - \rho^2)}{s} > 0.$$

The quadratic form in (17) is

$$A\Sigma_{T|S}A^{\mathsf{T}} = Q_0 - \frac{(a\rho + b)^2}{s} = \operatorname{Var}(aY + bV \mid S) =: Q_1,$$

and, explicitly, $M = \operatorname{Cov}\big((\widehat{Y}, S)\big) = \left[\begin{smallmatrix} Q_0 + n & a\rho + b \\ a\rho + b & s \end{smallmatrix}\right]$ with $\det M = s(Q_0 + n) - (a\rho + b)^2 = s(Q_1 + n)$, while $\det \operatorname{Cov}\big((T, S, \widehat{Y})\big) = s \det \Sigma_{T|S} \cdot n$, so $\det \Sigma_e = n \det \Sigma_{T|S}/(Q_1 + n)$ and Lemma 11 reads

$$I(T; \widehat{Y} \mid S) \ \ge\ \tfrac{1}{2}\log_2 \frac{\det \Sigma_{T|S}}{\det \Sigma_e} \ =\ \tfrac{1}{2}\log_2 \frac{Q_1 + n}{n}. \tag{22}$$

The distortion is a function of the same moments:

$$\mathbb{E}(Y - \widehat{Y})^2 = 1 - 2(a + b\rho) + Q_0 + n = h(a, b) + n, \qquad h(a, b) := (1 - a)^2 - 2(1 - a)b\rho + b^2, \tag{23}$$

and $h(a, b) = \operatorname{Var}\big((1 - a)Y - bV\big) \ge 0$. Therefore

$$L(D) \ \ge\ \inf\Big\{ B(a, b, n) := \tfrac{1}{2}\log_2 \tfrac{Q_1 + n}{n} \ :\ h(a, b) + n \le D,\ n > 0 \Big\}, \tag{24}$$

and it suffices to evaluate the right side; achievability below matches it.

Activity, coercivity, existence. At fixed (a, b) with $Q_1 > 0$, B is strictly decreasing in n ($\partial_n[\ln(Q_1 + n) - \ln n] = (Q_1 + n)^{-1} - n^{-1} < 0$), so the infimum is unchanged by imposing the active constraint $n = D - h(a, b)$. On the feasible set, (a, b) is confined to the compact set $\{h \le D\}$; since $\Sigma_{T|S} \succ 0$, the form $Q_1 = (a, b)\,\Sigma_{T|S}\,(a, b)^{\mathsf{T}}$ vanishes only at $(a, b) = (0, 0)$, where $h = 1 > D$, so by continuity $Q_1 \ge q_0 > 0$ on the feasible set. Hence $B \ge \frac{1}{2}\log_2\big((q_0 + n)/n\big) \to \infty$ as $n \to 0$, the sublevel sets of the reduced objective

$$\Phi(a, b) := \tfrac{1}{2}\log_2 \frac{Q_1(a, b) + n(a, b)}{n(a, b)}, \qquad n(a, b) = D - h(a, b),$$

are compact, and the infimum in (24) is attained at some (a, b) with $h(a, b) < D$: an interior stationary point of Φ.

The stationarity system is linear in (a, b) *at fixed* g. At a stationary point write $g := (Q_1 + n)/n$ and $\mu_c := (a\rho + b)/s$; note $g > 1$ because $Q_1 > 0$. The partial derivatives of the three quadratics are

$$\partial_a Q_0 = 2(a + b\rho), \qquad \partial_b Q_0 = 2(b + a\rho), \qquad \partial_a h = \partial_a Q_0 - 2, \qquad \partial_b h = \partial_b Q_0 - 2\rho,$$
$$\partial_a Q_1 = \partial_a Q_0 - 2\rho\mu_c, \qquad \partial_b Q_1 = \partial_b Q_0 - 2\mu_c,$$

hence the four gradient identities

$$\partial_a(Q_0 - h) = 2, \quad \partial_b(Q_0 - h) = 2\rho, \quad \partial_a(Q_1 - h) = 2(1 - \rho\mu_c), \quad \partial_b(Q_1 - h) = 2(\rho - \mu_c). \tag{25}$$

(Only the last two enter here; all four enter the weighted system of Section V.) With $n = D - h$, so $\partial_x n = -\partial_x h$, stationarity of $2 \ln 2 \cdot \Phi = \ln(Q_1 + n) - \ln n$ in $x \in \{a, b\}$ reads

$$\frac{\partial_x(Q_1 - h)}{Q_1 + n} + \frac{\partial_x h}{n} = 0, \qquad \text{that is,} \qquad \frac{1}{g}\,\partial_x(Q_1 - h) + \partial_x h = 0.$$

By (25) and $\partial_a h = 2(a + b\rho - 1)$, $\partial_b h = 2(a\rho + b - \rho)$, the two equations are

$$a + b\rho = 1 - \frac{1 - \rho\mu_c}{g}, \qquad a\rho + b = \rho - \frac{\rho - \mu_c}{g}. \tag{26}$$

Substituting $\mu_c = (a\rho + b)/s$ makes (26) a 2×2 linear system in (a, b) whose coefficients depend on g alone, and it solves in closed form. The second equation reads $gs\mu_c - \mu_c = \rho(g - 1)$, so with $k := gs - 1 > 0$,

$$\mu_c = \frac{\rho(g - 1)}{k}. \tag{27}$$

Subtracting ρ times the first equation of (26) from the second gives $(a\rho + b) - \rho(a + b\rho) = b(1 - \rho^2)$ on the left and $\mu_c(1 - \rho^2)/g$ on the right, so $b = \mu_c/g$; the first equation then yields $a = 1 - (1 - \rho\mu_c)/g - b\rho = 1 - 1/g$. The stationary channel coefficients at level g are therefore

$$1 - a = \frac{1}{g}, \qquad b = \frac{\mu_c}{g} = \frac{(g - 1)\rho}{gk}. \tag{28}$$

Reduction to the quadratic. At the values (28), $(1 - a)Y - bV = (Y - \mu_c V)/g$, so

$$h = \frac{1 - 2\rho\mu_c + \mu_c^2}{g^2}, \tag{29}$$

and, writing $aY + bV = Y - (Y - \mu_c V)/g$ and using the definition $a\rho + b = s\mu_c$,

$$Q_0 = 1 - \frac{2(1 - \rho\mu_c)}{g} + h, \qquad Q_1 = Q_0 - \frac{(a\rho + b)^2}{s} = 1 - \frac{2(1 - \rho\mu_c)}{g} + h - s\mu_c^2.$$

The definition of g and the active constraint combine to $(g - 1)(D - h) = Q_1$. Substituting the display for Q_1 and collecting the h terms ($h + (g - 1)h = gh = (1 - 2\rho\mu_c + \mu_c^2)/g$ by (29)), the terms in $\rho\mu_c$ cancel and

$$1 + \frac{\mu_c^2 - 1}{g} - s\mu_c^2 = (g - 1)D. \tag{30}$$

Finally substitute (27). Multiplying (30) by gk^2 and using $k^2\mu_c^2 = \rho^2(g - 1)^2$,

$$gk^2 + \rho^2(g - 1)^2 - k^2 - gs\rho^2(g - 1)^2 = g(g - 1)Dk^2.$$

Since $1 - gs = -k$, the second and fourth terms on the left combine to $-\rho^2(g - 1)^2 k$, so the left side is $(g - 1)k\big[k - \rho^2(g - 1)\big]$, and dividing by $(g - 1)k > 0$,

$$k - \rho^2(g - 1) = gDk \quad \Longleftrightarrow \quad Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2) = 0,$$

that is, $P(g) = 0$. Every interior stationary point of Φ therefore sits at a root of P exceeding 1.

Root bracketing and uniqueness. $P(1) = Ds - (D + s - \rho^2) + (1 - \rho^2) = (D - 1)(s - 1) = (D - 1)\tau^2 < 0$ for $D < 1$, $\tau^2 > 0$. The leading coefficient Ds is positive and $P(0) = 1 - \rho^2 > 0$, so P has two real roots, exactly one exceeding 1 (the other lies in $(0,1)$). Hence every stationary point has $g = g^\star$, and by (28) the stationary (a,b), and with it $n = Q_1/(g^\star - 1)$, is unique. The attained minimum of (24) is $\frac{1}{2}\log_2 g^\star$.

Achievability. The channel (21) has exactly the moments (28) with residual variance $n = Q_1/(g^\star - 1)$. Its distortion is $h + n$, and $(g^\star - 1)n = Q_1$ together with $P(g^\star) = 0$ is, by the chain of equivalences above run backward, the statement $h + n = D$. The channel is jointly Gaussian with (T, S), so the equality clause of Lemma 11 gives

$$I(T; \widehat{Y} \mid S) = \tfrac{1}{2}\log_2 \frac{Q_1 + n}{n} = \tfrac{1}{2}\log_2 g^\star.$$

Hence $L(D) = \frac{1}{2}\log_2 g^\star$. □

Corollary 14 (Anchors: the three classical corners)**.** *(i) At $\rho = 0$, $P(g) = (Dg - 1)(sg - 1)$ and $g^\star = 1/D$: $L(D) = \frac{1}{2}\log_2(1/D)$ is the classical rate–distortion function, attained with $b = 0$ by the classical reverse test channel. (ii) As $\tau^2 \to 0$, $L(D) \to \frac{1}{2}\log_2^+\big((1-\rho^2)/D\big)$, Gray's conditional rate–distortion function of the pair (Y, V) [4], matching Proposition 10. (iii) As $\rho^2 \to 1$, $L(D) \to \frac{1}{2}\log_2\big[(D+\tau^2)/(D(1+\tau^2))\big]$, Steinberg's scalar Gaussian formula at the induced correlation $\rho_{YS}^2 = 1/s$ [12].*

Proof. (i) $(Dg-1)(sg-1) = Dsg^2 - (D+s)g + 1 = P(g)\big|_{\rho=0}$. The roots are $1/D > 1$ and $1/s < 1$, so $g^\star = 1/D$. At this root (21) gives $b = 0$, $a = 1 - D$, $Q_1 = (1-D)^2$, and $\operatorname{Var} N = (1-D)^2/(1/D - 1) = D(1-D)$: the classical reverse channel.

(ii) At $s = 1$, $P(g) = Dg^2 - (D+1-\rho^2)g + (1-\rho^2) = (g-1)\big(Dg - (1-\rho^2)\big)$, with discriminant $\big(D - (1-\rho^2)\big)^2$, so the larger root at $s = 1$ is exactly $\max\{1, (1-\rho^2)/D\}$; the roots of P are continuous in s, and the limit follows. For $D < 1 - \rho^2$ the value is $\frac{1}{2}\log_2\big(\operatorname{Var}(Y \mid V)/D\big)$ with $\operatorname{Var}(Y \mid V) = 1 - \rho^2$, Gray's conditional function; for $D \ge 1 - \rho^2$ it is 0.

(iii) At $\rho^2 = 1$ the constant term of P vanishes: $P(g) = g\big(Dsg - (D+\tau^2)\big)$, with roots 0 and $(D+\tau^2)/(Ds)$; the latter exceeds 1 because $(D+\tau^2) - Ds = \tau^2(1-D) > 0$. By root continuity $g^\star \to (D+\tau^2)/(Ds)$ as $\rho^2 \to 1$. Steinberg's scalar function at unit source variance and side-information correlation ρ_{YS}^2 is $\frac{1}{2}\log_2\big[(1 - \rho_{YS}^2 + \rho_{YS}^2 D)/D\big]$ [12]; at $\rho_{YS}^2 = \rho^2/s \to 1/s$ this is $\frac{1}{2}\log_2\big[(\tau^2 + D)/(sD)\big]$, the same value. □

Remark 15 (Convergence rates at the anchors)**.** *Each anchor of Corollary 14 is approached at a linear rate, with an explicit first-order coefficient. In every case the anchor value is a simple root of P ($P'(g^\star) \neq 0$, evaluated below), so the root is differentiable in the perturbing parameter ϵ with $dg^\star/d\epsilon = -P_\epsilon/P'(g^\star)$, and $L = \frac{1}{2}\log_2 g^\star$ converts the derivative to the gap coefficient; the three evaluations are verified symbolically and numerically (Appendix C). All values are in bits; the factor $\ln 2$ converts the natural-log derivative to base two. (i) As $\rho^2 \to 0$ at fixed (τ^2, D), with $P'(1/D) = s - D > 0$,*

$$L(D) - \tfrac{1}{2}\log_2 \frac{1}{D} = -\frac{1-D}{2\ln 2\,(s-D)}\,\rho^2 + O(\rho^4):$$

the gap to the classical rate–distortion function closes linearly in ρ^2 with coefficient $-(1-D)/\big(2\ln 2\,(s-D)\big)$, from below (even a weakly correlated context lowers the conditional content; $L(D) \le R_{\min}(D)$ always, with equality at $\rho = 0$). (ii) As $\tau^2 \to 0$ at fixed (ρ^2, D) with $D < 1 - \rho^2$, with $P'\big((1-\rho^2)/D\big) = 1 - \rho^2 - D > 0$ at $s = 1$,

$$L(D) - \tfrac{1}{2}\log_2 \frac{1-\rho^2}{D} = \frac{\rho^2}{2\ln 2\,(1-\rho^2-D)}\,\tau^2 + O(\tau^4):$$

the gap to Gray's conditional function closes linearly in τ^2, from above, with coefficient $\rho^2/\big(2\ln 2\,(1-\rho^2-D)\big)$. The coefficient diverges as $D \uparrow 1 - \rho^2$, where the two roots of P collide at $g = 1$ and the linear rate fails; on the far branch $D > 1 - \rho^2$, where Gray's value is zero, the gap is $L(D)$ itself and closes linearly with coefficient $(1-D)/\big(2\ln 2\,(D - (1-\rho^2))\big)$. (iii) As $\rho^2 \to 1$ at fixed (τ^2, D), with $P'\big((D+\tau^2)/(Ds)\big) = D + \tau^2 > 0$,

$$L(D) - \tfrac{1}{2}\log_2 \frac{D+\tau^2}{Ds} = \frac{\tau^2(1-D)}{2\ln 2\,(D+\tau^2)^2}\,(1-\rho^2) + O\big((1-\rho^2)^2\big):$$

the gap to Steinberg's scalar formula closes linearly in $1-\rho^2$, from above, with coefficient $\tau^2(1-D)/\big(2\ln 2\,(D+\tau^2)^2\big)$.

V. The Rate–Content Region and Misalignment

The content-only endpoint of Theorem 13 does not come for free in rate. This section characterizes the full tradeoff between the two coordinates of (11) for the Gaussian pair source. Throughout, $\rho^2 \in (0,1)$, $\tau^2 > 0$, $0 < D < 1$, and h, Q_0, Q_1, μ_c are as in the proof of Theorem 13. In the frontier system below, $\alpha \in [0,1]$ weights rate against conditional content, and $\gamma_0 = n/(Q_0+n)$ and $\gamma_1 = n/(Q_1+n)$ are the two water levels the stationarity system balances: the frontier point at weight α is $\big(\tfrac{1}{2}\log_2(1/\gamma_0),\, \tfrac{1}{2}\log_2(1/\gamma_1)\big)$. The name is used in a narrow sense: γ_0 and γ_1 are called water levels because they play the inverse signal-plus-noise roles in the two mutual-information coordinates; unlike classical parallel-channel water-filling, in which independent subchannels receive separate allocations, the two levels here are coupled through one reproduction and one distortion constraint.

Theorem 16 (Core: the rate–content region)**.** *The closure of the single-letter rate–content pairs $\big(I(T;\widehat{Y}),\, I(T;\widehat{Y} \mid S)\big)$ over admissible test channels at distortion D, which Theorem 30 identifies with the operational region $\mathcal{RW}(D)$, is the closed convex set whose Pareto frontier is traced by $\alpha \in [0,1]$ as follows. Let $(a,b,n,\gamma_0,\gamma_1)$, with $n > 0$ and $\gamma_0,\gamma_1 \in (0,1)$, solve the stationarity system*

$$a = 1 - \alpha\gamma_0 - (1-\alpha)\gamma_1, \qquad \mu_c = \frac{a\rho}{s - (1-\alpha)\gamma_1}, \qquad b = (1-\alpha)\gamma_1\mu_c, \tag{31}$$

$$n = D - h(a,b), \qquad \gamma_0 = \frac{n}{Q_0+n}, \qquad \gamma_1 = \frac{n}{Q_1+n}. \tag{32}$$

The solution exists and is unique at every α (Proposition 18 below; at $\alpha = 0$ uniqueness also follows from Theorem 13), and the frontier point is

$$\big(R(\alpha), L(\alpha)\big) = \Big(\tfrac{1}{2}\log_2\frac{1}{\gamma_0},\ \tfrac{1}{2}\log_2\frac{1}{\gamma_1}\Big),$$

achieved by $\widehat{Y} = aY + bV + N$, $N \sim \mathcal{N}(0,n)$. Endpoints: $\alpha = 1$ recovers the classical reverse channel ($b = 0$, $a = 1-D$, $\gamma_0 = D$, $R = \tfrac{1}{2}\log_2(1/D)$); $\alpha = 0$ recovers the channel and value of Theorem 13 ($\gamma_1 = 1/g^\star$).

Proof. Convexity and scalarization. Both coordinates are convex in the test channel: Lemma 3 and the per-s decomposition (10) apply verbatim with the sum over s replaced by an integral against the law of S, the conditional laws $p(t \mid s)$ being furnished by disintegration through a regular conditional probability (S takes values in a standard Borel space). The achievable set of pairs at distortion D is therefore convex, closed, and upward closed in each coordinate, so every point of its Pareto frontier minimizes one of the weighted objectives

$$J_\alpha := \alpha\, I(T;\widehat{Y}) + (1-\alpha)\, I(T;\widehat{Y} \mid S), \qquad \alpha \in [0,1],$$

over admissible channels, by the supporting-hyperplane argument. At the interior weights each J_α-minimizer is a frontier point; at the endpoint weights $\alpha \in \{0,1\}$ this a priori fixes only one coordinate of the minimizer, and the value of the other coordinate is pinned by the uniqueness of the minimizing moments (Proposition 18 below; the argument is run in detail in the proof of Corollary 19 at the end of this section).

The two determinant identities. Fix an admissible channel, centered without loss, with regression coordinates (a,b,n), $n > 0$, as in the proof of Theorem 13; the Markov moment identity $\mathrm{Cov}(N', S) = 0$ holds as there. Lemma 11 applies twice to the same three moments, once with the conditioner S and once with S deleted:

$$I(T;\widehat{Y}) \ \geq\ \tfrac{1}{2}\log_2\frac{\det\Sigma_T}{\det\Sigma_{e0}} = \tfrac{1}{2}\log_2\frac{Q_0+n}{n}, \qquad I(T;\widehat{Y} \mid S) \ \geq\ \tfrac{1}{2}\log_2\frac{\det\Sigma_{T|S}}{\det\Sigma_e} = \tfrac{1}{2}\log_2\frac{Q_1+n}{n}, \tag{33}$$

where Σ_{e0} is the error covariance of the best linear estimate of T from $\widehat{Y}$ alone. The conditional identity is written out in the proof of Theorem 13. The rate identity, in full: with $\hat{c} := \mathbb{E}[T\widehat{Y}] = \Sigma_T(a,b)^{\mathsf{T}}$ and $\mathrm{Var}\,\widehat{Y} = Q_0 + n$,

$$\Sigma_{e0} = \Sigma_T - \frac{\hat{c}\,\hat{c}^{\mathsf{T}}}{Q_0+n}, \qquad \det\Sigma_{e0} = \det\Sigma_T\left(1 - \frac{\hat{c}^{\mathsf{T}}\Sigma_T^{-1}\hat{c}}{Q_0+n}\right) = \det\Sigma_T\,\frac{n}{Q_0+n},$$

by the rank-one determinant identity $\det(\Sigma - \hat{c}\hat{c}^{\mathsf{T}}/v) = \det\Sigma\,(1 - \hat{c}^{\mathsf{T}}\Sigma^{-1}\hat{c}/v)$ and $\hat{c}^{\mathsf{T}}\Sigma_T^{-1}\hat{c} = (a,b)\,\Sigma_T\,(a,b)^{\mathsf{T}} = Q_0$.

The weighted moment program. Combining, J_α is bounded below by

$$\min_{(a,b,n)}\ \varphi_\alpha(a,b,n) := \frac{\alpha}{2}\log_2\frac{Q_0+n}{n} + \frac{1-\alpha}{2}\log_2\frac{Q_1+n}{n} \quad \text{s.t.} \quad h(a,b) + n \leq D,\ n > 0. \tag{34}$$

As in Theorem 13's proof: φ_α is strictly decreasing in n at fixed $(a,b) \neq (0,0)$, so the distortion constraint is active at the optimum; on the feasible set $Q_0 \geq Q_1 \geq q_0 > 0$, so $\varphi_\alpha \to \infty$ as $n \to 0$ and the minimum is attained at an interior stationary point of $\varphi_\alpha(a,b,D-h(a,b))$.

The gradient identities reduce the first-order system linearly. At a stationary point set $\gamma_0 = n/(Q_0+n)$ and $\gamma_1 = n/(Q_1+n)$, both in $(0,1)$. For $x \in \{a,b\}$, using $\partial_x n = -\partial_x h$,

$$\alpha\,\frac{\partial_x(Q_0-h)}{Q_0+n} + (1-\alpha)\,\frac{\partial_x(Q_1-h)}{Q_1+n} + \frac{\partial_x h}{n} = 0;$$

multiplying by $n/2$ and inserting the four gradient identities (25):

$$x = a: \quad \alpha\gamma_0 + (1-\alpha)\gamma_1(1-\rho\mu_c) + (a+b\rho-1) = 0, \tag{35}$$

$$x = b: \quad \alpha\gamma_0\rho + (1-\alpha)\gamma_1(\rho-\mu_c) + (a\rho+b-\rho) = 0. \tag{36}$$

Subtract ρ times (35) from (36): the γ_0 terms cancel, the γ_1 terms give $(1-\alpha)\gamma_1\big[(\rho-\mu_c)-\rho(1-\rho\mu_c)\big] = -(1-\alpha)\gamma_1\mu_c(1-\rho^2)$, and the moment terms give $(a\rho+b)-\rho(a+b\rho) = b(1-\rho^2)$, so

$$b(1-\rho^2) = (1-\alpha)\gamma_1\mu_c(1-\rho^2) \quad\Longrightarrow\quad b = (1-\alpha)\gamma_1\mu_c,$$

dividing by $1-\rho^2 > 0$, which is in force since $\rho^2 \in (0,1)$ throughout this section. Equation (35) then reads $a+b\rho = 1-\alpha\gamma_0-(1-\alpha)\gamma_1+(1-\alpha)\gamma_1\rho\mu_c = 1-\alpha\gamma_0-(1-\alpha)\gamma_1+b\rho$, so $a = 1-\alpha\gamma_0-(1-\alpha)\gamma_1$. Finally $s\mu_c = a\rho+b = a\rho+(1-\alpha)\gamma_1\mu_c$ rearranges to $\mu_c = a\rho/\big(s-(1-\alpha)\gamma_1\big)$, the denominator being positive since $s > 1 > (1-\alpha)\gamma_1$. This is (31); (32) restates the activity of the constraint and the definitions of (γ_0,γ_1). Conversely, every solution of the system with $n > 0$ is a stationary point of the reduced program, and by Proposition 18 below the program is convex in moment coordinates with a unique minimizer, so the system's solution is unique and is that minimizer.

Achievability. The jointly Gaussian channel $\widehat{Y} = aY + bV + N$, $N \sim \mathcal{N}(0,n)$, with the optimal (a,b,n) realizes the same moments and, by the equality clause of Lemma 11, attains both bounds of (33) simultaneously: $I(T;\widehat{Y}) = \frac{1}{2}\log_2(1/\gamma_0)$ and $I(T;\widehat{Y}\mid S) = \frac{1}{2}\log_2(1/\gamma_1)$. Hence the value of (34) equals the minimum of J_α, the frontier point is as displayed, and the trace over $\alpha \in [0,1]$ is the entire Pareto boundary.

Endpoints. At $\alpha = 1$, (31) gives $b = 0$ and $a = 1-\gamma_0$; then $h = (1-a)^2 = \gamma_0^2$, $Q_0 = a^2$, and $\gamma_0(Q_0+n) = n$ forces $\gamma_0 a^2 = n(1-\gamma_0) = na$, so $n = \gamma_0 a$; combined with $n = D-\gamma_0^2$ this gives $\gamma_0(1-\gamma_0) = D-\gamma_0^2$, that is, $\gamma_0 = D$, $a = 1-D$, $n = D(1-D)$: the classical reverse channel. At $\alpha = 0$ the γ_0 terms drop from (35)–(36), which become (26) with $g = 1/\gamma_1$; Theorem 13 identifies $\gamma_1 = 1/g^\star$ and the channel (21). □

Uniqueness at every weight is a convexity statement in moment coordinates. It is cleanest, and no harder, in the whitened frame that Section VII also uses; we set it up once. Since $\Sigma_T \succ 0$ and $0 \prec \Sigma_{T|S} \preceq \Sigma_T$, there is a nonsingular W with

$$W\Sigma_T W^\mathsf{T} = I, \qquad W\Sigma_{T|S}W^\mathsf{T} = \Lambda = \operatorname{diag}(\lambda_1,\ldots,\lambda_p), \qquad 0 \prec \Lambda \preceq I \tag{37}$$

(factor $\Sigma_T = LL^\mathsf{T}$, diagonalize $L^{-1}\Sigma_{T|S}L^{-\mathsf{T}} = O\Lambda O^\mathsf{T}$, and set $W = O^\mathsf{T}L^{-1}$). Write $Z = WT$, so $\operatorname{Cov} Z = I$; the described variable becomes $Y = y_0^\mathsf{T}Z$ with $y_0 = W^{-\mathsf{T}}e_Y$ and $|y_0|^2 = e_Y^\mathsf{T}\Sigma_T e_Y = 1$. A channel with regression vector $(a,b)^\mathsf{T}$ on T has regression vector $c = W^{-\mathsf{T}}(a,b)^\mathsf{T}$ on Z, and

$$Q_0 = |c|^2, \qquad Q_1 = c^\mathsf{T}\Lambda c, \qquad h = |y_0-c|^2, \qquad \mathbb{E}(Y-\widehat{Y})^2 = |y_0-c|^2+n.$$

Here $p = 2$; Section VII uses $p = 1+r$.

Lemma 17 (Matrix convexity of the Schur ratio)**.** *Let $\mathcal{Z} \subseteq \mathbb{R}^q$ be convex, $a : \mathcal{Z} \to \mathbb{R}^p$ affine, and $\sigma : \mathcal{Z} \to (0,\infty)$ concave. Then*

$$F(z) := \frac{a(z)\,a(z)^\mathsf{T}}{\sigma(z)}$$

is matrix-convex: $F(z_t) \preceq (1-t)F(z_0) + tF(z_1)$ for all $z_0, z_1 \in \mathcal{Z}$, $t \in [0,1]$, $z_t = (1-t)z_0 + tz_1$.

Proof. For $\sigma > 0$ the Schur-complement criterion states

$$\begin{bmatrix} W & a \\ a^\mathsf{T} & \sigma \end{bmatrix} \succeq 0 \quad\Longleftrightarrow\quad W \succeq \frac{aa^\mathsf{T}}{\sigma},$$

so $F(z)$ is the smallest matrix admitted by the criterion at z. Fix z_0, z_1, t and abbreviate $a_i = a(z_i)$, $\sigma_i = \sigma(z_i)$, $F_i = F(z_i)$. The two positive semidefinite block matrices

$$\begin{bmatrix} F_0 & a_0 \\ a_0^{\mathsf T} & \sigma_0 \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} F_1 & a_1 \\ a_1^{\mathsf T} & \sigma_1 \end{bmatrix} \succeq 0$$

average, with weights $(1-t, t)$, to

$$\begin{bmatrix} (1-t)F_0 + tF_1 & a(z_t) \\ a(z_t)^{\mathsf T} & (1-t)\sigma_0 + t\sigma_1 \end{bmatrix} \succeq 0,$$

using affinity of a in the off-diagonal blocks. Concavity gives $\sigma(z_t) \geq (1-t)\sigma_0 + t\sigma_1$, and adding the positive semidefinite matrix $\operatorname{diag}\big(0, \sigma(z_t) - (1-t)\sigma_0 - t\sigma_1\big)$ raises the corner:

$$\begin{bmatrix} (1-t)F_0 + tF_1 & a(z_t) \\ a(z_t)^{\mathsf T} & \sigma(z_t) \end{bmatrix} \succeq 0.$$

By the criterion again, $(1-t)F_0 + tF_1 \succeq a(z_t)a(z_t)^{\mathsf T}/\sigma(z_t) = F(z_t)$. □

Proposition 18 (Uniqueness at every weight)**.** *In the whitened frame* (37)*, with $0 \prec \Lambda \preceq I$, $|y_0| = 1$, $D \in (0,1)$, and any $\alpha \in [0,1]$, the weighted program* (34) *(with $Q_0 = |c|^2$, $Q_1 = c^{\mathsf T}\Lambda c$, $h = |y_0 - c|^2$, minimized over (c, n)) has a unique minimizer, and every stationary point with $n > 0$ is that minimizer. Since the change of variables $c = W^{-\mathsf T}(a,b)^{\mathsf T}$ is a linear bijection, the same holds for the program in the coordinates of Theorem 16.*

Proof. Work in the moment coordinates $z = (c, v)$, $v = \operatorname{Var}\widehat{Y} = |c|^2 + n$, on the domain

$$\mathcal{D} = \big\{(c,v) : v > |c|^2,\ 1 - 2y_0^{\mathsf T}c + v \leq D\big\},$$

which is convex: $v > |c|^2$ is the strict epigraph of a convex function, and the distortion constraint $h + n = 1 - 2y_0^{\mathsf T}c + v \leq D$ is linear in z. On $\mathcal{D}$ define $\sigma := v - c^{\mathsf T}(I - \Lambda)c = Q_1 + n > 0$ and

$$B_0 = \tfrac{1}{2}\log_2 \frac{Q_0 + n}{n} = -\tfrac{1}{2}\log_2 \det\Big(I - \frac{cc^{\mathsf T}}{v}\Big),$$
$$B_1 = \tfrac{1}{2}\log_2 \frac{Q_1 + n}{n} = \tfrac{1}{2}\log_2 \det\Lambda - \tfrac{1}{2}\log_2\det\Big(\Lambda - \frac{(\Lambda c)(\Lambda c)^{\mathsf T}}{\sigma}\Big);$$

the first equality holds because $\det(I - cc^{\mathsf T}/v) = 1 - |c|^2/v = n/v$, the second because $\det\big(\Lambda - (\Lambda c)(\Lambda c)^{\mathsf T}/\sigma\big) = \det\Lambda\,\big(1 - c^{\mathsf T}\Lambda c/\sigma\big) = \det\Lambda \cdot n/\sigma$.

Convexity. Apply Lemma 17 twice. For B_0: $z \mapsto c$ is affine and $z \mapsto v$ is affine, hence concave and positive on $\mathcal{D}$, so $W_0(z) = cc^{\mathsf T}/v$ is matrix-convex. Moreover $I - W_0 \succ 0$ on $\mathcal{D}$, hence invertible: by the Schur criterion, $I \succ cc^{\mathsf T}/v$ is equivalent to $v - |c|^2 > 0$, and $v - |c|^2 = n > 0$ on $\mathcal{D}$ by construction. For B_1: $z \mapsto \Lambda c$ is affine and $z \mapsto \sigma = v - c^{\mathsf T}(I-\Lambda)c = Q_1 + n$ is concave (affine minus a convex quadratic, $I - \Lambda \succeq 0$) and positive on $\mathcal{D}$ ($Q_1 \geq 0$, $n > 0$), so $W_1(z) = (\Lambda c)(\Lambda c)^{\mathsf T}/\sigma$ is matrix-convex, $\Lambda - W_1$ is matrix-concave, and $\Lambda - W_1 \succ 0$ because $c^{\mathsf T}\Lambda c = Q_1 < Q_1 + n = \sigma$. The map $\Sigma \mapsto -\log_2\det\Sigma$ on positive definite matrices is convex and decreasing in the Löwner order; for the latter, if $0 \prec A \preceq B$ then $B^{-1/2}AB^{-1/2} \preceq I$ has eigenvalues in $(0,1]$, so $\det A/\det B = \det(B^{-1/2}AB^{-1/2}) \leq 1$, that is, $\det A \leq \det B$. Composing a convex antitone function with a matrix-concave map preserves convexity: for matrix-concave $F \succ 0$,

$$-\log\det F(z_t) \ \leq\ -\log\det\big((1-t)F(z_0) + tF(z_1)\big) \ \leq\ (1-t)\big(-\log\det F(z_0)\big) + t\big(-\log\det F(z_1)\big).$$

Hence B_0, B_1, and every $B_\alpha := \alpha B_0 + (1-\alpha)B_1$ are convex on $\mathcal{D}$.

Reduction to the active slice. On $\mathcal{D}$, feasibility forces $c \neq 0$: from $v > |c|^2 \geq 0$ and $1 - 2y_0^{\mathsf T}c + v \leq D$,

$$2\,y_0^{\mathsf T}c \ \geq\ 1 - D + v \ >\ 1 - D \ >\ 0. \tag{38}$$

At fixed $c \neq 0$ both B_0 and B_1 strictly decrease in v ($\partial_v[\ln v - \ln n] = v^{-1} - n^{-1} < 0$ and $\partial_v[\ln\sigma - \ln n] = \sigma^{-1} - n^{-1} < 0$, since $Q_0 \geq Q_1 > 0$), so every minimizer of B_α lies on the active slice $v = D - 1 + 2y_0^{\mathsf T}c$, an affine section of $\mathcal{D}$. On the slice B_α is convex, so its stationary points are exactly its global minimizers and the minimizer set is convex; coercivity as in Theorem 16's proof gives attainment.

Strictness. Suppose $z_0 \neq z_1$ are both minimizers on the slice, and let z_t run over the segment, along which v and c are affine in t. Then B_α is affine on the segment, and since B_0, B_1 are convex and their weighted sum is affine, each B_i carrying nonzero weight is affine on the segment. We use twice the following fact: for an affine positive definite matrix segment $\Sigma_t = (1-t)\Sigma_0 + t\Sigma_1$, the function $t \mapsto -\log\det\Sigma_t$ has second derivative $\mathrm{tr}\big[(\Sigma_t^{-1}\dot{\Sigma})^2\big] \geq 0$, which vanishes only if $\dot{\Sigma} = 0$, because $\Sigma_t^{-1/2}\dot{\Sigma}\Sigma_t^{-1/2}$ is symmetric and a symmetric matrix whose square has zero trace is zero. Moreover, if a matrix-concave F has $-\log\det F$ affine on the segment, then squeezing $-\log\det F(z_t) \leq -\log\det\big((1-t)F(z_0) + tF(z_1)\big) \leq$ the affine interpolant forces both inequalities to be equalities; the second forces $F(z_0) = F(z_1) =: \bar{F}$ by the fact just stated, and the first then forces $F(z_t) \succeq \bar{F}$ with $\det F(z_t) = \det \bar{F}$, hence $F(z_t) = \bar{F}$ for all t (if $A \succeq B \succ 0$ and $\det A = \det B$ then $A = B$).

Case $\alpha > 0$, where strictness comes from the forced constancy of the rank-one ratio W_0 along the segment: B_0 is affine, so $W_0(z_t) = c(t)c(t)^\mathsf{T}/v(t)$ is constant, a rank-one matrix $\bar{W}_0 \neq 0$ by (38). Then $c(t) = \theta(t)e$ for a fixed unit vector e, with θ affine (a coordinate of the affine c), and $\theta(t)^2 = v(t)\,(e^\mathsf{T}\bar{W}_0 e)$ with v affine. An affine function whose square is affine is constant, so θ, hence c, hence $v = D - 1 + 2y_0^\mathsf{T}c$, is constant: $z_0 = z_1$, a contradiction.

Case $\alpha < 1$, where strictness comes from the constancy of the rank-one ratio W_1 played against the concavity of σ: B_1 is affine, so $W_1(z_t) = (\Lambda c)(\Lambda c)^\mathsf{T}/\sigma$ is constant and nonzero. Then $\Lambda c(t) = \theta(t)e$ with θ affine, and $\theta(t)^2 = \sigma(t)\,(e^\mathsf{T}\bar{W}_1 e)$. Along the segment $\sigma(t) = v(t) - c(t)^\mathsf{T}(I-\Lambda)c(t)$ is a concave quadratic in t, while θ^2 is a convex quadratic; equality up to the positive constant $e^\mathsf{T}\bar{W}_1 e$ forces both to be affine, so θ is an affine function with affine square, hence constant. Then Λc is constant, c is constant (Λ nonsingular), and $z_0 = z_1$ as before, a contradiction.

Every $\alpha \in [0,1]$ falls in at least one case, so the minimizer is unique.

We record one further fact for the clean-context boundary. At $\alpha = 1$ the argument is Λ-free: the objective is B_0 alone, whose convexity (Lemma 17 with $a(z) = c$ and $\sigma = v$), strict decrease in v at $c \neq 0$, coercivity through $Q_0 \geq q_0 > 0$, and strictness case use only c, v, and the domain $\mathcal{D}$, never Λ. The uniqueness of the rate minimizer, equivalently of the moment triple $(a,b,n) = (1-D, 0, D(1-D))$, therefore holds for an arbitrary conditioner, including the boundary $\tau^2 = 0$, where $\Sigma_{T|S}$ is singular and the whitening (37) of the conditional covariance is unavailable. □

Corollary 19 (Core: misalignment opens a strict tradeoff)**.** *Within the standing model of Section III ($\tau^2 > 0$, $|\rho| < 1$, $0 < D < 1$), the frontier is nondegenerate if and only if $\rho^2 \in (0,1)$. When $\rho^2 \in (0,1)$: any channel attaining $L(D)$ has rate strictly above $R_{\min} = \frac{1}{2}\log_2(1/D)$, and any channel attaining $R_{\min}$ has conditional content strictly above $L(D)$. In particular, writing $L_R^\star(D)$ for the conditional content of the rate-optimal channel and $R_L^\star(D)$ for the rate of the content-optimal channel,*

$$L_R^\star(D) = \tfrac{1}{2}\log_2 \frac{(1-D)(1-\rho^2/s) + D}{D} \;>\; L(D),$$

the content of the classical reverse channel, and $R_L^\star(D) = \frac{1}{2}\log_2\big[(Q_0+n)/n\big]$ evaluated at the channel of Theorem 13 strictly exceeds $R_{\min}$. (Numerically, at $(\rho^2, \tau^2, D) = (0.75, 0.5, 0.3)$: $R_L^\star(D) - R_{\min} = 0.0400$ and $L_R^\star(D) - L(D) = 0.0349$ bits; Fig. 2.) When $\rho = 0$ the classical reverse channel attains both minima and the frontier is a single corner. The frontier also collapses in the limits $\rho^2 \to 1$ (where both endpoint excesses vanish; see the proof) and $\tau^2 \to \infty$ (where $L \to R$ coordinatewise); positivity of τ^2, by contrast, is a modeling assumption and not necessary for the tradeoff (Remark 20).

Proof. By Theorem 13, the moment program for the conditional coordinate alone has the unique minimizer $(a^\star, b^\star, n^\star)$ of (28), and $b^\star = (g^\star - 1)\rho/(g^\star k) \neq 0$ since $\rho \neq 0$, $g^\star > 1$, $k > 0$. By the $\alpha = 1$ endpoint computation in Theorem 16's proof (uniqueness by Proposition 18), the moment program for the rate coordinate alone has the unique minimizer $\big(1-D,\, 0,\, D(1-D)\big)$, with $b = 0$. The two minimizers are distinct.

Let a channel attain $L(D)$. Writing B_0, B_1 for the two bounds of (33) as functions of the channel's moments, $L(D) = I(T; \widehat{Y} \mid S) \geq B_1(\text{its moments}) \geq \min B_1 = L(D)$ forces equality throughout, so its moments minimize B_1 and equal $(a^\star, b^\star, n^\star)$. Its rate then satisfies $R \geq B_0(a^\star, b^\star, n^\star) > \min B_0 = R_{\min}$: strict, because B_0's minimizer is unique and differs from $(a^\star, b^\star, n^\star)$. Symmetrically, any rate-optimal channel carries the moments $(1-D, 0, D(1-D))$ and its content is at least B_1 there, which strictly exceeds $L(D)$; evaluating, $b = 0$ gives $Q_1 = (1-D)^2(1-\rho^2/s)$ and $n = D(1-D)$, so $(Q_1+n)/n = \big[(1-D)(1-\rho^2/s) + D\big]/D$, the displayed $L_R^\star(D)$.

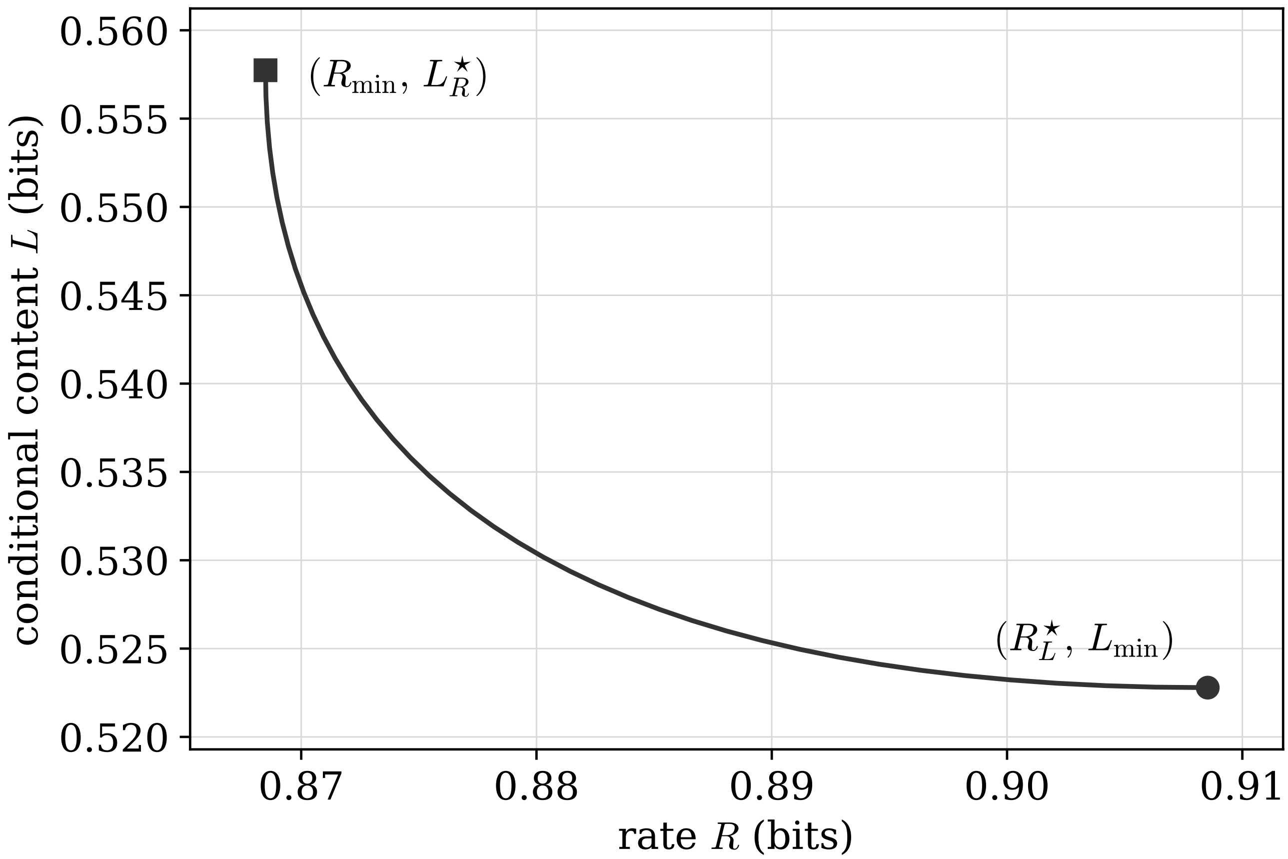

Fig. 2. The Pareto frontier of the rate–content region at $(\rho^2, \tau^2, D) = (0.75, 0.5, 0.3)$, traced by the two-water-level system of Theorem 16. The tradeoff is strict: the rate-optimal channel pays $L_R^\star(D) - L(D) = 0.0349$ bits of excess conditional content, and the content-optimal channel pays $R_L^\star(D) - R_{\min} = 0.0400$ bits of excess rate.

At $\rho = 0$, $b^\star = 0$ and $g^\star = 1/D$ (Corollary 14(i)) give $(a^\star, n^\star) = (1 - D, D(1 - D))$: the two minimizers coincide and the frontier is the corner $R_{\min} = L(D) = \frac{1}{2}\log_2(1/D)$. As $\rho^2 \to 1$ both endpoint excesses vanish, so the frontier collapses to a corner in the limit: by Corollary 14(iii), $L(D) \to \frac{1}{2}\log_2\big[(D + \tau^2)/(Ds)\big]$, and the displayed $L_R^\star(D)$ has the same limit, since $(1 - D)(1 - \rho^2/s) + D \to (1 - D)\tau^2/s + D = (D + \tau^2)/s$; for the rate side, the channel (21) has, as $\rho \to 1$, $g^\star \to (D + \tau^2)/(Ds)$ and

$$(a, b, n) \;\longrightarrow\; \Big(\frac{\tau^2(1 - D)}{D + \tau^2},\; \frac{D(1 - D)}{D + \tau^2},\; D(1 - D)\Big),$$

whence $a + b \to 1 - D$, $Q_0 \to (1 - D)^2$, and $R_L^\star(D) = \frac{1}{2}\log_2\big[(Q_0 + n)/n\big] \to \frac{1}{2}\log_2(1/D) = R_{\min}$; for $\rho \to -1$ only the sign of b flips, leaving Q_0 and $R_L^\star$ unchanged. As $\tau^2 \to \infty$, $\Sigma_{T|S} \to \Sigma_T$ makes the two coordinates, and so the two endpoints, coincide in the limit. □

Fig. 2 traces the frontier at the numerical instance of Corollary 19. The displayed excesses are representative of moderate parameters, not extremal. Over the box $\rho^2 \in [0.05, 0.95]$, $\tau^2 \in [\frac{1}{4}, 4]$, $D \in [0.05, 0.95]$, the endpoint excesses reach at most $R_L^\star(D) - R_{\min} = 0.1138$ bits, at $(\rho^2, \tau^2, D) = (0.5, 0.25, 0.5)$, and $L_R^\star(D) - L(D) = 0.0770$ bits, at $(0.615, 0.25, 0.385)$. Toward the clean-context boundary the rate excess is unbounded: at $D = 1 - \rho^2$ the content-optimal channel pins the reproduction to the context as $\tau^2 \to 0$ and its rate diverges, reaching, for instance, $R_L^\star(D) - R_{\min} = 1.537$ bits at $(0.5, 10^{-3}, 0.5)$; the content excess instead saturates there at $\frac{1}{2}\log_2(1 + \rho^2)$, the clean-boundary value of Remark 20 evaluated at $D = 1 - \rho^2$. All quoted values are computed by the verification script.

Remark 20 (The tradeoff persists at the clean-context boundary)**.** *The strict tradeoff extends to the boundary $\tau^2 = 0$ excluded by the standing model. Let $\tau^2 = 0$, so $S = V$, and let $\rho^2 \in (0,1)$, $0 < D < 1$. The minimal conditional content is*

$$L_{\min}(D) \;=\; \tfrac{1}{2}\log_2^+ \frac{1-\rho^2}{D}, \tag{39}$$

while every *channel attaining the minimal rate $R_{\min} = \frac{1}{2}\log_2(1/D)$, in particular the classical reverse channel $\widehat{Y}_R = (1-D)Y + W$ with $W \sim \mathcal{N}(0, D(1-D))$ independent of X, has conditional content exactly*

$$L_R^\star(D) \;=\; \tfrac{1}{2}\log_2 \frac{(1-D)(1-\rho^2)+D}{D} \;=\; \tfrac{1}{2}\log_2 \frac{1-\rho^2+D\rho^2}{D} \;>\; L_{\min}(D), \tag{40}$$

strictly, on all of $0 < D < 1$. In particular no channel attains $R_{\min}$ and $L_{\min}$ simultaneously: a clean context trades rate against content just as a noisy one does, for $D < 1-\rho^2$ with both quantities positive, and for $D \geq 1-\rho^2$ with $L_{\min} = 0$ but $L_R^\star > 0$.

Proof. The value (39). Write $Y = \rho V + Y'$ with $Y' \perp V$, $\operatorname{Var} Y' = 1-\rho^2$. Converse: for any admissible channel, $I(X;\widehat{Y} \mid V) \geq I(Y;\widehat{Y} \mid V) \geq R_{Y|V}(D) = \frac{1}{2}\log_2^+\big(\operatorname{Var}(Y \mid V)/D\big)$, Gray's conditional function [4], exactly as in Proposition 10(ii). Achievability for $D \leq 1-\rho^2$: with $\theta = D/(1-\rho^2)$, take $\widehat{Y} = \rho V + (1-\theta)Y' + N$, $N \sim \mathcal{N}\big(0, \theta(1-\theta)(1-\rho^2)\big)$: the distortion is $\theta^2(1-\rho^2) + \theta(1-\theta)(1-\rho^2) = D$, and

$$I(T;\widehat{Y} \mid V) = \tfrac{1}{2}\log_2 \frac{\operatorname{Var}(\widehat{Y} \mid V)}{\operatorname{Var}(\widehat{Y} \mid T)} = \tfrac{1}{2}\log_2 \frac{(1-\theta)^2(1-\rho^2)+\theta(1-\theta)(1-\rho^2)}{\theta(1-\theta)(1-\rho^2)} = \tfrac{1}{2}\log_2 \frac{1-\rho^2}{D}\,.$$

For $D \geq 1-\rho^2$, $\widehat{Y} = \rho V$ has distortion $1-\rho^2 \leq D$ and is V-measurable, so $I(T;\widehat{Y} \mid V) = 0$.

The value (40). For the reverse channel, $\operatorname{Var}(\widehat{Y}_R \mid V) = (1-D)^2 \operatorname{Var}(Y \mid V) + \operatorname{Var} W = (1-D)^2(1-\rho^2) + D(1-D)$ and $\operatorname{Var}(\widehat{Y}_R \mid T) = \operatorname{Var} W = D(1-D)$, so

$$L_R^\star(D) = \tfrac{1}{2}\log_2 \frac{(1-D)^2(1-\rho^2)+D(1-D)}{D(1-D)} = \tfrac{1}{2}\log_2 \frac{(1-D)(1-\rho^2)+D}{D}\,,$$

and $(1-D)(1-\rho^2)+D = 1-\rho^2+D\rho^2$.

Every rate-optimal channel carries the content (40). The rate moment program never references the conditioner, and at $\alpha = 1$ the uniqueness proof of Proposition 18 is Λ-free (the closing note there), so also at $\tau^2 = 0$ its unique minimizer is $(a,b,n) = (1-D, 0, D(1-D))$. Let a channel attain $R_{\min}$ at distortion D. The chain $R_{\min} = I(T;\widehat{Y}) \geq B_0(\text{its moments}) \geq \min B_0 = R_{\min}$ forces its moments to be that triple and forces equality in the max-entropy bound (18). Equality makes the linear-estimation error $E := T - J\widehat{Y}$, $J = \mathbb{E}[T\widehat{Y}]/\operatorname{Var}\widehat{Y}$, Gaussian and independent of $\widehat{Y}$. At these moments $\operatorname{Var}\widehat{Y} = (1-D)^2 + D(1-D) = 1-D$ and $\mathbb{E}[Y\widehat{Y}] = 1-D$, so the Y-row of J is 1 and $Y = \widehat{Y} + E_Y$ with E_Y Gaussian independent of $\widehat{Y}$ and Y standard Gaussian; by Cramér's decomposition theorem [9], $\widehat{Y}$ is Gaussian. Independent Gaussian vectors are jointly Gaussian, so $(T,\widehat{Y}) = (J\widehat{Y}+E, \widehat{Y})$ is jointly Gaussian with the pinned moments: the law of the reverse channel. Its conditional content is then determined by that law,

$$I(T;\widehat{Y} \mid V) = \tfrac{1}{2}\log_2 \frac{\operatorname{Var}(\widehat{Y} \mid V)}{\operatorname{Var}(\widehat{Y} \mid T)} = \tfrac{1}{2}\log_2 \frac{(1-D)-(1-D)^2\rho^2}{D(1-D)} = \tfrac{1}{2}\log_2 \frac{1-\rho^2+D\rho^2}{D}\,,$$

which is (40).

Strictness. For $D < 1-\rho^2$: $1-\rho^2+D\rho^2 > 1-\rho^2$ since $D\rho^2 > 0$, so $L_R^\star > L_{\min}$. For $1-\rho^2 \leq D < 1$: $L_{\min} = 0$, while $1-\rho^2+D\rho^2 > D \Leftrightarrow (1-\rho^2)(1-D) > 0$, so $L_R^\star > 0$. □

Corollary 21 (Core: L is not determined by the reduced marginal (Y,S))**.** *Two instances whose (Y,S) joint laws are equivalent up to bijective scaling can have different minimal conditional content: the instances $(\rho^2,\tau^2) = (\frac{1}{2}, 0)$ and $(\frac{3}{4}, \frac{1}{2})$ have $\rho_{YS}^2 = \frac{1}{2}$ in both cases, yet different L at both tested distortions; Table III lists the values, in both cases strictly below Steinberg's scalar formula evaluated on the reduced pair. So no functional of the (Y,S) joint law, in particular neither Steinberg's scalar formula evaluated on (Y,S) nor Gray's, can equal L. This is no departure from Steinberg's functional, whose source here is the full pair (Y,V); it shows the augmentation is load-bearing: the encoder's access to the clean context V enters the function.*

TABLE III

THE VALUES BEHIND COROLLARY 21: TWO INSTANCES WITH THE SAME REDUCED (Y, S) LAW ($\rho^2_{YS} = \frac{1}{2}$), IN BITS. EXACT FORMS FOR THE SECOND INSTANCE: $g^\star = (17 + \sqrt{229})/6$ AT $D = 0.1$ AND $(21 + \sqrt{261})/18$ AT $D = 0.3$ (DISPLAYED IN THE PROOF).

	$\mathrm{Corr}(Y,S) = \rho/\sqrt{s}$	L at $D = 0.1$	L at $D = 0.3$
Instance $(\rho^2, \tau^2) = (\frac{1}{2}, 0)$	$1/\sqrt{2}$	1.1610	0.3685
Instance $(\rho^2, \tau^2) = (\frac{3}{4}, \frac{1}{2})$	$1/\sqrt{2}$	1.2105	0.5228
Steinberg's scalar formula on the reduced pair	$1/\sqrt{2}$	1.2297	0.5577

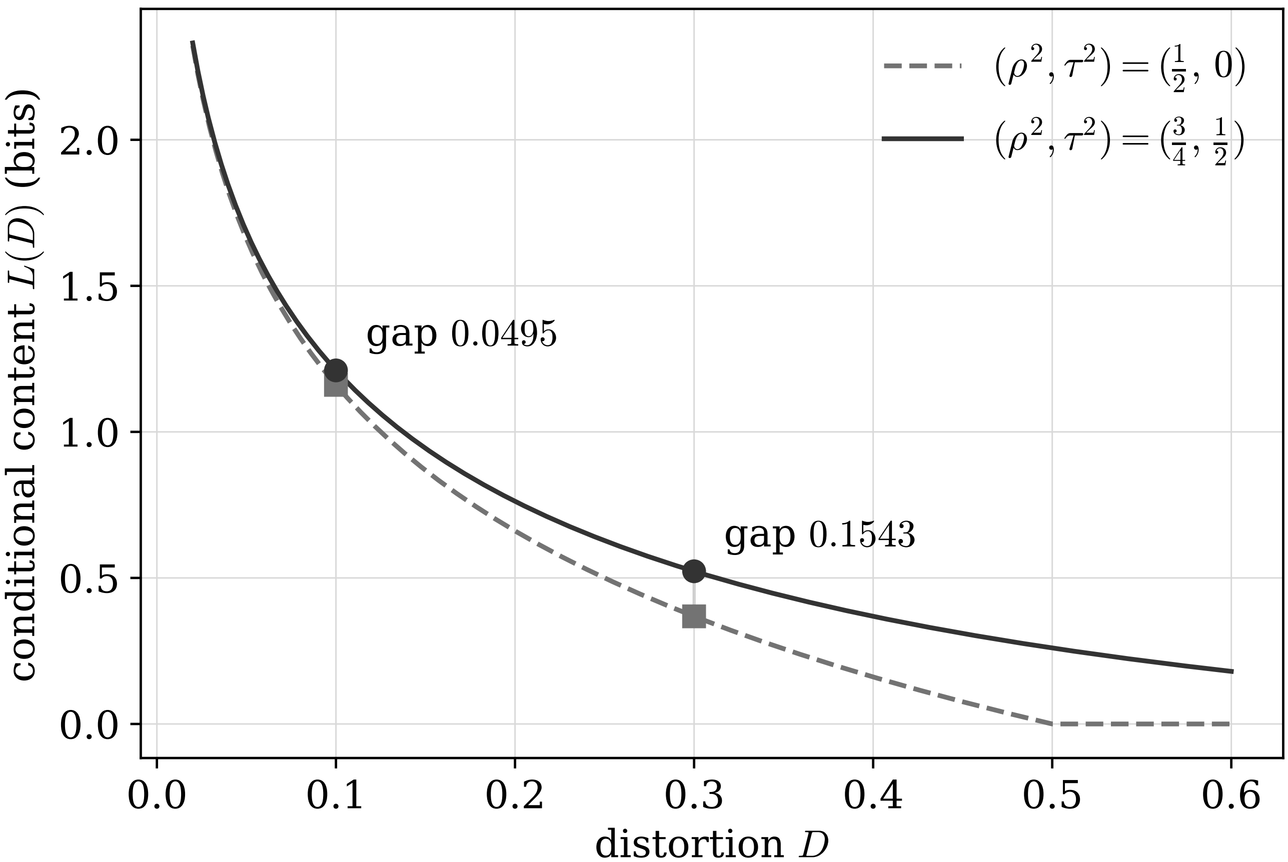

Fig. 3. Non-determination by the reduced marginal (Corollary 21): $L(D)$ against D for the two instances $(\rho^2, \tau^2) = (\frac{1}{2}, 0)$ (dashed, squares) and $(\frac{3}{4}, \frac{1}{2})$ (solid, circles), which share the reduced correlation $\mathrm{Corr}(Y, S) = \rho/\sqrt{s} = 1/\sqrt{2} = 0.7071$. The two functions differ at every distortion shown; at the tabulated levels (markers; Table III) the gap is 0.0495 bits at $D = 0.1$ and 0.1543 bits at $D = 0.3$.

Proof. The two instances share the (Y, S) joint law: in both, Y is standard and $\rho^2_{YS} = \mathrm{Cov}(Y, S)^2 / \mathrm{Var}\, S = \rho^2/s$ equals $\frac{1}{2}$, namely $(\frac{1}{2})/1$ and $(\frac{3}{4})/(\frac{3}{2})$, and rescaling S changes none of the quantities considered: a bijective scaling $S \mapsto \kappa S$, $\kappa \neq 0$, leaves the generated σ-algebra unchanged, $\sigma(\kappa S) = \sigma(S)$, so every conditional mutual information given S, and with it $L(D)$, is invariant, and the two instances may be normalized to a common (Y, S) law.

First instance, $(\rho^2, \tau^2) = (\frac{1}{2}, 0)$: the context is clean. This instance sits at the boundary $\tau^2 = 0$ excluded by the standing assumption $\tau^2 > 0$ of Section III, so its value is supplied not by Theorem 13 but directly by Proposition 10(ii), which is proved at $\tau^2 = 0$ (and which the $\tau^2 \to 0$ anchor of Corollary 14 matches in the limit). The instance gives

$$L = \tfrac{1}{2} \log_2 \frac{1}{2D}\,,$$

so $L = \frac{1}{2} \log_2 5 = 1.1610$ bits at $D = 0.1$, and $L = \frac{1}{2} \log_2 (5/3) = 0.3685$ bits at $D = 0.3$.

Second instance, $(\rho^2, \tau^2) = (\frac{3}{4}, \frac{1}{2})$: $s = \frac{3}{2}$, and (20) becomes, at $D = 0.1$,

$$P(g) = \tfrac{3}{20}g^2 - \tfrac{17}{20}g + \tfrac{1}{4}, \qquad g^\star = \frac{17 + \sqrt{229}}{6} = 5.3555, \qquad L = \tfrac{1}{2}\log_2 g^\star = 1.2105 \text{ bits},$$

and, at $D = 0.3$,

$$P(g) = \tfrac{9}{20}g^2 - \tfrac{21}{20}g + \tfrac{1}{4}, \qquad g^\star = \frac{21 + \sqrt{261}}{18} = 2.0642, \qquad L = \tfrac{1}{2}\log_2 g^\star = 0.5228 \text{ bits}.$$

The value of Steinberg's scalar formula on the reduced pair: an encoder observing Y alone pays the marginalized-class optimum of Section IV, the scalar-corner value $\frac{1}{2}\log_2\bigl[(1-\rho_{YS}^2+\rho_{YS}^2 D)/D\bigr]$ at $\rho_{YS}^2 = \frac{1}{2}$, that is, $\frac{1}{2}\log_2\bigl[(1+D)/(2D)\bigr]$: $\frac{1}{2}\log_2 5.5 = 1.2297$ bits at $D = 0.1$, and $\frac{1}{2}\log_2(13/6) = 0.5577$ bits at $D = 0.3$. Both instances lie strictly below it (1.1610, 1.2105 $<$ 1.2297 and 0.3685, 0.5228 $<$ 0.5577), as Proposition 10 requires.

Since the two instances have the same (Y, S) law after this normalization and different L, no functional of the (Y, S) joint law, in particular neither Steinberg's scalar formula evaluated on the pair nor Gray's under any reparameterization, can equal L. $\square$

The mechanism is worth stating plainly. Both instances have the same reduced correlation $\mathrm{Corr}(Y, S) = \rho/\sqrt{s} = 1/\sqrt{2}$, realized by different splits: entirely through a clean context in the first instance ($\rho^2 = \frac{1}{2}$, $\tau^2 = 0$), through a stronger context read in noise in the second ($\rho^2 = \frac{3}{4}$, $\tau^2 = \frac{1}{2}$). The function depends on (ρ^2, τ^2) separately, not only through the ratio ρ^2/s, because the encoder's channel acts on its observation of (Y, V), not on (Y, S): two sources with identical (Y, S) joint laws but different internal splits of the correlation admit different optimal channels. Fig. 3 plots the two functions across distortions; the gap at the tabulated levels is 0.0495 bits at $D = 0.1$ and 0.1543 bits at $D = 0.3$.

VI. When the Determinant Bound Is Attained

The conditional content of any admissible description is bounded below by a determinant expression, and it is natural to ask when the bound is exact. The question is sharpest for a joint description serving two described variables, and this section settles it there; the single-variable answer appears as a corollary inside the proof. In this section only, $Y = (Y_A, Y_B)$ denotes a pair of normalized described variables of the source, jointly Gaussian with the normalized context V, with an arbitrary correlation pattern; the third party again holds $S = V + U$, $U \sim \mathcal{N}(0, \tau^2)$, with X, U, and all channel randomness mutually independent. A joint channel produces $\hat{\mathbf{Y}} = (\widehat{Y}_A, \widehat{Y}_B)$ with $\hat{\mathbf{Y}} - X - S$ and must meet $\mathbb{E}(Y_A - \widehat{Y}_A)^2 \le D_A$ and $\mathbb{E}(Y_B - \widehat{Y}_B)^2 \le D_B$; write $L_{\min}(D_A, D_B) = \min I(X; \hat{\mathbf{Y}} \mid S)$.

Write Σ_Y for the covariance of the pair Y, $\beta = \mathrm{Cov}(Y, V) \in \mathbb{R}^2$, $\Sigma_{Y|V} = \Sigma_Y - \beta\beta^\mathsf{T}$, $\Sigma_{Y|S} = \Sigma_Y - \beta\beta^\mathsf{T}/(1+\tau^2)$, and $\Delta = \mathrm{diag}(D_A, D_B) \succ 0$. Define the *determinant lower bound*

$$\mathsf{B}_{\det} := \tfrac{1}{2}\log_2 \frac{\det \Sigma_{Y|S}}{\det \Delta},$$

which can be negative at large distortions. Every admissible channel obeys

$$L \;\ge\; I(Y; \hat{\mathbf{Y}} \mid S) \;=\; h(Y \mid S) - h(Y \mid \hat{\mathbf{Y}}, S) \;\ge\; \mathsf{B}_{\det}, \qquad \text{hence} \qquad L \;\ge\; \bigl[\mathsf{B}_{\det}\bigr]^+ := \max\{\mathsf{B}_{\det}, 0\}, \tag{41}$$

since $h(Y \mid S) = \frac{1}{2}\log_2\bigl((2\pi e)^2 \det \Sigma_{Y|S}\bigr)$, $h(Y \mid \hat{\mathbf{Y}}, S) \le \sum_{i\in\{A,B\}} h(Y_i - \widehat{Y}_i) \le \frac{1}{2}\log_2\bigl((2\pi e)^2 D_A D_B\bigr)$, and $L \ge 0$ always.[1] The exactness question below concerns $\mathsf{B}_{\det}$ itself: on the attainment branches of Theorem 23 the semidefinite conditions force $\det\Delta \le \det\Sigma_{Y|S}$, so there $\mathsf{B}_{\det} \ge 0$ and the two forms agree; where $\mathsf{B}_{\det} < 0$, attainment fails trivially since $L_{\min} \ge 0$.

Three facts from the machinery already built carry over. First, pair sufficiency extends verbatim: the resampling proof of Theorem 9 applies with $T = (Y_A, Y_B, V)$, since $S = V + U$ with U jointly independent of $(X, \text{all channel randomness})$ and the distortions are functionals of the pairs $(Y_i, \widehat{Y}_i)$; so the channel may be restricted to depend on X only through T, and then $I(X; \hat{\mathbf{Y}} \mid S) = I(T; \hat{\mathbf{Y}} \mid S)$. Second, the same mutual independence forces the Markov moment identity $\mathrm{Cov}(S, \hat{\mathbf{Y}}) = \mathrm{Cov}(V, \hat{\mathbf{Y}})$. Third, *for* $\tau^2 > 0$, jointly Gaussian channels $\hat{\mathbf{Y}} = AT + N$, $N \sim \mathcal{N}(0, \Sigma_N)$, exhaust the problem, by Lemma 11: for any admissible T-channel with $C = \mathrm{Cov}(T, \hat{\mathbf{Y}})$ and

[1] In the thermodynamic reading of Remark 6, the clipped form is the *work floor* on the reset work per symbol, $k_\mathrm{B} T \ln 2\,[\mathsf{B}_{\det}]^+$, the positive part reflecting that the bound carries no information where the determinant expression is negative.

$\Sigma_{\hat{Y}} = \mathrm{Cov}(\hat{\mathbf{Y}})$, set $A := C^{\mathsf{T}}\Sigma_T^{-1}$ and $\Sigma_N := \Sigma_{\hat{Y}} - A\Sigma_T A^{\mathsf{T}} \succeq 0$ (assume $\Sigma_T \succ 0$; degenerate instances reduce by dropping dependent context components; singular Σ_N is dispatched by Remark 12, whereby a singular residual either reveals a noiseless linear functional of T, making every bound trivial, or has a removable zero component). The lemma's positivity hypothesis $\mathrm{Cov}((T,S)) \succ 0$ holds exactly when $\tau^2 > 0$, since $\Sigma_{S|T} = \tau^2$; the boundary $\tau^2 = 0$, where $S = V$ is a coordinate of T and the joint covariance is singular, is treated separately by Lemma 22 below. The residual $\hat{\mathbf{Y}} - AT$ is uncorrelated with T, hence with V, and with U by mutual independence, hence with S, so for $\tau^2 > 0$

$$L \;\geq\; \tfrac{1}{2}\log_2 \frac{\det\bigl(A\Sigma_{T|S}A^{\mathsf{T}} + \Sigma_N\bigr)}{\det \Sigma_N}, \qquad \Sigma_{T|S} = \Sigma_T - \frac{\sigma_{TS}\sigma_{TS}^{\mathsf{T}}}{1+\tau^2}, \quad \sigma_{TS} = \Sigma_T e_V, \tag{42}$$

with equality for the jointly Gaussian channel realizing the same moments, and the distortion constraints $(e_i - A_{i\cdot})^{\mathsf{T}}\Sigma_T(e_i - A_{i\cdot}) + (\Sigma_N)_{ii} \leq D_i$ depend on the channel only through those moments.

$L_{\min}$ is attained. The feasible set

$$\mathcal{F} = \bigl\{(A,\Sigma_N) : \Sigma_N \succeq 0,\ (e_i - A_{i\cdot})^{\mathsf{T}}\Sigma_T(e_i - A_{i\cdot}) + (\Sigma_N)_{ii} \leq D_i,\ i \in \{A,B\}\bigr\}$$

is compact: the quadratic constraints confine the rows of A to ellipsoids ($\Sigma_T \succ 0$), the diagonal of Σ_N is bounded by (D_A, D_B), and the off-diagonal obeys $(\Sigma_N)_{AB}^2 \leq (\Sigma_N)_{AA}(\Sigma_N)_{BB} \leq D_A D_B$. The map $(A,\Sigma_N) \mapsto I(T; AT+N \mid S) = I\bigl(T;(AT+N,S)\bigr) - I(T;S)$ is lower semicontinuous on $\mathcal{F}$: the second term is a constant, the joint Gaussian law of $(T, AT+N, S)$ is weakly continuous in (A,Σ_N), and mutual information is lower semicontinuous under weak convergence of the joint law [8]. A minimizing sequence therefore has a cluster point in $\mathcal{F}$ at which the infimum is attained.

A plausible conjecture states the attainment condition for $\mathsf{B}_{\det}$ through the third party's conditional covariance $\Sigma_{Y|S}$; the truth involves the encoder-accessible $\Sigma_{Y|V}$, the same S-versus-V substitution as in Proposition 10. For the *unconditional* Gaussian vector source under individual distortion constraints, the exact tightness condition of the determinant (Hadamard-type) lower bound on the rate has recently been settled by Chen et al. [24]: the bound is tight precisely when a semidefinite condition of the form $\Delta \preceq \Sigma$ holds. Conditions (a) and (b) of Theorem 23 instantiate exactly that condition, on the conditional covariance $\Sigma_{Y|V}$ and on Σ_Y respectively; the new content here is which covariance carries it, namely the encoder-accessible one, decided by the conditioner's noise.

Lemma 22 (Clean-context reduction)**.** *Let $\tau^2 = 0$, so $S = V$ almost surely, and assume $\Sigma_c := \Sigma_{Y|V} \succ 0$. Write $\widetilde{Y} := Y - \beta V$, which is independent of V with covariance Σ_c, and let*

$$R_c(D_A, D_B) := \min\bigl\{I(\widetilde{Y};\hat{\mathbf{Y}}') :\ \mathbb{E}(\widetilde{Y}_i - \widehat{Y}_i')^2 \leq D_i,\ i \in \{A,B\}\bigr\}$$

be the unconditional minimum for the pair source $\widetilde{Y} \sim \mathcal{N}(0,\Sigma_c)$. Then

$$L_{\min}(D_A, D_B) \;=\; R_c(D_A, D_B),$$

and the determinant lower bound is attained at $\tau^2 = 0$, namely $L_{\min} = \mathsf{B}_{\det} = \frac{1}{2}\log_2(\det\Sigma_c/\det\Delta)$, if and only if $\Delta \preceq \Sigma_c$; otherwise $L_{\min} > \mathsf{B}_{\det}$ [23], [24].

Proof. Reduction of the objective. $\mathrm{Cov}(\widetilde{Y}, V) = \beta - \beta = 0$ and $(\widetilde{Y}, V)$ is jointly Gaussian, so $\widetilde{Y} \perp V$. The map $(Y,V) \leftrightarrow (\widetilde{Y}, V)$ is a linear bijection, so for every channel $I(T;\hat{\mathbf{Y}} \mid V) = I\bigl((\widetilde{Y},V);\hat{\mathbf{Y}} \mid V\bigr) = I(\widetilde{Y};\hat{\mathbf{Y}} \mid V)$.

Disintegration and recentering. By the per-s decomposition (as in Lemma 3, through a regular conditional probability), $I(\widetilde{Y};\hat{\mathbf{Y}} \mid V) = \int I_v\, dP_V(v)$ and $\mathbb{E}(Y_i - \widehat{Y}_i)^2 = \int D_i(v)\, dP_V(v)$ with $D_i(v) := \mathbb{E}\bigl[(Y_i - \widehat{Y}_i)^2 \mid V = v\bigr]$, where I_v is the mutual information of $(\widetilde{Y}, \hat{\mathbf{Y}})$ under the conditional law given $V = v$. Under that law $\widetilde{Y} \sim \mathcal{N}(0,\Sigma_c)$, independently of v. The deterministic shift $\hat{\mathbf{Y}}' := \hat{\mathbf{Y}} - \beta v$ changes neither I_v nor the errors, $Y_i - \widehat{Y}_i = \widetilde{Y}_i - \widehat{Y}_i'$, so each conditional ensemble is an admissible channel for the source $\mathcal{N}(0,\Sigma_c)$ at distortion pair $(D_A(v), D_B(v))$, whence $I_v \geq R_c\bigl(D_A(v), D_B(v)\bigr)$.

Convexity in the distortion pair. R_c is nonincreasing in each argument and jointly convex (mix the optimal channels of two distortion pairs; mutual information is convex in the channel at fixed input). By Jensen and monotonicity,

$$I(\widetilde{Y};\hat{\mathbf{Y}} \mid V) \;\geq\; \int R_c\bigl(D_A(v), D_B(v)\bigr) dP_V(v) \;\geq\; R_c\Bigl(\textstyle\int D_A\, dP_V, \int D_B\, dP_V\Bigr) \;\geq\; R_c(D_A, D_B).$$

Achievability. Let $p(\hat{\mathbf{y}}' \mid \tilde{y})$ attain $R_c(D_A, D_B)$; it exists by the S-deleted instance of the compactness and lower-semicontinuity argument above. Set $\hat{\mathbf{Y}} := \hat{\mathbf{Y}}' + \beta V$ with $\hat{\mathbf{Y}}'$ drawn from $p(\cdot \mid \widetilde{Y})$ using channel randomness independent of V. This is an admissible channel (a function of X and fresh noise; the Markov chain is trivial at $S = V$), its distortions are $\mathbb{E}(\widetilde{Y}_i - \widehat{Y}_i')^2 \le D_i$, and given $V = v$ the pair $(\widetilde{Y}, \hat{\mathbf{Y}})$ is $(\widetilde{Y}, \hat{\mathbf{Y}}' + \beta v)$, so $I(\widetilde{Y}; \hat{\mathbf{Y}} \mid V) = I(\widetilde{Y}; \hat{\mathbf{Y}}') = R_c(D_A, D_B)$. Hence $L_{\min} = R_c(D_A, D_B)$, attained.

The tightness condition. At $\tau^2 = 0$, $\Sigma_{Y|S} = \Sigma_c$, so $\mathsf{B}_{\det} = \frac{1}{2}\log_2(\det\Sigma_c / \det\Delta)$. Sufficiency: if $\Delta \preceq \Sigma_c$, take $\hat{\mathbf{Y}}' \sim \mathcal{N}(0, \Sigma_c - \Delta)$ and $Z \sim \mathcal{N}(0, \Delta)$ independent with $\widetilde{Y} = \hat{\mathbf{Y}}' + Z$ (the reverse channel; the required covariances are $\Sigma_c - \Delta \succeq 0$ and $\Delta \succ 0$, no further condition). Its distortions are D_i exactly, and since $\widetilde{Y} \mid \hat{\mathbf{Y}}' \sim \mathcal{N}(\hat{\mathbf{Y}}', \Delta)$,

$$I(\widetilde{Y}; \hat{\mathbf{Y}}') = h(\widetilde{Y}) - h(\widetilde{Y} \mid \hat{\mathbf{Y}}') = \tfrac{1}{2}\log_2\big((2\pi e)^2 \det\Sigma_c\big) - \tfrac{1}{2}\log_2\big((2\pi e)^2 \det\Delta\big) = \mathsf{B}_{\det}.$$

Necessity: suppose some channel attains $I(\widetilde{Y}; \hat{\mathbf{Y}}') = \mathsf{B}_{\det}$, and write $Z' := \widetilde{Y} - \hat{\mathbf{Y}}'$. In the chain

$$\mathsf{B}_{\det} = I(\widetilde{Y}; \hat{\mathbf{Y}}') = h(\widetilde{Y}) - h\big(Z' \mid \hat{\mathbf{Y}}'\big) \;\ge\; h(\widetilde{Y}) - h(Z') \;\ge\; h(\widetilde{Y}) - h(Z_A') - h(Z_B') \;\ge\; \mathsf{B}_{\det},$$

equality in conditioning-reduces-entropy, in subadditivity, and in the Gaussian maximum-entropy bound forces $Z' \perp \hat{\mathbf{Y}}'$ with $Z' \sim \mathcal{N}(0, \Delta)$ (the same three-step equality analysis spelled out in the necessity part of Theorem 23's proof below); then $\mathrm{Cov}(\hat{\mathbf{Y}}') = \Sigma_c - \Delta \succeq 0$, that is, $\Delta \preceq \Sigma_c$. Since R_c is attained, $L_{\min} > \mathsf{B}_{\det}$ whenever $\Delta \npreceq \Sigma_c$. □

Theorem 23 (Consequence: determinant-bound exactness)**.** *The determinant lower bound is attained, $L_{\min} = \mathsf{B}_{\det}$, iff*

$$\textit{(a) } \tau^2 = 0 \textit{ and } \Delta \preceq \Sigma_{Y|V}, \qquad \textit{or} \qquad \textit{(b) } \beta = 0 \textit{ and } \Delta \preceq \Sigma_Y.$$

In every other case the bound is strict, both on the misaligned branch ($\tau^2 > 0$ and $\beta \neq 0$, at every positive (D_A, D_B)) and outside the semidefinite conditions of (a) and (b). On the misaligned branch with $\Delta \prec \Sigma_{Y|V}$, the deficit obeys the sandwich

$$0 \;<\; L_{\min} - \mathsf{B}_{\det} \;\le\; \tfrac{1}{2}\log_2 \frac{\det\big(I - \Sigma_{Y|S}^{-1}\Delta\big)}{\det\big(I - \Sigma_{Y|V}^{-1}\Delta\big)} \;=\; \tfrac{1}{2}\,\mathrm{tr}\big[(\Sigma_{Y|V}^{-1} - \Sigma_{Y|S}^{-1})\Delta\big]\log_2 e + O(\|\Delta\|^2) \;\longrightarrow\; 0 \;\; (\Delta \to 0).$$

Map of the attainment proof: The proof tracks equalities through the chain (41) and reads off one structural conclusion at each. The clean and noisy branches split first: at $\tau^2 = 0$ the joint covariance of (T, S) is singular and the question reduces, by Lemma 22, to the unconditional pair source with covariance $\Sigma_{Y|V}$. At $\tau^2 > 0$ the chain is $L \ge I(Y; \hat{\mathbf{Y}} \mid S) = h(Y \mid S) - h(Z \mid \hat{\mathbf{Y}}, S) \ge h(Y \mid S) - h(Z) \ge h(Y \mid S) - h(Z_A) - h(Z_B) \ge \mathsf{B}_{\det}$ with $Z = Y - \hat{\mathbf{Y}}$, and attainment forces every inequality to equality. Equality in conditioning-reduces-entropy makes Z independent of $(\hat{\mathbf{Y}}, S)$; the finiteness this step needs is automatic along the chain, since $h(Z \mid \hat{\mathbf{Y}}, S) = -\infty$ would make the conditional content infinite rather than equal to $\mathsf{B}_{\det}$. Equality in subadditivity makes $Z_A \perp Z_B$. Equality in the Gaussian maximum-entropy bound makes each Z_i Gaussian with variance exactly D_i, the distortion constraints active. Together these pin the error law, $Z \sim \mathcal{N}(0, \Delta)$ independent of $(\hat{\mathbf{Y}}, S)$, and hence the moments $\mathrm{Cov}(\hat{\mathbf{Y}}) = \Sigma_Y - \Delta \succeq 0$, the semidefinite part of the conditions, and $\mathrm{Cov}(\hat{\mathbf{Y}}, V) = \beta$. No Gaussianity of the channel itself is used: the source (V, Y, S) is Gaussian, so $\mathbb{E}[V \mid Y, S]$ is linear, and the remaining equality $I(T; \hat{\mathbf{Y}} \mid S) = I(Y; \hat{\mathbf{Y}} \mid S)$, conditional independence of V and $\hat{\mathbf{Y}}$ given (Y, S), becomes a vanishing partial covariance; the block inversion displayed in the proof evaluates it as a positive multiple of $\tau^2 \Delta \Sigma_Y^{-1} \beta$, forcing $\tau^2 = 0$ or $\beta = 0$ and so producing the two branches (a) and (b). Throughout, $\Delta \succ 0$ and $\Sigma_T \succ 0$ are in force (zero distortions and degenerate covariances are excluded by the standing assumptions), and the clipping is immaterial here: on the attainment branches $\mathsf{B}_{\det} \ge 0$, and where $\mathsf{B}_{\det} < 0$ attainment fails trivially since $L_{\min} \ge 0$.

Proof. The branch $\tau^2 = 0$ is Lemma 22: there $\Sigma_{Y|S} = \Sigma_{Y|V}$, and $L_{\min} = \mathsf{B}_{\det}$ iff $\Delta \preceq \Sigma_{Y|V}$, which is condition (a). It remains to treat $\tau^2 > 0$, where $L_{\min}$ is attained by the compactness and lower-semicontinuity argument preceding Lemma 22.

Necessity at $\tau^2 > 0$: the equality analysis. Suppose a channel attains $L = \mathsf{B}_{\det}$. Write $Z := Y - \hat{\mathbf{Y}}$. The chain (41) reads, term by term,

$$\mathsf{B}_{\det} = L \;\ge\; I(Y; \hat{\mathbf{Y}} \mid S) = h(Y \mid S) - h\big(Z \mid \hat{\mathbf{Y}}, S\big) \;\ge\; h(Y \mid S) - h(Z) \;\ge\; h(Y \mid S) - h(Z_A) - h(Z_B) \;\ge\; \mathsf{B}_{\det},$$

using $h(Y \mid \hat{\mathbf{Y}}, S) = h(Z \mid \hat{\mathbf{Y}}, S)$ (translation by the conditioning variable), conditioning reduces entropy, subadditivity, and the Gaussian maximum-entropy bound $h(Z_i) \le \frac{1}{2}\log_2\bigl(2\pi e \operatorname{Var} Z_i\bigr) \le \frac{1}{2}\log_2(2\pi e D_i)$. Equality throughout forces: (i) $h(Z \mid \hat{\mathbf{Y}}, S) = h(Z)$, so $I\bigl(Z; (\hat{\mathbf{Y}}, S)\bigr) = 0$ and Z is independent of $(\hat{\mathbf{Y}}, S)$; (ii) $h(Z) = h(Z_A) + h(Z_B)$, so $Z_A \perp Z_B$; (iii) each Z_i is Gaussian with variance exactly D_i. Hence $Z \sim \mathcal{N}(0, \Delta)$ independent of $(\hat{\mathbf{Y}}, S)$. No replacement of the channel by a Gaussian one is needed: the equality analysis delivers the Gaussianity of the error directly, and only moments enter below. Consequences at the moment level: $\operatorname{Cov} Z = \Delta$, $\operatorname{Cov}(Z, \hat{\mathbf{Y}}) = 0$, $\operatorname{Cov}(Z, S) = 0$; also $\operatorname{Cov}(Z, U) = 0$ (Z is a function of X and channel noise, U is independent of both), so $\operatorname{Cov}(Z, V) = \operatorname{Cov}(Z, S) - \operatorname{Cov}(Z, U) = 0$ and $\operatorname{Cov}(\hat{\mathbf{Y}}, V) = \operatorname{Cov}(Y, V) - \operatorname{Cov}(Z, V) = \beta$. Moreover $\operatorname{Cov}(Y, Z) = \operatorname{Cov}(\hat{\mathbf{Y}} + Z, Z) = \Delta$, so

$$\operatorname{Cov}(\hat{\mathbf{Y}}) = \operatorname{Cov}(Y - Z) = \Sigma_Y - \Delta - \Delta + \Delta = \Sigma_Y - \Delta \succeq 0, \qquad \text{that is,} \qquad \Delta \preceq \Sigma_Y.$$

Necessity at $\tau^2 > 0$: the no-leakage condition and the block inversion. Equality of the first step of the chain, $L = I(T; \hat{\mathbf{Y}} \mid S) = I(Y; \hat{\mathbf{Y}} \mid S)$, gives $I(V; \hat{\mathbf{Y}} \mid Y, S) = 0$: V and $\hat{\mathbf{Y}}$ are conditionally independent given (Y, S). Because (V, Y, S) is jointly Gaussian, $\mathbb{E}[V \mid Y, S]$ is linear, and the law of total covariance turns conditional independence into a vanishing *partial* covariance: with $V_{\text{lin}} := \mathbb{E}[V \mid Y, S]$ and $\hat{\mathbf{Y}}_{\text{lin}}$ the best linear estimate of $\hat{\mathbf{Y}}$ from (Y, S),

$$\operatorname{pCov}\bigl(\hat{\mathbf{Y}}, V\bigr) := \operatorname{Cov}\bigl(\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{\text{lin}},\, V - V_{\text{lin}}\bigr) = \mathbb{E}\bigl[\operatorname{Cov}(\hat{\mathbf{Y}}, V \mid Y, S)\bigr] = 0,$$

where the cross term of the total-covariance decomposition vanishes identically because V_{lin} coincides with the conditional mean $\mathbb{E}[V \mid Y, S]$ for the jointly Gaussian triple (V, Y, S). Now evaluate pCov from the pinned moments. With $M := \operatorname{Cov}\bigl((Y, S)\bigr) = \bigl[\begin{smallmatrix} \Sigma_Y & \beta \\ \beta^\mathsf{T} & s \end{smallmatrix}\bigr]$ ($s = 1 + \tau^2$), $\operatorname{Cov}(\hat{\mathbf{Y}}, V) = \beta$, $\operatorname{Cov}\bigl(\hat{\mathbf{Y}}, (Y, S)\bigr) = [\Sigma_Y - \Delta,\ \beta]$, and $\operatorname{Cov}\bigl((Y, S), V\bigr) = [\beta; 1]$,

$$\operatorname{pCov}(\hat{\mathbf{Y}}, V) = \beta - [\Sigma_Y - \Delta,\ \beta]\, M^{-1} \begin{bmatrix} \beta \\ 1 \end{bmatrix} = \beta - \bigl([I_2,\ 0] - [\Delta,\ 0]\, M^{-1}\bigr) \begin{bmatrix} \beta \\ 1 \end{bmatrix} = [\Delta,\ 0]\, M^{-1} \begin{bmatrix} \beta \\ 1 \end{bmatrix},$$

where $[\Sigma_Y,\ \beta] M^{-1} = [I_2,\ 0]$ because $[\Sigma_Y,\ \beta]$ is the first block row of M. Solve $Mw = [\beta; 1]$ by block elimination: writing $w = (w_Y, w_S)$ and $\rho_V^2 := \beta^\mathsf{T} \Sigma_Y^{-1} \beta$ (the variance explained by the best linear predictor of V from Y; $\rho_V^2 < 1$ since $\Sigma_T \succ 0$),

$$\Sigma_Y w_Y + \beta w_S = \beta, \quad \beta^\mathsf{T} w_Y + s w_S = 1 \implies w_S = \frac{1 - \rho_V^2}{s - \rho_V^2}, \qquad w_Y = \frac{\tau^2}{s - \rho_V^2}\, \Sigma_Y^{-1} \beta,$$

with $s - \rho_V^2 \ge s - 1 = \tau^2 > 0$. Hence

$$\operatorname{pCov}(\hat{\mathbf{Y}}, V) = \Delta w_Y = \frac{\tau^2}{s - \rho_V^2}\, \Delta \Sigma_Y^{-1} \beta,$$

which vanishes iff $\tau^2 = 0$ or $\beta = 0$ ($\Delta \succ 0$ and Σ_Y^{-1} nonsingular act injectively). On the branch $\tau^2 > 0$ this forces $\beta = 0$. Combined with $\Delta \preceq \Sigma_Y$ above, attainment at $\tau^2 > 0$ implies condition (b); attainment at $\tau^2 = 0$ implies condition (a) by Lemma 22. Necessity is proved.

Sufficiency (a). Lemma 22.

Sufficiency (b). At $\beta = 0$, $Y \perp (V, U)$, so $\Sigma_{Y|S} = \Sigma_Y$. Take the reverse channel of Lemma 22's tightness step with Σ_Y in place of Σ_c: $\hat{\mathbf{Y}} \sim \mathcal{N}(0, \Sigma_Y - \Delta)$, $Z \sim \mathcal{N}(0, \Delta)$ independent, $Y = \hat{\mathbf{Y}} + Z$, which exists iff $\Delta \preceq \Sigma_Y$. The channel depends on X only through Y, with channel randomness independent of (V, U), so $\hat{\mathbf{Y}} \perp S$ and $\hat{\mathbf{Y}} \perp (V, S)$ given Y; hence $I(T; \hat{\mathbf{Y}} \mid S) = h(\hat{\mathbf{Y}} \mid S) - h(\hat{\mathbf{Y}} \mid T, S) = h(\hat{\mathbf{Y}}) - h(\hat{\mathbf{Y}} \mid Y) = I(Y; \hat{\mathbf{Y}}) = \frac{1}{2}\log_2\bigl(\det \Sigma_Y / \det \Delta\bigr) = \mathsf{B}_{\det}$ [23], [24].

Sandwich. At $\tau^2 > 0$ with $\Delta \prec \Sigma_{Y|V}$, take $Z \sim \mathcal{N}(0, \Delta)$ jointly Gaussian with $\operatorname{Cov}(Z, Y) = \Delta$, $\operatorname{Cov}(Z, V) = 0$; such a Z exists iff $\Delta - \Delta \Sigma_{Y|V}^{-1} \Delta \succeq 0 \Leftrightarrow \Delta \preceq \Sigma_{Y|V}$ (Schur complement). The channel $\hat{\mathbf{Y}} = Y - Z$ yields $L \le \frac{1}{2}\log_2[\det(\Sigma_{Y|S} - \Delta) / \det(\Delta - \Delta \Sigma_{Y|V}^{-1} \Delta)] = \mathsf{B}_{\det} + \frac{1}{2}\log_2[\det(I - \Sigma_{Y|S}^{-1}\Delta) / \det(I - \Sigma_{Y|V}^{-1}\Delta)]$; the ratio is ≥ 1 because $\Sigma_{Y|S} \succeq \Sigma_{Y|V}$ (congruence by $\Delta^{1/2}$ keeps both factors in the PSD order where $\det$ is monotone), and $\to 1$ as $\Delta \to 0$. Strictness (> 0) is the necessity direction, using that $L_{\min}$ is attained at $\tau^2 > 0$ (the compactness and lower-semicontinuity argument above).

The single-variable corollary. For a single described variable, on the active regime $D \le \operatorname{Var}(Y \mid V)$, the determinant bound $\frac{1}{2}\log_2(\operatorname{Var}(Y \mid S)/D)$ is attained iff $\rho\tau = 0$: substituting $g_f = (s - \rho^2)/(Ds)$ into the quadratic

$P(g)$ of Theorem 13, equation (20), whose larger root $g^\star$ gives $L(D) = \frac{1}{2}\log_2 g^\star$, yields $P(g_f) = -\rho^2\tau^2/s$, which is < 0 (hence $g^\star > g_f$ strictly) exactly when $\rho\tau \neq 0$. (The regime qualifier is the single-variable shadow of condition (a): for $\tau = 0$, $D > 1-\rho^2$ the clipped bound $[\mathsf{B}_{\rm det}]^+ = 0$ and $L = 0$ meet trivially while $g_f < 1$ becomes the quadratic's *smaller* root.) $\square$

VII. Vector Context

The frontier survives a vector conditioner. Let $S = \Gamma^\mathsf{T} X + U \in \mathbb{R}^r$, $U \sim \mathcal{N}(0, \Sigma_U)$, $\Sigma_U \succ 0$, with X, U, and all channel randomness mutually independent. Write $V := \Gamma^\mathsf{T} X \in \mathbb{R}^r$ and $T := (Y, V) \in \mathbb{R}^{1+r}$, and assume $\Sigma_T \succ 0$; degenerate instances reduce by dropping linearly dependent context components. Reproductions are scalar without loss of optimality by the data-processing reduction of Section III, which is generic in S.

Theorem 24 (Extension: vector context)**.** *Let $0 < D < 1$. (i)* Sufficiency: *it is without loss of optimality to let the channel depend on X only through T, and then $I(X; \widehat{Y} \mid S) = I(T; \widehat{Y} \mid S)$. (ii)* Whitened form: *with W, $Z = WT$, $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_{1+r})$, $0 \prec \Lambda \preceq I$, and the unit vector y_0 of* (37) *at $p = 1+r$, every admissible channel with regression vector $c = \mathbb{E}[Z\widehat{Y}]$ and residual variance $n > 0$ satisfies*

$$R \;\geq\; \tfrac{1}{2}\log_2 \frac{|c|^2 + n}{n}, \qquad L \;\geq\; \tfrac{1}{2}\log_2 \frac{c^\mathsf{T}\Lambda c + n}{n}, \qquad \mathbb{E}(Y - \widehat{Y})^2 = |y_0 - c|^2 + n, \tag{43}$$

with both bounds attained simultaneously by the jointly Gaussian channel $\widehat{Y} = c^\mathsf{T} Z + N$, $N \sim \mathcal{N}(0, n)$. (iii) The frontier*, traced by $\alpha \in [0,1]$: with $\gamma_0 = n/(|c|^2 + n)$ and $\gamma_1 = n/(c^\mathsf{T}\Lambda c + n)$, the Pareto-optimal channel is the unique solution with $n > 0$ of the generalized two-water-level system*

$$c = \big(1 - \alpha\gamma_0 - (1-\alpha)\gamma_1\big)\big[\big(1 - (1-\alpha)\gamma_1\big)I + (1-\alpha)\gamma_1\Lambda\big]^{-1} y_0, \qquad n = D - |y_0 - c|^2, \tag{44}$$

per-mode explicit given the two scalars (γ_0, γ_1), and the frontier point is $\big(\frac{1}{2}\log_2(1/\gamma_0), \frac{1}{2}\log_2(1/\gamma_1)\big)$. For $r = 1$ the program coincides with that of Theorem 16 under the linear bijection $c = W^{-\mathsf{T}}(a,b)^\mathsf{T}$, so the two systems trace the same frontier. The $\alpha = 0$ endpoint gives the vector-context function $L(D)$ as a one-dimensional root problem in γ_1.

Proof. (i) In the resampling proof of Theorem 9, the only property of S used is that its conditional law given $(X, \widehat{Y})$, and given $(X, \widehat{Y}')$, is the fixed kernel $p(s \mid t, \hat{y}) = p_U(s - v)$, depending on x only through t. This holds verbatim for the vector $S = V + U$ with $U \sim \mathcal{N}(0, \Sigma_U)$ jointly independent of X and the channel randomness. Steps (a) through (d) of that proof are unchanged.

(ii) *Whitening.* First, $\Sigma_{T|S} = \Sigma_T - \Sigma_{TS}\Sigma_S^{-1}\Sigma_{ST} \succ 0$, where $\Sigma_S = \Sigma_V + \Sigma_U$ is the conditioner's covariance (V is not renormalized in this section): positivity of $\operatorname{Cov}((T,S))$ holds because $\Sigma_T \succ 0$ and $\Sigma_{S|T} = \operatorname{Cov}(\Gamma^\mathsf{T} X \mid T) + \Sigma_U \succeq \Sigma_U \succ 0$, and the Schur complement of a positive definite matrix is positive definite. Also $\Sigma_{T|S} \preceq \Sigma_T$. The congruence (37) therefore exists at $p = 1+r$ with $0 \prec \Lambda \preceq I$, and $\operatorname{Cov} Z = I$ makes y_0 a unit vector: $Y = y_0^\mathsf{T} Z$, $|y_0|^2 = e_Y^\mathsf{T}\Sigma_T e_Y = 1$.

Moments and the two bounds. Fix an admissible channel, centered without loss, with $c = \mathbb{E}[Z\widehat{Y}]$ and $n = \operatorname{Var}\widehat{Y} - |c|^2$; the case $n = 0$ is dispatched by Remark 12 (the reproduction is then a noiseless linear functional of T, so either every bound is trivial or the reproduction is null). The residual $N' = \widehat{Y} - c^\mathsf{T} Z$ is uncorrelated with Z, hence with T and with its subvector V; and it is uncorrelated with U by mutual independence, hence with $S = V + U$: componentwise, $\operatorname{Cov}(N', S) = 0$. Lemma 11 (the two determinant lower bounds, with equality for the jointly Gaussian channel matching the second moments) applies with $A = c^\mathsf{T} W$: $A\Sigma_T A^\mathsf{T} = c^\mathsf{T} W\Sigma_T W^\mathsf{T} c = |c|^2$ and $A\Sigma_{T|S}A^\mathsf{T} = c^\mathsf{T}\Lambda c$, giving the two bounds of (43). The distortion evaluation is exact because Z is white: $\mathbb{E}(Y - \widehat{Y})^2 = 1 - 2y_0^\mathsf{T} c + (|c|^2 + n) = |y_0 - c|^2 + n$. Equality for the Gaussian channel is Lemma 11's equality clause.

(iii) *The weighted program.* As in Theorem 16: both coordinates are convex in the channel, so the frontier is the trace of the weighted minimizations, and the weighted objective is bounded below by the moment program

$$\min_{c,n} \; \frac{\alpha}{2}\log_2 \frac{Q_0 + n}{n} + \frac{1-\alpha}{2}\log_2 \frac{Q_1 + n}{n}, \qquad Q_0 = |c|^2, \quad Q_1 = c^\mathsf{T}\Lambda c, \quad |y_0 - c|^2 + n \leq D, \; n > 0,$$

whose value the Gaussian channel of (ii) attains. The objective is strictly decreasing in n at fixed $c \neq 0$, so the distortion constraint is active at the optimum. Feasibility forces $2y_0^\mathsf{T} c \geq 1 - D + v > 1 - D > 0$ as in (38), so

$c \neq 0$ and $Q_1 \geq \lambda_{\min}|c|^2$ is bounded away from zero on the feasible set, giving coercivity as $n \to 0$ and attainment at an interior stationary point of the reduced objective in c, with $n = D - |y_0 - c|^2$.

The first-order system. At such a point, with $h = |y_0 - c|^2$, $\nabla h = -2(y_0 - c)$, $\nabla n = 2(y_0 - c)$, $\nabla Q_0 = 2c$, and $\nabla Q_1 = 2\Lambda c$, stationarity of $\alpha \ln(Q_0 + n) + (1-\alpha)\ln(Q_1 + n) - \ln n$ reads, after multiplying by $n/2$,

$$\alpha\gamma_0\big(c + (y_0 - c)\big) + (1-\alpha)\gamma_1\big(\Lambda c + (y_0 - c)\big) - (y_0 - c) = 0.$$

The first parenthesis collapses, $c + (y_0 - c) = y_0$; collecting the c terms,

$$\big[\big(1 - (1-\alpha)\gamma_1\big)I + (1-\alpha)\gamma_1\Lambda\big]c = \big(1 - \alpha\gamma_0 - (1-\alpha)\gamma_1\big)y_0,$$

which is (44). The bracket is invertible: each diagonal entry $\big(1-(1-\alpha)\gamma_1\big) + (1-\alpha)\gamma_1\lambda_j$ is a convex combination of 1 and $\lambda_j > 0$, since $(1-\alpha)\gamma_1 \in [0,1)$. Uniqueness of the minimizer, and hence of the system's solution with $n > 0$, at every $\alpha \in [0,1]$ is Proposition 18 at $p = 1+r$ (the weighted program is convex in the moment coordinates with a unique minimizer, and every stationary point with $n > 0$ is that minimizer); its proof is dimension-free. Achievability of the frontier point is (ii).

For $r = 1$ the bijection $c = W^{-\mathsf{T}}(a,b)^{\mathsf{T}}$ carries (Q_0, Q_1, h) and the constraint to their Section IV forms, so the two weighted programs are identical with the same unique minimizer, and the two stationarity systems are equivalent parameterizations of it. At $\alpha = 0$, (44) reads $c = (1-\gamma_1)\big[(1-\gamma_1)I + \gamma_1\Lambda\big]^{-1}y_0$; substituting into $n = D - |y_0 - c|^2$ and $\gamma_1 = n/(c^{\mathsf{T}}\Lambda c + n)$ leaves one scalar equation in $\gamma_1 \in (0,1)$, the vector-context generalization of the quadratic (20). □

VIII. The Binary Solution

The binary case is in this paper for two reasons: it shows that the misalignment phenomenon and the strict rate–content tradeoff are not Gaussian artifacts, the frontier being fully characterized here too (Remark 28), and it shows the functional is explicitly computable outside the Gaussian family. The case closes completely and, unlike its Gaussian counterpart, with an elementary self-contained proof.

Let (Y, V) be a doubly symmetric binary pair (fair marginals, $\Pr[Y \neq V] = p \in (0, \frac{1}{2})$), context V, side information $S = V \oplus W$ with $W \sim \mathrm{Bern}(q)$, $q \in [0, \frac{1}{2}]$, independent. The description is the reproduction $\hat{Y} \in \{0,1\}$ itself (richer descriptions only cost more: for any stored Z from which the decoder computes $\hat{Y} = f(Z)$, $I(Y,V;Z \mid S) \geq I(Y,V;\hat{Y} \mid S)$ by the data-processing inequality), Hamming distortion $\Pr[\hat{Y} \neq Y] \leq D$. Write $x * q = x(1-q) + (1-x)q$ and $\ell(x) = \ln\frac{1-x}{x}$.

Lemma 25 (Convexity over the shared channel)**.** *$L(C) := I(Y,V;\hat{Y} \mid S)$ is convex in the channel $C = P(\hat{y} \mid y, v)$, and the Hamming distortion is linear in C.*

Proof. The Markov chain $\hat{Y} - (Y,V) - S$ means that one channel C is shared across the values of S, so $L(C) = \sum_s P(s)\, I_s(C)$, where $I_s(C)$ is the mutual information of the channel C under the input law $P(Y,V \mid S = s)$. The input laws depend on s, but each I_s is convex in C at its own fixed input law, so the nonnegative combination is convex in C. Linearity of the distortion is immediate. □

Lemma 26 (The tilt root: existence, interiority, uniqueness)**.** *Fix $p \in (0, \frac{1}{2})$, $q \in (0, \frac{1}{2}]$, $D \in (0, \frac{1}{2})$. Parameterize the constraint segment $(1-p)d_0 + pd_1 = D$, $(d_0, d_1) \in (0,1)^2$, by $d_0 \in \big(\max\{0, \frac{D-p}{1-p}\},\ \frac{D}{1-p}\big)$, an open subinterval of $(0,1)$, and define the tilt residual*

$$\psi(d_0) := \ell(d_0) - \ell\big(d_1(d_0)\big) - 2(1-2q)\,\ell\big(u(d_0)\big),$$

$$d_1(d_0) = \frac{D - (1-p)d_0}{p}, \qquad u = a * q, \qquad a = 2(1-p)d_0 + p - D.$$

Then ψ has exactly one zero on the open segment, and the family value $L(d_0)$ attains its minimum over the closed segment exactly there. The same statements hold at $q = 0$ when $D < p$.

Proof. Work in nats; $L(d_0) = H(u) - (1-p)H(d_0) - pH\big(d_1(d_0)\big)$ with H the natural-log binary entropy. Differentiating with $H'(x) = \ell(x)$, $u'(d_0) = 2(1-p)(1-2q)$, and $d_1'(d_0) = -(1-p)/p$,

$$L'(d_0) = 2(1-p)(1-2q)\,\ell(u) - (1-p)\,\ell(d_0) + (1-p)\,\ell(d_1) = -(1-p)\,\psi(d_0).$$

Existence and interiority. Boundary behavior. At the left endpoint: if $D \le p$ the endpoint is $d_0 = 0$ and $\ell(d_0) \to +\infty$; if additionally $D = p$ then also $d_1 \to 1$ and $-\ell(d_1) \to +\infty$, the same sign; if $D > p$ the endpoint is $d_0 = (D-p)/(1-p) > 0$ with $d_1 \to 1$, so $-\ell(d_1) \to +\infty$ with the other terms bounded. The u term stays bounded in every case: $u = a * q \in [q, 1-q]$ with $q > 0$. Hence $\psi \to +\infty$ at the left endpoint. At the right endpoint $d_0 = D/(1-p) < 1$ (since $1 - p > \frac{1}{2} > D$): $d_1 \to 0$, so $-\ell(d_1) \to -\infty$ with the other terms bounded, and $\psi \to -\infty$. ψ is continuous on the open segment, so it has a zero there by the intermediate value theorem, and trivially every zero is interior.

Zeros are minimizers. The family channel is affine in (d_0, d_1), hence affine in d_0 along the segment, and L is convex in the channel (Lemma 25), so L is convex on the segment. For a convex differentiable function, $L'(d_0) = 0$ at an interior point makes that point a global minimizer; conversely $L' \to -\infty$ at the left endpoint and $L' \to +\infty$ at the right (the sign flips of ψ), so the minimum of L over the closed segment is attained in the interior and only at zeros of ψ.

Uniqueness. The minimizer set of the convex L is a closed interval on which L is constant. Suppose it contains two distinct points. Then $L'' \equiv 0$ on a nondegenerate subinterval. But L is real analytic on the open segment: d_1 and a are affine in d_0 with values in $(0,1)$, $u = a(1-2q) + q \in (0,1)$, and H is real analytic on $(0,1)$. By the identity theorem, $L'' \equiv 0$ on the whole connected open segment, so L' is constant there, contradicting $L' \to -\infty$ at the left endpoint. Hence the minimizer is unique, and since the zeros of ψ are exactly the interior stationary points of L, which are exactly its minimizers, ψ has exactly one zero.

At $q = 0$ with $D < p$ the same argument applies verbatim: the left endpoint is $d_0 = 0$ with $d_1(0) = D/p < 1$, and $u = a \in (p - D, p + D) \subset (0,1)$ stays bounded away from 0 and 1, so the boundary blow-ups and the analyticity are as before. □

Fig. 4 shows the objective, the tilt residual, and the rate–content frontier at representative instances.

Theorem 27 (Extension: the binary function)**.** *For $D \in (0, \frac{1}{2})$, the conditional coordinate $L(D) = \min I(Y, V; \hat{Y} \mid S)$ over channels $P(\hat{Y} \mid Y, V)$ (Markov $\hat{Y} - (Y, V) - S$) with $\Pr[\hat{Y} \neq Y] \le D$ is attained in the two-parameter symmetric family $\hat{Y} = Y \oplus E$, $\Pr[E = 1 \mid Y \oplus V = j] = d_j$, and equals*

$$L(D) \;=\; h_2(u) - (1-p)\, h_2(d_0) - p\, h_2(d_1), \qquad u = a * q, \quad a = (1-p)d_0 + p(1-d_1),$$

where, for $q \in (0, \frac{1}{2}]$ with any $D \in (0, \frac{1}{2})$, and for $q = 0$ with $D < p$, the pair (d_0, d_1) is the unique *solution, interior to the constraint segment (Lemma 26), of the active constraint $(1-p)d_0 + pd_1 = D$ together with the* tilt equation

$$\ell(d_0) - \ell(d_1) \;=\; 2(1-2q)\, \ell(u).$$

At the boundary point $q = 0$, $D = p$, the minimum $L = 0$ is attained at the segment endpoint $(d_0, d_1) = (0, 1)$, the reproduction $\hat{Y} = V$. (In the excluded sliver $q = 0$, $D > p$, the constraint is inactive and $L(D) = 0$, attained non-uniquely by reproductions measurable in V alone; the tilt system still has a root there, the degenerate point $d_1 = 1 - d_0$, which evaluates to the correct value 0. For $q > 0$ the constraint is active on all of $(0, \frac{1}{2})$: L is convex, nonincreasing, positive by the last claim below, and 0 at $D = \frac{1}{2}$, hence strictly decreasing.) On the same channel the rate coordinate is $R = 1 - (1-p)h_2(d_0) - p\, h_2(d_1)$ and the discount is exactly $R - L = 1 - h_2(u) = I(\hat{Y}; S)$ (the chain-rule identity $I(Y, V; \hat{Y}) - I(Y, V; \hat{Y} \mid S) = I(\hat{Y}; S)$, valid under the Markov chain, in closed form: $a = \Pr[\hat{Y} \neq V]$, $u = \Pr[\hat{Y} \neq S]$). Anchors: at $q = 0$ the system gives $a = (p - D)/(1 - 2D)$ and $L = h_2(p) - h_2(D)$ (Gray's conditional rate–distortion function [4], for $D \le p$); at $q = \frac{1}{2}$ it gives $d_0 = d_1 = D$ and $L = 1 - h_2(D)$ (the marginal rate–distortion function; the context is useless exactly when the side information is). For $q \in (0, \frac{1}{2}]$ and $D < \frac{1}{2}$, $L(D) > 0$.

Proof. Reduction and convexity. Binary $\hat{Y}$ is definitional (DPI above); convexity of L in the shared channel and linearity of the distortion are Lemma 25.

Symmetrization. The global bit flip $\sigma : (y, v, s, \hat{y}) \mapsto (1-y, 1-v, 1-s, 1-\hat{y})$ preserves the joint law of (Y, V, S): the pair (Y, V) is doubly symmetric with fair marginals, so $(1-Y, 1-V) \overset{d}{=} (Y, V)$, and $W \sim \mathrm{Bern}(q)$ is independent of (Y, V), so $S = V \oplus W$ flips with V. The flip also preserves the distortion, and $L(C^\sigma) = L(C)$; averaging C with C^σ preserves distortion and, by convexity, does not increase L. So an optimum exists among σ-symmetric channels, and these are exactly the family $\{(d_0, d_1)\}$: σ-symmetry forces $\Pr[\hat{Y} \neq Y \mid y, v]$ to depend only on $y \oplus v$.

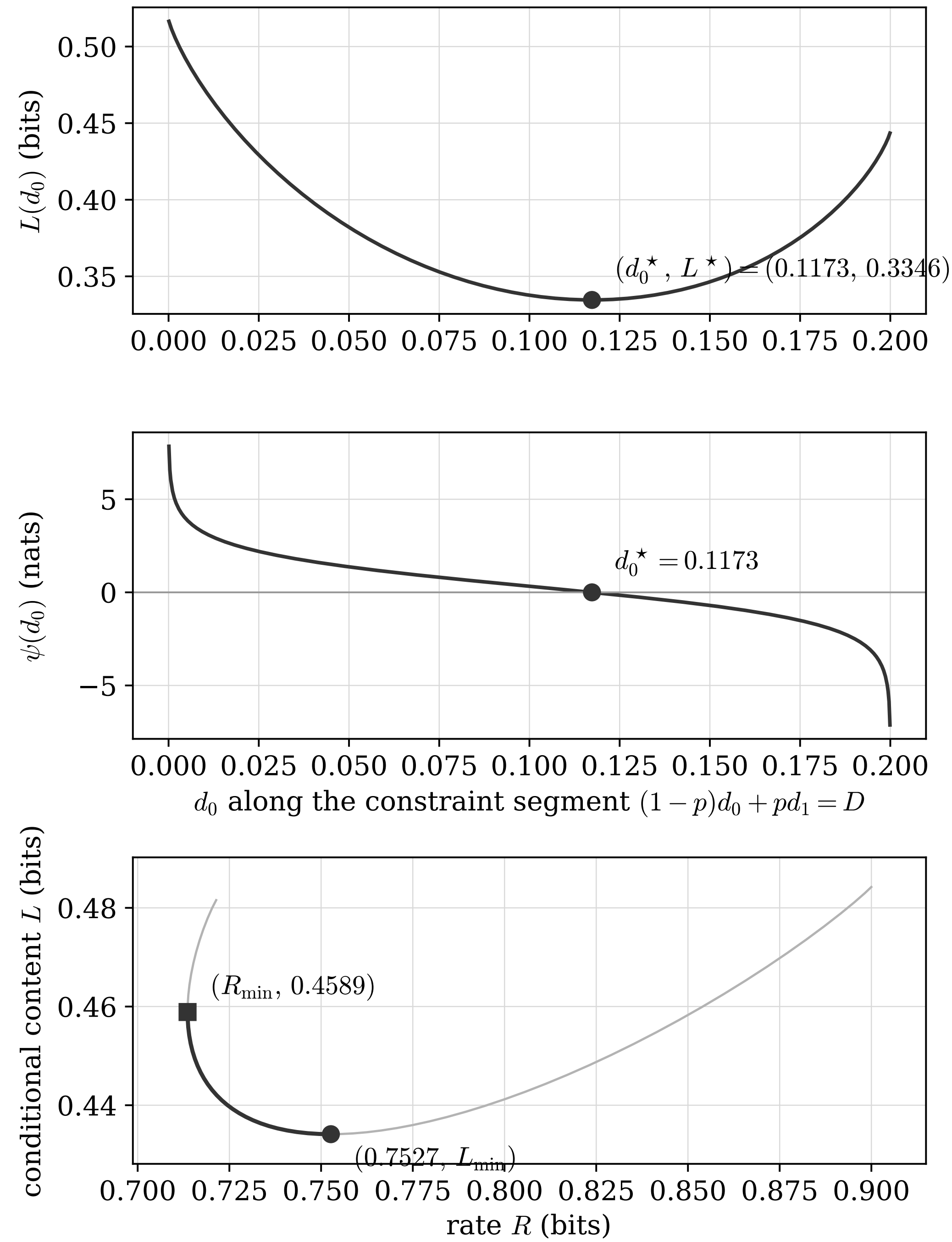

Fig. 4. The binary objective $L(d_0)$ (top) and the tilt residual $\psi(d_0)$ (middle) along the constraint segment $(1-p)d_0 + pd_1 = D$ at $(p, q, D) = (0.25, 0.15, 0.15)$: the residual crosses zero exactly once, at $d_0^\star = 0.1173$ ($d_1^\star = 0.2480$), and L attains its unique minimum 0.3346 bits there, as Lemma 26 asserts. Bottom: the binary rate–content frontier at $(p, q, D) = (0.1, 0.1, 0.05)$, the curve $\big(R(d_0), L(d_0)\big)$ along the segment (Remark 28), with the Pareto arc from the tilt root $d_0^\star = 0.0282$ (the L-minimizer, $L_{\min} = 0.4341$ bits) to $d_0 = D$ (the R-minimizer, $R_{\min} = 1 - h_2(D) = 0.7136$ bits) drawn bold. The tradeoff is strict: the L-optimal point pays 0.0391 bits of excess rate and the R-optimal point pays 0.0248 bits of excess conditional content.

Closed form. On the family, $H(\hat{Y} \mid Y, V) = (1-p)h_2(d_0) + p\,h_2(d_1)$, and $\hat{Y} \oplus S = J \oplus E \oplus W$ $(J = Y \oplus V)$ with $\Pr[J \oplus E = 1] = a$. The step $H(\hat{Y} \mid S) = h_2(a * q)$ needs $\hat{Y} \oplus S \perp S$, which holds because the joint law of $(\hat{Y}, S)$ is σ-invariant (σ-invariant source, σ-symmetric channel), forcing equal conditional flip probabilities given either value of S; the same symmetry gives $H(\hat{Y}) = 1$, hence R.

Tilt equation. Substituting $d_1 = (D-(1-p)d_0)/p$ makes $a = 2(1-p)d_0 + p - D$ affine in d_0; differentiating L and using $h_2'(x) = \ell(x)$ (nats) gives $(1-p)[2(1-2q)\ell(u) - \ell(d_0) + \ell(d_1)] = 0$. Since L restricted to the affine family slice is convex (restriction of a convex function), interior stationarity is sufficient, not merely necessary.

Anchors and positivity. $q = \frac{1}{2}$: $u = \frac{1}{2}$, the tilt equation forces $d_0 = d_1 = D$. $q = 0$: substitution of Gray's backward channel; equality of values is elementary algebra. Positivity, in two branches over the family (which suffices by attainment): if $a \neq \frac{1}{2}$, then $h_2(u) > h_2(a) \geq (1-p)h_2(d_0) + p\,h_2(1-d_1) = (1-p)h_2(d_0) + p\,h_2(d_1)$, where the strict step holds because $u = a * q$ is strictly closer to $\frac{1}{2}$ for $q \in (0, \frac{1}{2}]$ and the second step by concavity; so $L > 0$. If $a = \frac{1}{2}$, then $u = \frac{1}{2}$, $h_2(u) = 1$, and $L = 1 - (1-p)h_2(d_0) - p\,h_2(d_1) \geq 1 - h_2(D) > 0$ by concavity at the distortion. □

Reproduction alphabets larger than binary are already covered by the data-processing reduction that opens this section; no separate argument is needed for them. The unconditional Lagrangian bounds, which assume no convexity, numerically re-confirm that reduction and the family-plus-tilt optimum.

Remark 28 (The binary rate–content frontier)**.** *The symmetric family exhausts not only the conditional coordinate but the entire rate–content frontier at distortion D. The rate coordinate $R = I(Y, V; \hat{Y})$ is convex in the shared channel by the argument of Lemma 25 with the trivial conditioner, so every weighted objective $\alpha R + (1-\alpha)L$, $\alpha \in [0, 1]$, is convex in the channel. The global bit flip σ of Theorem 27's proof preserves the joint law of (Y, V, S), the distortion, and both coordinates: the flip is a law-preserving bijective relabeling of both arguments of each mutual information, so $R(C^\sigma) = R(C)$ and $L(C^\sigma) = L(C)$. Averaging a channel with its flip therefore preserves the distortion and, by convexity, does not increase any weighted objective, and larger reproduction alphabets are covered by the data-processing reduction in both coordinates. Every weighted optimum is thus attained in the family $\{(d_0, d_1)\}$, and the distortion constraint may be taken active there: both coordinates are nonnegative and vanish at $d_0 = d_1 = \frac{1}{2}$, so by convexity the objective does not increase along the straight path from any point with distortion slack toward $(\frac{1}{2}, \frac{1}{2})$, while the distortion, affine in (d_0, d_1) with positive weights, rises continuously to $\frac{1}{2} > D$ and meets the budget on the way. The Pareto frontier of the binary rate–content region at distortion D is therefore the lower-left envelope of the curve*

$$d_0 \longmapsto \big(R(d_0),\, L(d_0)\big), \qquad R(d_0) = 1 - (1-p)\,h_2(d_0) - p\,h_2\big(d_1(d_0)\big),$$

along the constraint segment. On the segment, R is strictly convex with its unique minimum at $d_0 = d_1 = D$ (strict concavity of $(1-p)h_2(d_0) + p\,h_2(d_1)$ and the stationarity $\ell(d_0) = \ell(d_1)$), where $R = 1 - h_2(D)$, the marginal rate–distortion function; L has its unique minimum at the tilt root $d_0^\star$ (Lemma 26). For $q \in (0, \frac{1}{2})$ the two minimizers are distinct, since $\psi(D) = -2(1-2q)\,\ell\big(u(D)\big) < 0$ (there $u(D) < \frac{1}{2}$ because $a(D) = p + D(1-2p) < \frac{1}{2}$), so $d_0^\star < D$: on $[d_0^\star, D]$ the curve is a strict tradeoff arc, L increasing and R decreasing, and outside that arc both coordinates worsen. At $q = \frac{1}{2}$ the two minimizers coincide at $d_0 = D$ and the frontier is a single corner. Fig. 4 traces the frontier at a representative instance.

The nearest published binary calculation is the binary Heegard–Berger problem with encoder side information and common reconstruction of Lu, Xu, Qian, and Wang [19], in which the encoder observes side information that may differ from the decoder's. Under the natural relabeling of their source and side-information variables, that model appears closely related to the conditional endpoint considered here; Appendix D fixes the relabeling and identifies which endpoint results are inherited. Two further common-reconstruction calculations are decoder-side-information ones. Steinberg's binary example [12] evaluates the constrained functional for a binary symmetric pair: the decoder's side information is the source through a binary symmetric channel, the distortion is Hamming, the optimal reverse test channel is a BSC of crossover D independent of the pair correlation, and the value is $h_2(\rho_0 * D) - h_2(D)$ at side-channel crossover ρ_0. Ahmadi, Tandon, Simeone, and Poor [14] computed common-reconstruction functions for a binary source with *erased* side information under Hamming and erasure distortion, in point-to-point, Heegard–Berger, and cascade configurations; the point-to-point value is $p_e(1 - h_2(D))$ at erasure rate p_e. Both computations live in the decoder-side-information setting: S sits at the decoder, the encoder observes the source alone, and the common-reconstruction constraint disciplines how the decoder may use S. Theorem 27 differs structurally. Here S is held by a third party that never decodes, the encoder observes the pair (Y, V), and the constraint binds the unconditional Hamming error $\Pr[\hat{Y} \neq Y]$. The function is anchored at Gray's conditional function at $q = 0$ and at the marginal rate–distortion function at $q = \frac{1}{2}$, an interpolation with no counterpart in the decoder-side calculations; whether its $q = 0$ conditional endpoint coincides with the value of [19] under the relabeling is settled in Appendix D. An encoder restricted to Y alone faces exactly the constrained problem of the pair (Y, S), a binary symmetric pair with crossover $p * q$, and pays Steinberg's value there (the marginalization identity of Section IV is alphabet-free); the tilt equation is the price of pair access.

IX. Discussion and Conclusion

This paper characterized the minimal conditional content of a description whose encoder observes the pair (Y, V) while the conditioner S stays with a third party: the closed form of Theorem 13 for Y and V scalar linear functionals of an arbitrary-dimensional jointly Gaussian source, the exact rate–content region of Theorem 16 with its two-water-level frontier and the misalignment dichotomy of Corollary 19, the determinant-bound attainment condition of Theorem 23, the vector-context extension of Theorem 24, and the complete binary solution of Theorem 27. The function degenerates to the classical rate–distortion function at $\rho = 0$ and to Gray's conditional function as $\tau^2 \to 0$, and meets Steinberg's scalar Gaussian formula at the $\rho^2 \to 1$ boundary. The structural moral is Corollary 21: the function is not determined by the reduced marginal (Y, S) but by what the encoder can see, so conditional description content is a property of the encoding configuration, not of the source and side-information pair alone.

The thermodynamic reading is that of Remark 6: $L(D)$ is the minimal ideal erasure work per symbol, in units of $k_{\mathrm{B}}T \ln 2$, paid by a reset mechanism that retains S. The region states when storing cheaply and erasing cheaply are compatible: at $\rho = 0$ one channel minimizes both coordinates, and under misalignment ($\rho^2 \in (0, 1)$, $\tau^2 > 0$) the two costs strictly trade off.

The same asymmetry, descriptions read by parties whose requirements they were not sized to, extends beyond source coding. In companion empirical work by the author, the operational failure rate of decisions made from stale or mismatched descriptions plays the role that L plays here as an information price. The misaligned regime $\tau^2 > 0$ is exactly the regime such measurements probe; we claim no equivalence beyond this correspondence of roles.

Four problems are open. The exact rate–content region for a joint description serving two described variables is not settled; its stationarity system is known in necessary-condition form only. The analogous function for a described variable of dimension greater than one is open. Stationary sources, and the process and spectral versions of the function, are not treated here. The finite-blocklength behavior of the pair (R_n, L_n) is unexplored.

Everything in this paper is single-letter and memoryless, Gaussian and binary. Every closed form is verified by the independent numerical checks of Appendix C.

Appendix A
Single-Letterization

For completeness, we justify the last inequalities in (12) and (13). For the rate coordinate,

$$\begin{aligned} I(X^n; \widehat{X}^n) &= H(X^n) - H(X^n \mid \widehat{X}^n) \\ &= \sum_{i=1}^{n} H(X_i) - \sum_{i=1}^{n} H(X_i \mid X^{i-1}, \widehat{X}^n) \\ &\geq \sum_{i=1}^{n} \Big[H(X_i) - H(X_i \mid \widehat{X}_i) \Big] \\ &= \sum_{i=1}^{n} I(X_i; \widehat{X}_i). \end{aligned} \tag{45}$$

For the content coordinate,

$$\begin{aligned} I(X^n;\widehat{X}^n \mid S^n) &= H(X^n \mid S^n) - H(X^n \mid \widehat{X}^n, S^n) \\ &= \sum_{i=1}^n H(X_i \mid S_i) - \sum_{i=1}^n H(X_i \mid X^{i-1}, \widehat{X}^n, S^n) \\ &\geq \sum_{i=1}^n \Big[H(X_i \mid S_i) - H(X_i \mid \widehat{X}_i, S_i) \Big] \\ &= \sum_{i=1}^n I(X_i;\widehat{X}_i \mid S_i). \end{aligned} \tag{46}$$

Appendix B
The Gaussian Operational Theorem

This appendix proves the operational region theorem for the jointly Gaussian source of Section III, replacing the customary appeal to an unspecified abstract-alphabet extension. The source letter is $X = (Y, V)$, jointly Gaussian; the conditioner is S as in Section III or Section VII; the distortion is $d(x, \hat{y}) = (y - \hat{y})^2$, unbounded; codes, R_n, D_n, and $L_n = H(M \mid S^n)/n$ are as in Section II, and achievability is Definition 1.

Two devices are fixed first. The *saturating dyadic quantizers*: for $k \geq 1$, let q_k partition $\mathbb{R}$ into the bounded cells $[j2^{-k}, (j+1)2^{-k})$, $-4^k \leq j < 4^k$, together with the two saturating tail cells $(-\infty, -2^k)$ and $[2^k, \infty)$; for a vector conditioner $S \in \mathbb{R}^r$, apply q_k to each coordinate, so the cells are rectangles. The partitions are nested, every cell edge of q_k being an edge of q_{k+1}, so the generated σ-fields $\sigma(q_k(S))$ increase, and their union generates the Borel σ-field of S, since the dyadic intervals do. Write $S_k := q_k(S)$. The second device is an integrability bound for the quantized pair.

Lemma 29 (Cell bound)**.** *Let $(\widehat{Y}, S)$ be jointly Gaussian, $\widehat{Y}$ scalar and centered, $S \in \mathbb{R}^r$, with $\Sigma_{S|\widehat{Y}} \succ 0$, and fix k. For every cell A of the product quantizer q_k there is a constant $C_A < \infty$ such that*

$$\left| \ln \frac{\Pr(S \in A \mid \widehat{Y} = \hat{y})}{\Pr(S \in A)} \right| \;\leq\; C_A\big(1 + \hat{y}^2\big) \qquad \textit{for every } \hat{y} \in \mathbb{R}.$$

Consequently the information density of the pair $(\widehat{Y}, S_k)$ is bounded in magnitude by a fixed quadratic in $\hat{y}$, and has finite moments of all orders.

Proof. The conditional law of S given $\widehat{Y} = \hat{y}$ is Gaussian with mean $m\hat{y}$ linear in $\hat{y}$ and fixed covariance $\widetilde{\Sigma} := \Sigma_{S|\widehat{Y}} \succ 0$; the marginal of S is Gaussian with covariance $\Sigma_S \succ 0$.

Scalar S, bounded cells. For a bounded cell $A = [u, v)$, the ratio of the two integrals lies between the infimum and the supremum over A of the integrand ratio $r(u', \hat{y}) := f_{S|\widehat{Y}}(u' \mid \hat{y})/f_S(u')$, so $|\ln(\Pr(A \mid \hat{y})/\Pr(A))| \leq \sup_{u' \in A} |\ln r(u', \hat{y})|$. Here

$$\ln r(u', \hat{y}) = \ln \frac{\sigma_S}{\widetilde{\sigma}} - \frac{(u' - m\hat{y})^2}{2\widetilde{\sigma}^2} + \frac{u'^2}{2\sigma_S^2},$$

a quadratic in $(u', \hat{y})$; taking the supremum over the bounded range $u' \in [u, v)$ leaves a quadratic in $\hat{y}$ with coefficients determined by the cell endpoints, which is the claimed bound.

Scalar S, tail cells. For $A = [2^k, \infty)$, $\Pr(A \mid \hat{y}) = \overline{\Phi}\big((2^k - m\hat{y})/\widetilde{\sigma}\big)$ and $\Pr(A) = \overline{\Phi}(2^k/\sigma_S)$, a constant, where $\overline{\Phi}$ is the standard Gaussian tail. The elementary bound $|\ln \overline{\Phi}(z)| \leq C(1+z^2)$ holds on all of $\mathbb{R}$: for $z \leq 0$, $\overline{\Phi}(z) \in [\frac{1}{2}, 1)$; for $0 < z \leq 1$, $\overline{\Phi}(z) \in [\overline{\Phi}(1), \frac{1}{2})$; and for $z > 1$ the Mills-ratio bounds $z\varphi(z)/(1+z^2) \leq \overline{\Phi}(z) \leq \varphi(z)/z$ give $|\ln \overline{\Phi}(z)| \leq z^2/2 + \ln\sqrt{2\pi} + \ln(2z)$. With $z = (2^k - m\hat{y})/\widetilde{\sigma}$, $z^2 \leq 2(4^k + m^2\hat{y}^2)/\widetilde{\sigma}^2$ is a quadratic in $\hat{y}$, and the marginal term is a constant; the mirror cell is symmetric.

Vector S, rectangular cells. For probabilities in $(0, 1]$, $|\ln(P/Q)| \leq |\ln P| + |\ln Q|$, so it suffices to bound $|\ln \Pr(S \in A \mid \hat{y})|$ (the marginal term is a constant). The rectangle A contains the bounded rectangle A' obtained by clipping each infinite side to length one at its finite end, and

$$\Pr(S \in A \mid \hat{y}) \;\geq\; \Pr(S \in A' \mid \hat{y}) \;\geq\; \mathrm{vol}(A')\, \frac{\exp\!\big(-\frac{1}{2}\sup_{u \in A'}(u - m\hat{y})^{\mathsf{T}}\widetilde{\Sigma}^{-1}(u - m\hat{y})\big)}{(2\pi)^{r/2}\det(\widetilde{\Sigma})^{1/2}},$$

where the supremum over the bounded A' is a quadratic in $\hat{y}$ (here m is the vector regression coefficient). Since also $\ln \Pr(S \in A \mid \hat{y}) \le 0$, the claimed bound follows.

Consequence. The information density of $(\widehat{Y}, S_k)$ at $(\hat{y}, j)$ is $\ln[\Pr(S_k = j \mid \hat{y})/\Pr(S_k = j)]$; taking $C := \max_j C_{A_j}$ over the finitely many cells bounds it by $C(1+\hat{y}^2)$, which has finite moments of all orders under the Gaussian law of $\widehat{Y}$. □

In the application below, $\Sigma_{S|\widehat{Y}} \succeq \Sigma_U \succ 0$ automatically, because $S = V + U$ with U independent of $(X, \widehat{Y})$.

Theorem 30 (Extension: the Gaussian operational region)**.** *For the Gaussian source and every* $0 < D < 1$,

$$\mathcal{RW}(D) = \bigcup_{p(\hat{y}|x):\, \mathbb{E}(Y-\widehat{Y})^2 \le D} \big\{(R,L) : R \ge I(X;\widehat{Y}),\ L \ge I(X;\widehat{Y} \mid S)\big\},$$

and the union is unchanged when restricted to jointly Gaussian test channels. The nontrivial range is $0 < D < 1$*, and the compactness step of the proof uses* $D < 1$*. For* $D \ge 1$ *the result follows from the zero reproduction, which meets the distortion constraint with* $R = L = 0$*; the point* $D = 0$ *follows as the extended-real limiting case, both coordinates diverging. Throughout,* L_n *is a coding quantity, the normalized conditional entropy of the stored index; its thermodynamic reading is confined to Remark 6, and no step of the proof depends on it.*

Proof. Converse. The chain of Theorem 5's converse (the discrete operational region theorem) holds verbatim with differential entropies for the continuous source. M is discrete, so $\log|\mathcal{M}_n| \ge H(M) \ge I(X^n; M) \ge I(X^n; \widehat{Y}^n)$ and $H(M \mid S^n) \ge I(X^n; M \mid S^n) \ge I(X^n; \widehat{Y}^n \mid S^n)$ are unchanged. The single-letterization of Appendix A holds with h in place of H: for the jointly Gaussian source all the differential entropies involved are finite (the unconditional ones are Gaussian values, and each conditional one differs from an unconditional one by a mutual information pinned between 0 and $H(M) \le \log|\mathcal{M}_n| < \infty$, since $\widehat{Y}^n$ is a function of the finite-valued M), the i.i.d. chain rules are unchanged, and conditioning reduces differential entropy. The per-letter induced channels satisfy the Markov constraint by the i.i.d. structure, exactly as in the finite case; the mixture channel $\bar{q}$ is defined by averaging the induced conditional laws, its distortion is D_n, and the convexity step is Lemma 3 (convexity of both coordinates in the test channel) with the per-s sum replaced by the integral against the law of S (regular conditional probabilities, as in Theorem 16's proof). Hence (R_n, L_n) lies in the displayed union at distortion D_n.

Closedness and the passage from D_n to D. By the restriction step below, the union at any distortion level $D' \in (0,1)$ equals the union over jointly Gaussian test channels, and these are parameterized by their regression vector and residual variance: in the whitened coordinates of (37) (the simultaneous congruence $\Sigma_T \to I$, $\Sigma_{T|S} \to \Lambda$), by pairs $\theta = (c, n)$ with $Q_0 = |c|^2$, $Q_1 = c^\mathsf{T} \Lambda c$, distortion $|y_0 - c|^2 + n \le D'$, and coordinates $\big(\frac{1}{2}\log_2 \frac{Q_0+n}{n},\ \frac{1}{2}\log_2 \frac{Q_1+n}{n}\big)$. The distortion constraint confines θ to a compact set: $|y_0 - c|^2 \le D'$ is a closed ball and $0 \le n \le D'$; and feasibility forces $c \ne 0$ there, since $c = 0$ gives distortion at least $|y_0|^2 = 1 > D'$. Both coordinates are lower semicontinuous on that set with values in $(0, \infty]$, finite and continuous where $n > 0$ and diverging as $n \to 0$ at $c \ne 0$. Now let (R_j, L_j) lie in the union at levels D'_j with $\limsup_j D'_j \le D$ and $(R_j, L_j) \to (R, L)$ finite. Pick feasible θ_j whose quadrant contains (R_j, L_j); the θ_j eventually lie in the compact set at level $D + \delta$ for each $\delta > 0$, so a subsequence converges to some θ^*, feasible at D by continuity of the distortion in θ. Lower semicontinuity gives $\frac{1}{2}\log_2\big[(Q_0+n)/n\big](\theta^*) \le \liminf_j \frac{1}{2}\log_2\big[(Q_0+n)/n\big](\theta_j) \le R$, and likewise for the conditional coordinate, so (R, L) lies in θ^*'s quadrant, hence in the union at D. This establishes both closedness at fixed distortion and right-continuity in the distortion level, and with $\limsup D_n \le D$ places every limit point of (R_n, L_n) in the union at D.

Restriction to Gaussian channels. For any admissible channel, Lemma 11 (Gaussian exhaustion: the two determinant lower bounds, met with equality by the jointly Gaussian channel matching the second moments) produces the jointly Gaussian channel with the same second moments and the same distortion, whose coordinates $I(X;\widehat{Y})$ and $I(X;\widehat{Y} \mid S)$ are componentwise no larger than the original channel's. Its quadrant therefore contains the original channel's quadrant, so the union over all channels equals the union over jointly Gaussian channels. It thus suffices to prove achievability for a jointly Gaussian test channel.

Achievability, Step A: quantizing the conditioner. Fix a jointly Gaussian admissible channel $p(\hat{y} \mid x)$ with distortion at most D, and fix $\epsilon > 0$. Along the saturating dyadic quantizers,

$$I(\widehat{Y}; S_k) \uparrow I(\widehat{Y}; S) \qquad (k \to \infty) : \tag{47}$$

the sequence is nondecreasing because the partitions are nested (data processing), and its supremum is $I(\widehat{Y};S)$ by monotone convergence of mutual information under σ-fields increasing to the Borel σ-field of S [7, Ch. 2]: mutual information is the supremum of the mutual informations of finite measurable partitions of the two sample spaces, and every finite Borel partition of the S-space is approximated, up to sets of arbitrarily small measure, by partitions measurable with respect to some $\sigma(S_k)$. Choose k with

$$I(\widehat{Y};S_\Delta) \;\ge\; I(\widehat{Y};S)-\epsilon, \qquad S_\Delta := S_k. \tag{48}$$

S_Δ is a function of S, so the Markov chain $\widehat{Y}-X-S_\Delta$ holds, and by the chain-rule identity used throughout $(I(X;\widehat{Y})-I(X;\widehat{Y}\mid S')=I(\widehat{Y};S')$ for any conditioner S' with $\widehat{Y}-X-S')$,

$$I(X;\widehat{Y}\mid S_\Delta)=I(X;\widehat{Y})-I(\widehat{Y};S_\Delta) \;\le\; I(X;\widehat{Y}\mid S)+\epsilon. \tag{49}$$

Achievability, Step B: coding against the quantized conditioner. The pair source is now (X,S_Δ) with continuous X and finite S_Δ. Build the code of Theorem 5's achievability proof (covering, binning, and expurgation, as in the discrete theorem) for this pair: a codebook of $2^{n(I(X;\widehat{Y})+2\epsilon)}$ codewords drawn from the $\widehat{Y}$-marginal with quadratic-distortion covering, a uniform random binning at rate $I(X;\widehat{Y}\mid S_\Delta)+3\epsilon$, and the side-information proof device decoding M from (B,S_Δ^n). Two classical probabilistic ingredients are imported. The covering step is Gaussian quadratic-distortion achievability for an arbitrary jointly Gaussian test channel, the Gaussian source coding theorem with the backward-channel argument [2]. The side-information step, the joint typicality of the selected codeword with the memoryless side information (the Markov lemma) together with the packing bound for wrong codewords in a bin, is the abstract-alphabet analysis of Wyner [6]. That analysis needs only that the per-letter information densities of the pairs $(X,\widehat{Y})$ and $(\widehat{Y},S_\Delta)$ be integrable, so that they obey the law of large numbers and the density-typical sets carry the standard change-of-measure bounds. Both conditions hold here: the $(X,\widehat{Y})$ density is a quadratic polynomial in $(x,\hat{y})$ for the jointly Gaussian pair, and the $(\widehat{Y},S_\Delta)$ density is bounded by a fixed quadratic in $\hat{y}$, with finite moments of all orders, by Lemma 29. Everything else, in particular the expurgation selecting one deterministic (codebook, binning) pair meeting the distortion and the conditional-entropy requirements simultaneously, is the argument written out in Theorem 5's proof, verbatim (the distortion excess on the covering-failure event is controlled by the uniformly integrable quadratic distortion of the Gaussian pair in place of $d_{\max}$: encode the failure event by the zero codeword, whose per-letter distortion has fixed finite mean, so a vanishing failure probability contributes $o(1)$ to D_n by Cauchy–Schwarz). The selected code has rate $I(X;\widehat{Y})+2\epsilon$, distortion at most $D+\epsilon+o(1)$, and

$$H(M\mid S_\Delta^n) \;\le\; n\big(I(X;\widehat{Y}\mid S_\Delta)+3\epsilon+o(1)\big). \tag{50}$$

Achievability, Step C: back to the unquantized conditioner. S_Δ^n is a function of S^n, so $\sigma(S_\Delta^n)\subseteq\sigma(S^n)$ and conditioning on the finer σ-algebra cannot increase entropy:

$$H(M\mid S^n) \;\le\; H(M\mid S_\Delta^n) \;\le\; n\big(I(X;\widehat{Y}\mid S_\Delta)+3\epsilon+o(1)\big) \;\le\; n\big(I(X;\widehat{Y}\mid S)+4\epsilon+o(1)\big), \tag{51}$$

by (50) and (49).

Collect. For every $\epsilon>0$ the triple $\big(I(X;\widehat{Y})+2\epsilon,\; I(X;\widehat{Y}\mid S)+4\epsilon,\; D+\epsilon\big)$ is achievable; a diagonal sequence $\epsilon_j\downarrow 0$ of codes achieves, in the $\limsup$ sense of Definition 1, the corner $\big(I(X;\widehat{Y}),I(X;\widehat{Y}\mid S)\big)$ at distortion D, and the quadrant above the corner is immediate from the definition. Together with the converse and the restriction step, this proves the theorem. □

The imports are stated for the record: Gaussian quadratic-distortion covering [2] and Wyner's abstract-alphabet side-information analysis [6] are classical, predate this work, and are logically independent of the present results; every other step is proved here or in Section II.

Appendix C
Numerical Verification

Each closed form and each displayed matrix identity was checked independently of the derivations, symbolically by exact computer algebra and numerically against direct optimization over test-channel moments, twice over: by the author's harness, written alongside the proofs, and by a separately commissioned re-derivation, archived with the repository. All checks pass. The load-bearing scalar algebra is additionally machine-checked in two further

TABLE IV
INDEPENDENT CHECKS OF THE CLOSED FORMS AND PROOF IDENTITIES.

Check	Method	Result
Determinant identity of Lemma 11	symbolic; 200 random instances at p=3, r=2, m=2	exact; max rel. error 2.5×10^{-13}
Closed form $L(D) = \frac{1}{2}\log_2 g^\star$, quadratic reduction, and achieving channel	symbolic; direct minimization (6 instances); multistart at $\alpha = \frac{1}{2}$	exact; max deviation 3.9×10^{-16} bits; spread 6.1×10^{-9}
Frontier at $\alpha \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$; vector-context system (44) at $r = 2$; the six values of Corollary 21	direct weighted and brute-force minimization; exact surds	residuals $\leq 3.7 \times 10^{-9}$; four-decimal values confirmed
Global optimality in Theorem 27	Lagrangian bounds, no convexity assumed, reproduction alphabets $2, 4, 6$	worst residual 3.4×10^{-14}
Anchor convergence rates (Remark 15)	symbolic implicit differentiation at the simple root; numeric gap ratios down to $\epsilon = 10^{-6}$	coefficients exact; ratios within 8.0×10^{-4} of 1
Algebraic core (quadratic, roots, anchors, surds, binary anchor)	Lean 4/Mathlib, `CRContext.lean`	builds clean; zero `sorry`
Determinant identities, stationarity solution, reduction, gradients	MATLAB Symbolic (R2026a), `matlab_checks.m`	11/11 symbolic pass

engines, a Lean 4/Mathlib module built clean with zero `sorry` and the MATLAB Symbolic toolbox. The matrix-general exhaustion lemma, the measure-theoretic steps (single-letterization, covering and binning, disintegration), the matrix-convexity lemma, the uniqueness proposition, and the binary theorem's symmetrization and positivity arguments are verified by written proof and the numerical harnesses, not by the proof assistant. The scripts, their archived run outputs, and the full check inventory accompany the manuscript sources at the repository of the title footnote (`VERIFICATION.md` in the paper directory). Table IV summarizes the main checks.

APPENDIX D
COMPARISON WITH THE BINARY COMMON-RECONSTRUCTION FUNCTION OF LU *et al.*

Lu, Xu, Qian, and Wang [19] solve a binary Heegard–Berger problem with encoder side information and a common-reconstruction constraint. Their setting is matched to the conditional endpoint of Section VIII by the relabeling

$$X_{\mathrm{Lu}} \leftrightarrow Y, \qquad S_{\mathrm{enc}} \leftrightarrow V, \qquad S_{\mathrm{dec}} \leftrightarrow S,$$

under which their encoder-observed side information plays the role of the encoder's context V and their decoder side information plays the role of the third party's copy S. At the conditional endpoint (content only) the present functional is $\min I(T; \widehat{Y} \mid S)$ under $\widehat{Y} - T - S$ with Hamming distortion on Y, the object their construction evaluates for the doubly symmetric binary source, so under this relabeling their setting appears to reproduce the conditional optimization here. The formula-by-formula identification of their reported value with $L(D)$ at $q = 0$, and the delineation of which endpoint results are inherited, are to be completed against the primary source; the conference article is closed access and the identification is not asserted until it is checked. The joint (R, L) frontier and the tilt parametrization of Remark 28, which price both coordinates at once, are not to our knowledge given by [19] and are the part of the binary contribution least affected by this comparison.

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